RITEEC 2025
Electrical Engineering

1st International Conference on

RECENT INNOVATIONS AND TRENDS IN ELECTRICAL & ELECTRONICS ENGINEERING AND COMPUTING

May 22-23, 2025

SOUVENIR CUM BOOK OF ABSTRACT

Organized by
Department of Electrical Engineering
National Institute of Technology Patna

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2025 First International Conference on Recent Innovations and Trends in Electrical & Electronics Engineering and Computing (RITEEC)

RITEEC-2025
NIT Patna

May 22-23, 2025

Department of Electrical Engineering

National Institute of Technology Patna

https://sites.google.com/nitp.ac.in/riteec-2025

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WELCOME NOTE FROM THE DIRECTOR NIT PATNA

राष्ट्रीय प्रौद्योगिकी संस्थान पटना
NATIONAL INSTITUTE OF TECHNOLOGY PATNA
(Institute of National Importance under Ministry of Education, Govt. of India)
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प्रो. प्रदीप कु. जैन/*Prof. Pradip K. Jain*
निदेशक/*Director*

15th May, 2025

Message

It is a pleasure to know that the Department of Electrical Engineering, NIT Patna, is organising the International Conference on "Recent Innovations and Trends in Electrical & Electronics Engineering and Computing (RITEEC 2025)", from 22nd to 23rd May, 2025.

In today's rapidly evolving technological landscape, the fields of Electrical and Electronics Engineering and Computing are at the forefront of innovation, shaping the future of industries and society at large. This conference is a wonderful opportunity put forth by NIT Patna and it will facilitate great learning opportunities to the academicians, scientists, students and industry persons who are eager to explore the convergence of cutting-edge technologies in these domains and will enable participants to stay updated with the latest advancements in these areas.

I extend my sincere appreciation to the organizing team, speakers and all participants for their enthusiasm in making this event a grand success. I wish RITEEC 2025 all the very best and hope that it will be a memorable and enriching experience for everyone involved.

(P. K. Jain)

WELCOME NOTE FROM THE ORGANIZING COMMITTEE

This International conference on Recent Innovations and Trends in Electrical & Electronics Engineering and Computing (RITEEC 2025) is organized by the Department of Electrical Engineering, National Institute of Technology, Patna, Bihar, India. RITEEC 2025 is scheduled to be in hybrid mode between 22nd May 2025 and 23rd May 2025. This conference will feature a comprehensive technical program including several special sessions and workshops. The goal of RITEEC 2025 is to provide an outstanding forum for academicians, researchers, practitioners, policymakers and students to exchange ideas, techniques and tools, raise awareness, and share experiences related to all practical and theoretical aspects of intelligence and innovations in engineering.

RITEEC 2025 focuses on cutting-edge topics in electrical systems, electronics engineering, and computing technologies, covering areas such as power systems, power electronics, control systems and applications, renewable energy, signal processing, VLSI, circuits and systems, embedded systems, industrial instrumentation and automation, communication technologies, artificial intelligence, IoT, Database Management Systems, Bio Inspired Computing, Image Processing etc. Participants share research findings, present new methodologies, and discuss challenges and opportunities shaping the future of these interdisciplinary fields. .

The conference had received papers from IITs, NITs, reputed organizations within and outside the country. The papers have been selected after a rigorous two-stage review process to ensure the quality of the conference. The conference received 657 submissions out of which 148 papers were accepted and registered for presentation. The acceptance rate of this conference is *22.5%*. The conference programme has been divided into seventeen technical Sessions. The session chairs have been invited from across different IITs, NITs, and reputed government institutes. The Conference is privileged to have four Plenary talks by the renowned experts; *Prof. Narayana Prasad Padhy,* Director, MNIT Jaipur, *Prof. Raj Senani, NSUT, Delhi, Prof. Shantanu Mishra, IIT Delhi* and *Dr. Sanjay Yadav*, Former Chief Scientist and Prof. Physical Sciences, AcSIR, CSIR-National Physical Laboratory (CSIR-NPL), New Delhi, Vice President, MSI and USI, New Delhi.

The organizing committee extends its heartfelt gratitude to the Director, NIT Patna for his patronage towards organization of the conference. We also extent our thankfulness to the Deans, Registrar, and the other staff members of NIT Patna to help us in organizing this conference.

We are grateful to the dignitaries of the conference; *Prof. Narayana Prasad Padhy, Prof. Raj Senani, Prof. Shantanu Mishra,* and *Dr. Sanjay Yadav* for agreeing to deliver the plenary talks for this conference. We sincerely thank technical advisory committee, technical program committee, reviewers and the Ph.D. and M.Tech. volunteers who extended very sincere efforts along with the organizers. Lastly it is our pleasure to thank the participants who have submitted and will be deliberating their research works in this conference.

We earnestly hope that the participants will find the conference very enriching and fruitful for finding directions in research and collaborations. The organizing committee once again welcomes you in this initiative to take part in the deliberations on possibilities of collaborative research and innovative ideas.

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CONFERENCE PROGRAM

1st International Conference on Recent Innovations and Trends in Electrical & Electronics Engineering and Computing (Hybrid Mode)

(RITEEC– 2025)

22-23 May, 2025

DAY 1: Thursday, May 22, 2025	
Time	**Program**
09:30 - 10:30	***Inauguration Ceremony (Hybrid Mode)***
	Web Link: https://teams.microsoft.com/l/meetup-join/19%3ameeting_MzgzMmVhYWUtMjNkYS00MWQ5LThkNDEtNTc4MjI1OTFlYjQ3%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d
10:30 - 11:00	***High Tea***
11:00 - 11:45	**Plenary Talk 1: Prof. N. P. Padhy** **Title: Empowering Renewable Energy Integration: A Field Deployment Experience** **Venue: Vishveshwarya Hall, NIT Patna**
11:45 - 12:30	**Plenary Talk 2: Prof. Raj Senani** **Title: Analog Integrated Circuits and Signal Processing: A brief Historical Overview** **Venue: Vishveshwarya Hall, NIT Patna**
12:30 - 14:30	***Lunch***

Date: 22 May 2025 14:30 - 16:30	***Day 1 | Technical Session-1 (TS1)***

Date: 22 May 2025 Time(14:30 – 16:30)	**TS1.1 - Control and Automation**
Mode: Offline	**Venue: Vishveshwarya Hall, NIT Patna**

14:30 – 14:45 (TS1.1.1)	***Paper Id***: *426* ***Title:*** *Design of Multi-stage Fractional-Order controller for Load Frequency Control of Islanded Multi-Microgrid System* ***Author Name***:*Manish Kumar* ***Email Id***: *phdee10054.23@bitmesra.ac.in*
14:45 – 15:00 (TS1.1.2)	***Paper Id:*** *51* ***Title:*** *DenseNet-Powered Textline Segmentation in Degraded Tamil Manuscripts for Digital Preservation* ***Author Name:*** *Dr.P.Prakash* ***Email Id:*** *prakashp_mit@annauniv.edu*
15:00 – 15:15 (TS1.1.3)	***Paper Id:*** *607* ***Title:*** *Evaluation of Tea Quality Marker using Equivalent Circuit of Voltammetric System: A Statistical and Machine Learning Approach* ***Author Name:*** *Dr. Sanjeev Kumar* ***Email Id:*** *sanjnitp@gmail.com*
15:15 – 15:30 (TS1.1.4)	***Paper Id:*** *202* ***Title:*** *Determination of Proximity to Voltage Instability Under Severe Line Outage Using Customized Artificial Neural Network* ***Author Name:*** *Shrikant Durgadaspant Naik* ***Email Id:*** *naiksd@rknec.edu*
15:30 – 15:45 (TS1.1.5)	***Paper Id:*** *350* ***Title:*** *An Accurate fuel level measurement system under sloshing conditions using data fusion approach* ***Author Name:*** *Prakash Ranjan* ***Email Id:*** *ranjanprakash966@gmail.com*
15:45 – 16:00 (TS1.1.6)	***Paper Id:*** *540* ***Title:*** *Variable structural control based on ADRC for unstable non-minimum phase processes* ***Author Name:*** *Anshu Priya* ***Email Id:*** *anshup.ph23.ee@nitp.ac.in*
16:00 – 16:15 (TS1.1.7)	***Paper Id:*** *545* ***Title:*** *A New Solar-Powered EV Charging with Grid Integration and Bidirectional Power Transfer* ***Author Name:*** *Ram Kumar* ***Email Id:*** *ramahk92@gmail.com*

Date: 22 May 2025 **Time(14:30 – 16:30)**	**TS1.2 - Power Electronics and Drives**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_N2FhZjVjYjUtYjhhYy00NTBiLWFlN2QtNGFlNjQzMjdlYWVi%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

14:40 – 14:50 (TS1.2.2)	***Paper Id:*** *30* ***Title:*** *Ultra-Gain Single -Inductor Boost DC-DC Converter* ***Author Name:*** *Vadthya Jagan* ***Email Id:*** *jagan.iitr@gmail.com*
14:50 – 15:00 (TS1.2.3)	***Paper Id:*** *84* ***Title:*** *EV Energy Management: Forecasting Energy Consumption and Range Using Machine Learning* ***Author Name:*** *Mondru Chiranjeevi* ***Email Id:*** *chiru.mondru@gmail.com*
15:00 – 15:10 (TS1.2.4)	***Paper Id:*** *103* ***Title:*** *Enhanced Dynamic Performance of a Solar-Powered SRM Drive through Advanced Control Strategies* ***Author Name:*** *Jeevarathinam P R* ***Email Id:*** *prjeevarathinam20@gmail.com*
15:10 – 15:20 (TS1.2.5)	***Paper Id:*** *186* ***Title:*** *Modelling of PMSM for Performance Evaluations using ANSYS Maxwell RMxprt* ***Author Name:*** *Dr. D D Padma Priya* ***Email Id:*** *padmapriyadesai@gmail.com*
15:20 – 15:30 (TS1.2.6)	***Paper Id:*** *240* ***Title:*** *Advancing SEPIC Converter Design for Optimal Performance in Power Electronics* ***Author Name:*** *Rahul Kumar* ***Email Id:*** *rahulk.ph21.ee@nitp.ac.in*
15:30 – 15:40 (TS1.2.7)	***Paper Id:*** *280* ***Title:*** *A Design of Highly Efficient PV Based Wireless Charger for EV at different atmospheric conditions* ***Author Name:*** *Preety Kumari* ***Email Id:*** *2023pgeeps002@nitjsr.ac.in*
15:40 – 15:50 (TS1.2.8)	***Paper Id:*** *302* ***Title:*** *Speed Control of Solar Powered BLDC Motor Using PI Controlled Boost Converter* ***Author Name:*** *Dronadula Lakshmi Charishma* ***Email Id:*** *rajanikandipati@gmail.com*
15:50 – 16:00 (TS1.2.9)	***Paper Id:*** *324* ***Title:*** *Power Factor Correction Of Bidirectional Wireless Power Transfer For EV Application* ***Author Name:*** *Roshan Saju* ***Email Id:*** *rs8854@srmist.edu.in*

Date: 22 May 2025 Time(14:30 – 16:30)	TS1.3 - Electrical Power Systems and Renewable Energy
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_YzZjNDYyNTItZTVlMy00YjFlLWJlYWEtMTFkY2E5YjMyYThi%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

14:30 – 14:40 (TS1.3.1)	***Paper Id:*** *10* ***Title:*** *Modified Self-tuned Filter based Control for Single-Phase Microgrid System* ***Author Name:*** *Pratham pundir* ***Email Id:*** *ranapratham176@gmail.com*
14:40 – 14:50 (TS1.3.2)	***Paper Id:*** *75* ***Title:*** *A Hybrid Machine Learning Approach for Fault Detection and Performance Prediction in PV Systems* ***Author Name:****NIKHIL KUSHWAHA* ***Email Id:****nikhil_2k18phdee16@dtu.ac.in*
14:50 – 15:00 (TS1.3.3)	***Paper Id:*** *79* ***Title:*** *Adaptive Differential Evolution Strategy to Improve AGC Performance of HVDC integrated Power System* ***Author Name:*** *Udaya Kumar Moyyi* ***Email Id:*** *gtchsekhar@gmail.com*
15:00 – 15:10 (TS1.3.4)	***Paper Id:*** *82* ***Title:*** *Performance Evaluation of a Grid-Connected Offshore Wind Farm Using Load Flow, Short-Circuit, and Transient Stability Analysis* ***Author Name:*** *A. Rathinavel* ***Email Id:*** *ra6705@srmist.edu.in*
15:10 – 15:20 (TS1.3.5)	***Paper Id:*** *83* ***Title:*** *Adaptive Protection Scheme for Hybrid Microgrid: Case Study on Fault Analysis and Relay Coordination in IEEE 9-Bus System* ***Author Name:*** *Raaz Haneef* ***Email Id:*** *rh7928@srmist.edu.in*
15:20 – 15:30 (TS1.3.6)	***Paper Id:*** *89* ***Title:*** *Techno-Economic analysis of renewable system integrated with AC generator in remote location* ***Author Name:*** *Vinoth John Prakash* ***Email Id:*** *vjp.tiglfg@gmail.com*
15:30 – 15:40 (TS1.3.7)	***Paper Id:*** *101* ***Title:*** *Reliability Assessment of a Grid-Connected Solar Photovoltaic System Considering a Parallel Backup Subsystem* ***Author Name:*** *SUDHANSHU* ***Email Id:*** *shudansurathi@gmail.com*
15:40 – 15:50 (TS1.3.8)	***Paper Id:*** *108* ***Title:*** *Modeling and Optimization of Emerging Hybrid Power Systems* ***Author Name:*** *Dr. V. Sandeep* ***Email Id:*** *sandeep@nitandhara.ac.in*
15:50 – 16:00 (TS1.3.9)	***Paper Id:*** *113* ***Title:*** *Effective Hybrid Deep Learning Model for Wind Power Prediction* ***Author Name:*** *Bikash Chandra Saha* ***Email Id:*** *bikash.eee@citranchi.ac.in*
16:00 – 16:10 (TS1.3.10)	***Paper Id:*** *116* ***Title:*** *Power quality and stability Analysis of a Renewable Micro Grid using Small-Signal Stability Analysis with PQ Based Artificial Neural Network Control Scheme* ***Author Name:*** *Prakash Chandra Patra* ***Email Id:*** *pcpatra@gmail.com*

Date: 22 May 2025 **Time(14:30 – 16:30)**	**TS1.4 - Control Systems and Automation**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_ZGQ0ZTkxNmQtM2NkNC00ZTJkLThlOGMtM2NiMDgxYjI1YmUz%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

14:30 – 14:40 (TS1.4.1)	***Paper Id:*** *78* ***Title:*** *Stability Region Based Graphical Methods for Controller Design: An Outline Survey* ***Author Name:*** *Md Imran Kalim* ***Email Id:*** *1821ee14@iitp.ac.in*
14:40 – 14:50 (TS1.4.2)	***Paper Id:*** *117* ***Title:*** *FireBot: AI-Powered Autonomous Fire-Fighting Robot with Real-Time Detection and GSM Alerts* ***Author Name:*** *Harshad S* ***Email Id:*** *harshad.subramaniyam03@gmail.com*
14:50 – 15:00 (TS1.4.3)	***Paper Id:*** *128* ***Title:*** *Optimization of Robotic Gripper Configurations Using Kinematic Modelling and Genetic Algorithms for Enhanced Stability and Performance* ***Author Name:*** *Rajalakshmi Murugesan* ***Email Id:*** *rajemtech@gmail.com*
15:00 – 15:10 (TS1.4.4)	***Paper Id:*** *150* ***Title:*** *Bidirectional Indian Sign Language Translator* ***Author Name:*** *Vipin Kamble* ***Email Id:*** *kurakulasaisreeakshaya@gmail.com*
15:10 – 15:20 (TS1.4.5)	***Paper Id:*** *154* ***Title:*** *Automatic Detection of Diabetic Foot Ulcers: Enhancing Early Diagnosis Using Deep Learning* ***Author Name:*** *Nandana Suresh* ***Email Id:*** *nandanasuresh806@gmail.com*
15:20 – 15:30 (TS1.4.6)	***Paper Id:*** *166* ***Title:*** *Hand Gesture-Controlled Robotic System for Cleaning Applications: A Hybrid Approach Using Grey Wolf Optimizer and EfficientNet* ***Author Name:*** *Roshan Chitranshi* ***Email Id:*** *roshan.ugi010@gmail.com*
15:30 – 15:40 (TS1.4.7)	***Paper Id:*** *168* ***Title:*** *Synergistic Stabilization of n-dimensional Fractional Order Chaotic Systems via Fixed Power Reaching Principle Based Fractional Order Sliding Mode Control* ***Author Name:*** *Deepika* ***Email Id:*** *deepika0323@fot.du.ac.in*
15:40 – 15:50 (TS1.4.8)	***Paper Id:*** *195* ***Title:*** *Autonomous Marine Oil Collector: A Robotic Solution for Efficient Oil Spill Cleanup* ***Author Name:*** *Shreenivas Margamwar* ***Email Id:*** *shreenivasmargamwar@gmail.com*
15:50 – 16:00 (TS1.4.9)	***Paper Id:*** *201* ***Title:*** *Data Driven Control of Nonlinear System with Noisy Measurements* ***Author Name:*** *Aastha Yadav* ***Email Id:*** *dnyadav196174@gmail.com*
16:00 – 16:10 (TS1.4.10)	***Paper Id:*** *210* ***Title:*** *Smart Speed Control of Single Phase Induction Motor via Wi-Fi Enabled ESP8266* ***Author Name:*** *Pranav Kumar Rowlo* ***Email Id:*** *vtu17929@veltech.edu.in*

Date: 22 May 2025 **Time(14:30 – 16:30)**	**TS1.5 - Emerging Technologies in Electrical, Electronics, and Computing**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_NDBlNTU0ZmItMWQ3YS00NWE3LWExNGMtNjg4Mjk3MzVhYjcz%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

14:30 – 14:40 (TS1.5.1)	***Paper Id:*** *11* ***Title:*** *Theoretical analysis of CZTGS/CZTS Tandem Solar Cell using SCAPS- 1D* ***Author Name:*** *Jeepa K J* ***Email Id:*** *jeepa2353@gmail.com*
14:40 – 14:50 (TS1.5.2)	***Paper Id:*** *18* ***Title:*** *Priority-Based Resource Allocation in UAV-Based Telemedicine Systems for Real-Time Critical Patient Care* ***Author Name:*** *Kalpana Naidu* ***Email Id:*** *kalpana@nitw.ac.in*
14:50 – 15:00 (TS1.5.3)	***Paper Id:*** *22* ***Title:*** *Few-Shot Bio-Medical Image Analysis Using Diffusion Model and Modified Explainable AI.* ***Author Name:*** *Tejas Wasekar* ***Email Id:*** *rajkumar.tejas22@vit.edu*
15:00 – 15:10 (TS1.5.4)	***Paper Id:*** *25* ***Title:*** *Enhanced U-Net Architecture with Scale-Pyramid Dynamic Atrous Convolution and Attention Mechanism for Precise Metal Artifact Segmentation in Dental CBCT Imaging* ***Author Name:*** *Pradeep T S* ***Email Id:*** *pradeep.tvm@gmail.com*
15:10 – 15:20 (TS1.5.5)	***Paper Id:*** *31* ***Title:*** *A Survey on Ayurvedic Plant Identification Using Image Processing and Machine Learning Techniques* ***Author Name:*** *Nisha Bhade* ***Email Id:*** *nisha.bhade21@pccoepune.org*
15:20 – 15:30 (TS1.5.6)	***Paper Id:*** *50* ***Title:*** *Strengthening High-Performance Computing Systems with Advanced AI-Driven Cybersecurity Solutions for Enhanced Protection* ***Author Name:*** *Iqra Rizvi* ***Email Id:*** *rizviiqra6@gmail.com*
15:30 – 15:40 (TS1.5.7)	***Paper Id:*** *81* ***Title:*** *Cancer Cell Detection using Terahertz Metamaterial Absorber Based Biosensor* ***Author Name:*** *PRAVIN KUMAR SAMANTA* ***Email Id:*** *pravin.samanta95@gmail.com*
15:40 – 15:50 (TS1.5.8)	***Paper Id:*** *95* ***Title:*** *Voltage Stability Assessment in Distributed Generation Photovoltaic System using Stability Index Computation* ***Author Name:*** *Rajakumar Palanisamy* ***Email Id:*** *rajkumareee164@gmail.com*
15:50 – 16:00 (TS1.5.9)	***Paper Id:*** *119* ***Title:*** *Detection of 9 varieties of Skin Cancer using a Deep Learning Ensemble* ***Author Name:*** *Shipra* ***Email Id:*** *shipra@bitmesra.ac.in*

Date: 22 May 2025 **Time(14:30 – 16:30)**	**TS1.6 - Emerging Technologies in Electrical, Electronics, and Computing**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_OTdmYjljNGItZjE1NS00ZWI5LThlOWMtNDU0YTk5MzNiMmFi%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

14:30 – 14:40 (TS1.6.1)	***Paper Id:*** *228* ***Title:*** *Comparison and Analysis between Line Start Permanent Magnet Synchronous Motor and Axial Flux Permanent Magnet Synchronous Motor as a Drive system in Electric Vehicle* ***Author Name:*** *Payel Bhattacharya* ***Email Id:*** *payel54@gmail.com*
14:40 – 14:50 (TS1.6.2)	***Paper Id:*** *234* ***Title:*** *Autism Spectrum Disorder Identification from Face Images using Densenet 121* ***Author Name:*** *K. Anuradha* ***Email Id:*** *anu47radha@gmail.com*
14:50 – 15:00 (TS1.6.3)	***Paper Id:*** *282* ***Title:*** *Classification of Prostate Cancer with Deep Learning Ensemble Model* ***Author Name:*** *Samana Jafri* ***Email Id:*** *samana84@gmail.com*
15:00 – 15:10 (TS1.6.4)	***Paper Id:*** *289* ***Title:*** *A Novel Deep Learning Framework for Automated Sickle Cell Disease Detection* ***Author Name:*** *Abinash Kodamsingh* ***Email Id:*** *ak4500@srmist.edu.in*
15:10 – 15:20 (TS1.6.5)	***Paper Id:*** *290* ***Title:*** *Underwater Acoustic Source Localization using Time-Difference of Arrival (TDOA) With Kalman Filter* ***Author Name:*** *Chikkam Ajay* ***Email Id:*** *ajaychajaych745@gmail.com*
15:20 – 15:30 (TS1.6.6)	***Paper Id:*** *307* ***Title:*** *Doubly Even Order Magic Square (DEOMS) based Reconfiguration scheme for the Mitigation of Power Loss in Solar Photovoltaic Systems* ***Author Name:*** *Karthick B* ***Email Id:*** *bkarthick39@gmail.com*
15:30 – 15:40 (TS1.6.7)	***Paper Id:*** *322* ***Title:*** *Optimal Allocation of Electric Vehicle Charging Station and Distributed Generation: An Application of Hunter Prey Optimizer Algorithm* ***Author Name:*** *Sivakumar J* ***Email Id:*** *janshivajee@gmail.com*
15:40 – 15:50 (TS1.6.8)	***Paper Id:*** *347* ***Title:*** *Implication of CNN Model in IoT Based Heterogeneous Distributed System* ***Author Name:*** *Gitanjali Shrivastava* ***Email Id:*** *gitanjali.subscriptions@gmail.com*
15:50 – 16:00 (TS1.6.9)	***Paper Id:*** *395* ***Title:*** *FPGA Implementation Of 32-Bit Arithmetic Logic Unit Using Reversible Logic Gates* ***Author Name:*** *Kommalapati Rajesh* ***Email Id:*** *rajeshkanna2050@gmail.com*

Date: 22 May 2025 **Time(14:30 – 16:30)**	**TS1.7 - Circuits, Systems, VLSI and Nanoelectronics**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_OTE4OGQxODgtYmRlMy00ZjM5LTkwMmEtNmJiOWMwNTMyZjcx%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

14:30 – 14:40 (TS1.7.1)	***Paper Id:*** *13* ***Title:*** *Design of FIFO Synchronizer for Efficient Clock Domain Crossing using Verilog HDL* ***Author Name:*** *Keshav Sharma* ***Email Id:*** *shakeshav@gmail.com*
14:40 – 14:50 (TS1.7.2)	***Paper Id:*** *15* ***Title:*** *New Explicit Current Output Single Capacitance Controlled Oscillators* ***Author Name:*** *Puneeth S* ***Email Id:*** *puneeth.2021ece0033@presidencyuniversity.in*
14:50 – 15:00 (TS1.7.3)	***Paper Id:*** *17* ***Title:*** *Detection and Classification of Brain Tumors in MRI using CNN and VGG-16* ***Author Name:*** *Akshay S* ***Email Id:*** *akshayshantharam26@gmail.com*
15:00 – 15:10 (TS1.7.4)	***Paper Id:*** *40* ***Title:*** *Convolutional Neural Networks and Mel-Frequency Cepstral Coefficients Based Image and Audio Processing for Pest Detection in Precision Agriculture* ***Author Name:*** *Bhavana M* ***Email Id:*** *bhavana.m.pisay@gmail.com*
15:10 – 15:20 (TS1.7.5)	***Paper Id:*** *52* ***Title:*** *Harnessing Eyebrow Features to Determine Stress: An Integrated ML-DL Framework* ***Author Name:*** *Akshay S* ***Email Id:*** *akshayshantharam26@gmail.com*
15:20 – 15:30 (TS1.7.6)	***Paper Id:*** *63* ***Title:*** *FPGA Implementation FIR Filter Using Self Tunable Approximate Computing* ***Author Name:*** *S Harini Sriram* ***Email Id:*** *harinisriram@ritrjpm.ac.in*
15:30 – 15:40 (TS1.7.7)	***Paper Id:*** *127* ***Title:*** *FPGA based Vending Machine with Excess Amount Return Feature* ***Author Name:*** *Sweta Kumari* ***Email Id:*** *sweta.kumari@bitmesra.ac.in*
15:40 – 15:50 (TS1.7.8)	***Paper Id:*** *137* ***Title:*** *Single VCII Based Biquadratic Filter Configuration* ***Author Name:*** *Kirti Kaharwar* ***Email Id:*** *kirtikaharwar_2k21phdee05@dtu.ac.in*
15:50 – 16:00 (TS1.7.9)	***Paper Id:*** *140* ***Title:*** *Multi-Input Single Output Electronically Tunable Universal Filter Using Two Commercially Available ICs LT1228* ***Author Name:*** *Vivek Kumar* ***Email Id:*** *vivekgnz@gmail.com*

Date: 22 May 2025 **Time(14:30 – 16:30)**	**TS1.8 - Circuits, Systems, VLSI and Nanoelectronics**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_YTE4OWE2NjctMjUyNy00Njk2LThiZDEtNjlmODc3ZDEyMGE0%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

14:30 – 14:40 (TS1.8.1)	***Paper Id:*** ***144*** ***Title:*** *A CFOA based Inverse Band Pass Filter and its application as PID Controller* ***Author Name:*** *Kaushal Kumar* ***Email Id:*** *1320kaushal@gmail.com*
14:40 – 14:50 (TS1.8.2)	***Paper Id:*** ***147*** ***Title:*** *VDCC-Based Voltage Mode Universal Filter* ***Author Name:*** *Ayush Bhardwaj* ***Email Id:*** *ayushb998@gmail.com*
14:50 – 15:00 (TS1.8.3)	***Paper Id:*** ***207*** ***Title:*** *Scalable Hardware Optimization and Performance Comparison of AES Encryption Variants Across Different Technology Nodes* ***Author Name:*** *Shreenivas G. Margamwar* ***Email Id:*** *shreenivasmargamwar@gmail.com*
15:00 – 15:10 (TS1.8.4)	***Paper Id:*** ***283*** ***Title:*** *Real time implementation of deep learning framework for medical data analysis* ***Author Name:*** *Pradeep Enukonda* ***Email Id:*** *enukondapradeep@gmail.com*
15:10 – 15:20 (TS1.8.5)	***Paper Id:*** ***288*** ***Title:*** *Feature Extraction of ECG Signals based on Polyphase Filter with Distributed Arithmetic* ***Author Name:*** *Santhosh Kumar S* ***Email Id:*** *sudhanyp@srmist.edu.in*
15:20 – 15:30 (TS1.8.6)	***Paper Id:*** ***353*** ***Title:*** *An Automatic Air Quality Monitoring and Purifying System* ***Author Name:*** *Bodige Meghana* ***Email Id:*** *23wh5a0205@bvrithyderabad.edu.in*
15:30 – 15:40 (TS1.3.7)	***Paper Id:*** ***370*** ***Title:*** *Performance Comparision Of FIR Filter Using Different Adders In 90nm CADENCE CMOS Technology* ***Author Name:*** *Mamidi Srihitha Patel* ***Email Id:*** *Srihithapatelmamidi04@gmail.com*
15:40 – 15:50 (TS1.8.8)	***Paper Id:*** ***386*** ***Title:*** *Voltage Mode Active Filters using Current Feedback Operational Amplifiers* ***Author Name:*** *Arunav Dhaka* ***Email Id:*** *arunav.dhaka@gmail.com*

DAY 2: Friday, May 23, 2025	
10:00 - 11:00	**Plenary Talk 3: Prof. Shantanu Mishra** **Title: EV Charging Infrastructure: Design and Challenges** **Venue: Vishveshwarya Hall, NIT Patna** **Joining Link:** https://teams.microsoft.com/l/meetup-join/19%3ameeting_YWE0OGJlMWEtMGJlNy00ODFlLWFlMWMtNzAzYWVkYTk1NDJm%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

Date: 23 May 2025 11:15- 13:15	Day 2 | Technical Session-2 (TS2)
Date: 23 May 2025 Time(11:15 – 13:15)	TS2.1 - Electrical Power Systems and Renewable Energy
Mode: Offline	Venue: Vishveshwarya Hall, NIT Patna
11:15 – 11:30 (TS2.1.1)	***Paper Id:*** *306* ***Title:*** *ViLLA: Vision-Language Layout Analyzer for Floor Plan Analysis* ***Author Name:*** *Sabari Nathan* ***Email Id:*** *sabarinathantce@gmail.com*
11:30 – 11:45 (TS2.1.2)	***Paper Id:*** *352* ***Title:*** *Performance analysis of GG-NMOS for ESD Applications* ***Author Name:*** *Prabhat Kumar* ***Email Id:*** *prabhat.ece.2203007@iiitbh.ac.in*
11:45 – 12:00 (TS2.1.3)	***Paper Id:*** *513* ***Title:*** *Implications of Height, Spacing, and Shading for Agrivoltaics System* ***Author Name:*** *Shanza Neda Hussain* ***Email Id:*** *A.Ghosh@exeter.ac.uk*
12:00 – 11:55 (TS2.1.4)	***Paper Id:*** *542* ***Title:*** *Analysis of Mutual Inductance of Circular Rogowski coil using Full Current Theory* ***Author Name:*** *Venkatamaheshbabu Lella* ***Email Id:*** *lellamahesh257@gmail.com*
11:55 – 12:05 (TS2.1.5)	***Paper Id:*** *575* ***Title:*** *Performance Analysis of a Novel Three-point Interpolated DFT using Chebyshev Polynomial Approximation* ***Author Name:*** *sweta prasad* ***Email Id:*** *swetaprasad.1698@gmail.com*
12:05 – 12:15 (TS2.1.6)	***Paper Id:*** *03* ***Title:*** *Implementation of a Single VDBA-Based Electronically Tunable Universal Filter Using Commercial ICs* ***Author Name:*** *Tvisha Pushkar* ***Email Id:*** *tvishapushkar01@gmail.com*

Date: 23 May 2025 Time(11:15 – 13:15)	TS2.2 - Emerging Technologies in Electrical, Electronics, and Computing
Mode: Offline	Venue: 4th Floor (EE Extension New Building)

11:15 – 11:30 (TS2.2.1)	***Paper Id:*** *609* ***Title:*** *Kinematic and Dynamic Modeling of a 4-DOF Lower-Limb Exoskeleton for Mobility Assistance* ***Author Name:Ram Kumar*** ***Email Id:*** *ramahk92@gmail.com*
11:30 – 11:45 (TS2.2.2)	***Paper Id:*** *487* ***Title:*** *Load Frequency Control in Four-area Interconnected Power System Using Fuzzy PI Controller with Penetration of Renewable Energies* ***Author Name:*** *Praveen Mishra* ***Email Id:*** *praveenm.phd22.ee@nitp.ac.in*
11:45 – 12:00 (TS2.2.3)	***Paper Id:*** *626* ***Title:*** *A Survey on Sensors Technology Used for Traffic Monitoring and Safety* ***Author Name:*** *Aaditi Kumari* ***Email Id:*** *imaaditipancham@gmail.com*
12:00 – 12:15 (TS2.2.4)	***Paper Id:*** *24* ***Title:*** *Automated Fish Disease Detection Using Transfer Learning for Sustainable Aquaculture Management* ***Author Name:*** *Sasithradevi A* ***Email Id:*** *sasithradevi.a@vit.ac.in*
12:15 – 12:30 (TS2.2.5)	***Paper Id:*** *37* ***Title:*** *Crayfish Optimization based Tuning of Fractional PI Controller for Blood Pressure Regulation* ***Author Name:*** *Akanksha Dwivedi* ***Email Id:*** *akanksha_2221ee02@iitp.ac.in*
12:30 – 12:45 (TS2.2.6)	***Paper Id:*** *85* ***Title:*** *Hiking Optimization Algorithm based Optimal Tuning of PI controller for Mechanical Ventilators* ***Author Name:*** *Pritam Raj* ***Email Id:*** *pritam_2101ee56@iitp.ac.in*
12:45 – 13:00 (TS2.2.7)	***Paper Id:*** *141* ***Title:*** *Voltage Mode Third-order Quadrature Sinusoidal Oscillators using UVC* ***Author Name:*** *Tvisha Pushkar* ***Email Id:*** *tvishapushkar01@gmail.com*
13:00 – 13:15 (TS2.2.8)	***Paper Id:*** *351* ***Title:*** *Design of PI Controller for Artificially Ventilated Human Respiratory System using Crayfish Optimization* ***Author Name:*** *Akanksha Dwivedi* ***Email Id:*** *akanksha_2221ee02@iitp.ac.in*

Date: 23 May 2025 **Time(11:15 – 13:15)**	**TS2.3 - Power Electronics and Drives**
Mode: Online	
Joining Link	**https://teams.microsoft.com/l/meetup-join/19%3ameeting_MTk1YTc2YzAtNzAwOS00OTFlLThjMGYtNzcwNDg2ZDNiZjJl%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d**

11:15 – 11:25 (TS2.3.1)	***Paper Id:*** *410* ***Title:*** *Artificial Intelligence for Transformer Fault Detection: A Comprehensive Review of Dissolved Gas Analysis Techniques* ***Author Name:*** *Swati R. Ghutke* ***Email Id:*** *Swatigajbhiye026@gmail.com*
11:25 – 11:35 (TS2.3.2)	***Paper Id:*** *497* ***Title:*** *Exploring the Modulation Methods for Unleashing Asymmetric Multilevel Inverter Potential for Renewable Energy Systems* ***Author Name:*** *Dr. R. Seyezhai Ramalingam* ***Email Id:*** *seyezhair@ssn.edu.in*
11:35 – 11:45 (TS2.3.3)	***Paper Id:*** *510* ***Title:*** *LMS Adaptive Filter based Three-Phase Grid Interactive Solar Photovoltaic System* ***Author Name:*** *Dheeraj Kumar* ***Email Id:*** *kumar.dheeraj.burnwal@gmail.com*
11:45 – 11:55 (TS2.3.4)	***Paper Id:*** *530* ***Title:*** *A High-Gain Interleaved Boost Converter for Fuel Cell Vehicles with PI-Based Voltage Regulation* ***Author Name:*** *Gaddala Anusha* ***Email Id:*** *gunsha216@gmail.com*
11:55 – 12:05 (TS2.3.5)	***Paper Id:*** *576* ***Title:*** *Impact Assessment on Power Quality Parameters at Various Stages of Induction Motor Drive Hardware Setup* ***Author Name:*** *Dheeraj Kumar* ***Email Id:*** *kumar.dheeraj.burnwal@gmail.com*
12:05 – 12:15 (TS2.3.6)	***Paper Id:*** *601* ***Title:*** *Analysis and experimental validation of real time EV loading equivalent to electrical loading through motor generator set in laboratory* ***Author Name:*** *Uditveer Rajput* ***Email Id:*** *uditveer09@gmail.com*
12:15 – 12:25 (TS2.3.7)	***Paper Id:*** *618* ***Title:*** *High-Gain DC-DC Converter: Performance Analysis for Electric Vehicle Charging Applications* ***Author Name:*** *Vadivelan R* ***Email Id:*** *vr7989@srmist.edu.in*
12:25 – 12:35 (TS2.3.8)	***Paper Id:*** *32* ***Title:*** *Fuzzy Based Power Decoupler Method for Regulating Electric Springs to Improve Power Factor and Stability* ***Author Name:*** *Vadthya Jagan* ***Email Id:*** *jagan.iitr@gmail.com*

Date: 23 May 2025 **Time(11:15 – 13:15)**	**TS2.4 - Electrical Power Systems and Renewable Energy**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_MTQ3YTQyZjktZmE5MC00ODQ5LTlkYWMtMjQ1Nzc2NmZkMTI3%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

11:15 – 11:25 (TS2.4.1)	***Paper Id:*** *118* ***Title:*** *A Machine Learning Framework for Mode Identification in Power Systems with Corrupted PMU Data* ***Author Name:*** *Mani Shankar Kumar* ***Email Id:*** *manishankar800020@gmail.com*
11:25 – 11:35 (TS2.4.2)	***Paper Id:*** *123* ***Title:*** *Comparative Study of Islanding Detection for A Single-Phase Photovoltaic System Using Machine Learning* ***Author Name:*** *Sasmita Behera* ***Email Id:*** *sbehera_eee@vssut.ac.in*
11:35 – 11:45 (TS2.4.3)	***Paper Id:*** *145* ***Title:*** *Smart integration of Wind and BESS enabled Solar Power to Grid* ***Author Name:*** *Shivani* ***Email Id:*** *shivani.ee21-25@recabn.ac.in*
11:45 – 11:55 (TS2.4.4)	***Paper Id:*** *149* ***Title:*** *Reliability Assessment of Radial Distribution Network with Probabilistic Failure of Protection Devices* ***Author Name:*** *Manish Kumar Madhav* ***Email Id:*** *manishh.madhav@gmail.com*
11:55 – 12:05 (TS2.4.5)	***Paper Id:*** *165* ***Title:*** *Development of Solar Energy Conversion System for Linear Load Application* ***Author Name:*** *Sreedevi Kunumalla* ***Email Id:*** *sreedevikunumalla@gmail.com*
12:05 – 12:15 (TS2.4.6)	***Paper Id:*** *196* ***Title:*** *Study of Reliability of Power Grid Integrated with Distributed Renewable Energy Sources: A Review* ***Author Name:*** *Sneha K. Awaze* ***Email Id:*** *sneha.awaze@gmail.com*
12:15 – 12:25 (TS2.4.7)	***Paper Id:*** *198* ***Title:*** *Application of Hunter Prey Optimizer Algorithm for Multi-objective DG Allocation* ***Author Name:*** *Rajakumar Palanisamy* ***Email Id:*** *rajkumareee164@gmail.com*
12:25 – 12:35 (TS2.4.8)	***Paper Id:*** *203* ***Title:*** *Compensating V & F Deviations Through Transient Droop Characteristics Enabled Microsource Inverters* ***Author Name:*** *Sameer Bhambri* ***Email Id:*** *sameerbhambri@nitdelhi.ac.in*
12:35 – 12:45 (TS2.4.9)	***Paper Id:*** *262* ***Title:*** *Hybrid Fuzzy-ANN Control Strategies for Power Quality Improvement using DSTATCOM* ***Author Name:*** *Chintapenta Manaswi* ***Email Id:*** *manaswi1509@gmail.com*
12:45 – 12:55 (TS2.4.10)	***Paper Id:*** *314* ***Title:*** *Frequency Analysis of Highly Penetrated Renewable Islanded Microgrid using Cascaded Controller* ***Author Name:*** *Bisaya Bhoi* ***Email Id:*** *bbhoi_ee@vssut.ac.in*
12:55 – 13:05 (TS2.4.11)	***Paper Id:*** *329* ***Title:*** *A comparative analysis of SMC, STSMC with NN for grid stability* ***Author Name:*** *Vyshnavi Thammishetty* ***Email Id:*** *vyshnavishetty1106@gmail.com*

Date: 23 May 2025 **Time(11:15 – 13:15)**	**TS2.5 - Electrical Power Systems and Renewable Energy**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_YmNhNmVkMmEtNDFjMC00ZWIwLTgyMjQtZWIzOGVlZTljZGIy%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

11:15 – 11:25 (TS2.5.1)	***Paper Id:*** *388* ***Title:*** *Smart Coupler Alignment System Using Swerve Drive for Wireless Drone Charging Station* ***Author Name:*** *Venkatesan R* ***Email Id:*** *venki56eee@gmail.com*
11:25 – 11:35 (TS2.5.2)	***Paper Id:*** *420* ***Title:*** *Load Frequency Control of a Single Area Distributed Generation System Considering CS Optimized Cascade Controller* ***Author Name:*** *Ainala Prem Kumar* ***Email Id:*** *ainalapremkumar@gmail.com*
11:35 – 11:45 (TS2.5.3)	***Paper Id:*** *476* ***Title:*** *Mitigating Hot Spot Effect on PV array using Reconfiguration Technique* ***Author Name:*** *Jaya Bharati* ***Email Id:*** *jayab.ph23.ee@nitp.ac.in*
11:45 – 11:55 (TS2.5.4)	***Paper Id:*** *503* ***Title:*** *Fault Location Scheme for AC Microgrid Based on Magnitude of Transient Voltage* ***Author Name:*** *Mahesh Ahuja* ***Email Id:*** *mmhesh@gmail.com*
11:55 – 12:05 (TS2.5.5)	***Paper Id:*** *538* ***Title:*** *An Intelligent Grid Monitoring System Utilizing Machine Learning for High Impedance Fault Detection* ***Author Name:*** *Lekshmi R Chandran* ***Email Id:*** *lekshmichandran@am.amrita.edu*
12:05 – 12:15 (TS2.5.6)	***Paper Id:*** *543* ***Title:*** *Improving PV Module Efficiency: An IOT-Driven Dust Cleaning System* ***Author Name:*** *Harshit Tiwari* ***Email Id:*** *harshittiwari0014@gmail.com*
12:15 – 12:25 (TS2.5.7)	***Paper Id:*** *596* ***Title:*** *ENHANCED FAULT DETECTION IN TWO-TERMINAL TRANSMISSION LINES USING ITD* ***Author Name:*** *Kandula Harshitha* ***Email Id:*** *harshithakandula9@gmail.com*
12:25 – 12:35 (TS2.5.8)	***Paper Id:*** *606* ***Title:*** *ARTIFICIAL NEURAL NETWORK BASED CHARGE SCHEDULING FOR ELECTRIC VEHICLE IN MICRO-GRID* ***Author Name:*** *NITISH BHARDWAJ* ***Email Id:*** *phdee10003.21@bitmesra.ac.in*
12:35 – 12:45 (TS2.5.9)	***Paper Id:*** *617* ***Title:*** *Optimal Power Flow using RIME Algorithm* ***Author Name:*** *ImranKhan S* ***Email Id:*** *imrankhan.mtech@gmail.com*
12:45 – 12:55 (TS2.5.10)	***Paper Id:*** *516* ***Title:*** *High efficient multi port converter based wireless EV charging system powered by renewable energy sources* ***Author Name:*** *Venugopal R* ***Email Id:*** *vr4964@srmist.edu.in*

Date: 23 May 2025 **Time(11:15 – 13:15)**	**TS2.6 - Control Systems and Automation**
Mode: Online	
Joining Link	**https://teams.microsoft.com/l/meetup-join/19%3ameeting_MDM1N2YzOWMtN2FlMC00M2MxLWExYTctMjVkZWMzZjQ2NTcy%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d**

11:15 – 11:25 (TS2.6.1)	***Paper Id:*** *237* ***Title:*** *Design and Implementation of Fractional Order PID Controller for Controlling Regulation in Blood Pressure* ***Author Name:*** *Abhishek Jain* ***Email Id:*** *abhishekjain.281988@gmail.com*
11:25 – 11:35 (TS2.6.2)	***Paper Id:*** *253* ***Title:*** *A Machine Learning Approach For Helmet Detection Implementation And Bike Speed Control System* ***Author Name:*** *Panchumarthy Kamalesh Karthik* ***Email Id:*** *kamaleshkarthik199@gmail.com*
11:35 – 11:45 (TS2.6.3)	***Paper Id:*** *269* ***Title:*** *FOPI Controller Design for Blood Pressure Regulation using Hippopotamus Optimization Algorithm* ***Author Name:*** *Gaurav Gupta* ***Email Id:*** *gaurav_2101ee31@iitp.ac.in*
11:45 – 11:55 (TS2.6.4)	***Paper Id:*** *293* ***Title:*** *PSO Based Gene Expression Network Modelling and PID Control of Sars-Cov2 Biopathway for Enhancing Drug Target Identification* ***Author Name:*** *Nivendu Nandy* ***Email Id:*** *nivendunandy@gmail.com*
11:55 – 12:05 (TS2.6.5)	***Paper Id:*** *331* ***Title:*** *Performance Analysis of Autonomous Emergency Braking System Using CAN* ***Author Name:*** *Jayasudha* ***Email Id:*** *jayasudhat11@gmail.com*
12:05 – 12:15 (TS2.6.6)	***Paper Id:*** *455* ***Title:*** *Hazy Image Quality Enhancement Using Deep Neural Networks: A Comparison between Traditional Methods and Deep Learning Approaches* ***Author Name:*** *Mustafa Jamal Shahbaz* ***Email Id:*** *jsblue2007@uomustansiriyah.edu.iq*
12:15 – 12:25 (TS2.6.7)	***Paper Id:*** *573* ***Title:*** *Design of Multi Stage FOPD(1+PI) Controller for Stabilization of Magnetic Levitation Plant* ***Author Name:*** *Harideo Ram* ***Email Id:*** *phdee10053.22@bitmesra.ac.in*
12:25 – 12:35 (TS2.6.8)	***Paper Id:*** *614* ***Title:*** *Automatic Generation Control in PV-Thermal Power System using PID and FOPID controllers* ***Author Name:*** *Jm Sadique Ahamed* ***Email Id:*** *jennathu.beevi@crescent.education*
12:35 – 12:45 (TS2.6.9)	***Paper Id:*** *339* ***Title:*** *Design of Gain Scheduled Reinforcement Learning algorithm based Controller for Conical Tank* ***Author Name:*** *N Santhosh* ***Email Id:*** *santhosh_m230867ee@nitc.ac.in*

Date: 23 May 2025 **Time(11:15 – 13:15)**	**TS2.7 - Emerging Technologies in Electrical, Electronics, and Computing**
Mode: Online	
Joining Link	**https://teams.microsoft.com/l/meetup-join/19%3ameeting_NjAwYzRiNGMtNzgyOC00MGIzLWE5ZjYtNjAyZTVkNTEzZTFm%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d**

11:15 – 11:25 (TS2.7.1)	***Paper Id:*** *473* ***Title:*** *The interaction of charged particles near the cathode surface and its effect on the voltage drop and ionization energy in the plasma arc* ***Author Name:*** *Hasan Abedulhadi Kadhim* ***Email Id:*** *hasanalsagad123456@gmail.com*
11:25 – 11:35 (TS2.7.2)	***Paper Id:*** *474* ***Title:*** *The Impact of Pre-processing Techniques on Facial Recognition Performance Using Customized CNN architecture* ***Author Name:*** *Ali Habeeb Hadi Al-Amili* ***Email Id:*** *alihh@uomustansiriyah.edu.iq*
11:35 – 11:45 (TS2.7.3)	***Paper Id:*** *481* ***Title:*** *A Comprehensive Analysis of Malware Detection on Android Using Machine Learning* ***Author Name:*** *Dr. S. P. Santhoshkumar* ***Email Id:*** *spsanthoshkumar16@gmail.com*
11:45 – 11:55 (TS2.7.4)	***Paper Id:*** *483* ***Title:*** *Enhancing Medical Imaging through CNN-Based Super-Resolution* ***Author Name:*** *Sugguna Seshagiri Rao* ***Email Id:*** *seshagiri.23phd7151@vitap.ac.in*
11:55 – 12:05 (TS2.7.5)	***Paper Id:*** *486* ***Title:*** *Optimized Solar Energy Utilization for EV Charging: ANN-Based MPPT with Levenberg-Marquardt Algorithm* ***Author Name:*** *Sivakumar J* ***Email Id:*** *dr.balasubramaniam.pm@hit.edu.in*
12:05 – 12:15 (TS2.7.6)	***Paper Id:*** *539* ***Title:*** *YOLOv8-Based Electric Vehicle Charging Port Detection for Smart Infrastructure* ***Author Name:*** *Pridhav Krishna Anoop* ***Email Id:*** *pa9370@srmist.edu.in*
12:15 – 12:25 (TS2.7.7)	***Paper Id:*** *563* ***Title:*** *Localization of underwater moving object using Distributed neighbourhood network for Underwater Internet of network Things* ***Author Name:*** *T. Srinivasa Reddy* ***Email Id:*** *tippireddyr.phd22.ec@nitp.ac.in*
12:25 – 12:35 (TS2.7.8)	***Paper Id:*** *590* ***Title:*** *Multimodal Iris Feature Extraction and Recognition Using Gabor Filters, Wavelets and Local Binary Patterns* ***Author Name:*** *Rahila N A* ***Email Id:*** *rahilanaphd@gmail.com*

Date: 23 May 2025 **Time(11:15 – 13:15)**	**TS2.8 - Emerging Technologies in Electrical, Electronics, and Computing**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_OTk3NTMxOWMtNjhmZC00NWQxLWI5YzAtZjliODYyYzBlOThh%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

11:15 – 11:25 (TS2.8.1)	***Paper Id:*** *598* ***Title:*** *CAN Bus System for Real-Time Automotive Sensor Monitoring and Remote Web Visualization* ***Author Name:*** *Sanket Bawistale* ***Email Id:*** *bawistalesa@rknec.edu*
11:25 – 11:35 (TS2.8.2)	***Paper Id:*** *621* ***Title:*** *Advances in Image-Based Disease Detection: A Survey on Deep Transfer Learning Approaches* ***Author Name:*** *Sohail Akhtar* ***Email Id:*** *sohail876akhtar@gmail.com*
11:35 – 11:45 (TS2.8.3)	***Paper Id:*** *634* ***Title:*** *An Intrusion Detection System Using Deep Learning for Cyber Threat Detection in IoT Networks* ***Author Name:*** *Anantha Venkata Sai Kola* ***Email Id:*** *ananthasaikola@gmail.com*
11:45 – 11:55 (TS2.8.4)	***Paper Id:*** *635* ***Title:*** *Improved Breast Cancer Detection using optimization-based Feature Selection Methods* ***Author Name:*** *Pravin Kumar Samanta* ***Email Id:*** *pravin.samantafet@kiit.ac.in*
11:55 – 12:05 (TS2.8.5)	***Paper Id:*** *105* ***Title:*** *A Light-Weight CNN-Transformer Based Framework for Dynamic Sign Language Recognition* ***Author Name:*** *Ajay Kumar Naik G* ***Email Id:*** *ak24ecr1r04@student.nitw.ac.in*
12:05 – 12:15 (TS2.8.6)	***Paper Id:*** *378* ***Title:*** *Non-Invasive EEG-Driven Autonomous Wheelchair* ***Author Name:*** *Kishor Kumar Kumaravel* ***Email Id:*** *kishor.receive@gmail.com*
12:15 – 12:25 (TS2.8.7)	***Paper Id:*** *475* ***Title:*** *The Effect of Concentration and Voltage on the Voltage drop in the Plasma Layers near the Cathode Surface in Arc Plasma* ***Author Name:*** *Banaz Hasheem Hassan* ***Email Id:*** *banazhasheem@uomustansiriyah.edu.iq*
12:25 – 12:35 (TS2.8.8)	***Paper Id:*** *532* ***Title:*** *HRV-based Stress Detection* ***Author Name:*** *Anupam Kumar* ***Email Id:*** *anupam.ec@nitp.ac.in*
12:35 – 12:45 (TS2.8.9)	***Paper Id:*** *397* ***Title:*** *A Renewable Energy Integrated IoT-Based Battery Management System for Electric Mobility* ***Author Name:*** *Venugopal R* ***Email Id:*** *balajic@srmist.edu.in*

Date: 23 May 2025 **Time (11:15 – 13:15)**	**TS2.9 - Circuits, Systems, VLSI and Nanoelectronics**
Mode: Online	
Joining Link	https://teams.microsoft.com/l/meetup-join/19%3ameeting_NzZmZGU3ZGMtN2U2My00ZDVlLTk3ZWYtY2NhMTNlZjdmOWQ0%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d

11:15 – 11:25 (TS2.9.1)	***Paper Id:*** *393* ***Title:*** *A Novel Method for User Authentication Using EEG Signals for Enhanced Security* ***Author Name:*** *Harini Manivelan* ***Email Id:*** *harinim5915@gmail.com*
11:25 – 11:35 (TS2.9.2)	***Paper Id:*** *472* ***Title:*** *A Comparative Analysis on Nanowire FET based Devices for improved performance* ***Author Name:*** *Neha Patel* ***Email Id:*** *nehap.vl.23@nitj.ac.in*
11:35 – 11:45 (TS2.9.3)	***Paper Id:*** *484* ***Title:*** *Audio Amplifier With Active Noise Cancellation in RLS Algorithm on FPGA* ***Author Name:*** *Manish Anil Bhavsar* ***Email Id:*** *bhavsarmanish1234@gmail.com*
11:45 – 11:55 (TS2.9.4)	***Paper Id:*** *494* ***Title:*** *Minimizing Delay, Power and PDP in Ternary Logic Circuits: A GNRFET-RRAM Approach* ***Author Name:*** *Shaik Ruksana* ***Email Id:*** *munawar.ece@srecnandyal.edu.in*
11:55 – 12:05 (TS2.9.5)	***Paper Id:*** *515* ***Title:*** *Impact of Structural and Material Variations on the Performance of Junctionless Nanowire Field Effect Transistor,A Comprehensive Analysis* ***Author Name:*** *Dipjyoti Borah* ***Email Id:*** *dipjyotiborah2016@gmail.com*
12:05 – 12:15 (TS2.9.6)	***Paper Id:*** *529* ***Title:*** *Structural Analysis of GATE-ALL-AROUND NANOWIRE TFET* ***Author Name:*** *Ankit Kumar* ***Email Id:*** *ankitk.vl.23@nitj.ac.in*
12:15 – 12:25 (TS2.9.7)	***Paper Id:*** *553* ***Title:*** *CFOA-based Second Order Voltage Mode Active Filter Configuration* ***Author Name:*** *Piyush Kumar* ***Email Id:*** *piyushkumar48966@gmail.com*
12:25 – 12:35 (TS2.9.8)	***Paper Id:*** *593* ***Title:*** *Two Port Octagonal MIMO Antenna for Ku and K-Band Applications* ***Author Name:*** *Adireddy Naveen* ***Email Id:*** *adireddynaveen90142@gmail.com*
12:35 – 12:45 (TS2.9.9)	***Paper Id:*** *619* ***Title:*** *Ensemble-Based Adaptive Filtering for Speech Enhancement in Car Noise Conditions* ***Author Name:*** *Arpit Sharma* ***Email Id:*** *arpitniraliya306@gmail.com*

Date: 23 May 2025 ***13:15 – 14:30***	***Lunch***
14:30 – 15:15	**Plenary Talk 4: Dr. Sanjay Yadav** **Title: Linkages of National and International Networking of Quality Infrastructure for Nation Building** **Venue: Vishveshwarya Hall, NIT Patna**
15:15 – 16:15	***Valedictory (Hybrid Mode)*** **Weblink: https://teams.microsoft.com/l/meetup-join/19%3ameeting_NzFmYTRkOGMtOWQ1NC00NjdhLTk4ODUtZTVkYmQxNWYzYTYw%40thread.v2/0?context=%7b%22Tid%22%3a%221e7b661a-52b5-4acb-bb8c-107e607ef854%22%2c%22Oid%22%3a%2299a8e042-586d-49ac-bc39-9ea4ed3c4f8c%22%7d**
16:15 – 17:00	***High Tea***

Plenary Lecture - 1

Day - 1 (22.05.2025), 11:00-11:45

Empowering Renewable Energy Integration: A Field Deployment Experience

Prof. Narayana Prasad Padhy
Director, MNIT Jaipur

This project embodies the core objective of the Mission Innovation Challenge on smart grids, seeking to construct cost-effective, resilient, and renewable-powered grids while reducing fossil fuel dependency. It focuses on India's transition to renewable energy, primarily wind and solar, which contribute significantly to the nation's energy portfolio. Recognizing the complexities that arise with the rapid integration of solar energy into distribution systems, the project proposes a standardized approach of incorporating energy storage systems at strategic points within the distribution network. This initiative, carried out at the Indian Institute of Technology Roorkee (IITR), serves as a practical demonstration, illustrating the possibilities and benefits of integrating energy storage into real-world applications. The primary challenge involves optimizing the allocation of energy storage systems considering various factors. The project employs mixed-integer linear programming to identify the ideal size and location of batteries within the distribution network. Furthermore, the study explores two-layer control strategies, ensuring voltage stability and facilitating seamless transitions between grid-connected and islanded mode to enhance reliability. In addressing the critical issue of frequency regulation, project leverages energy storage and inverters to mimic the functions of traditional generators. By optimizing five locations for energy storage and validating control strategies through Power Hardware in Loop (PHIL) simulations, this project aims to showcase the practicality and benefits of smart grid technologies. Two battery energy storage systems (BESS) have already been implemented on the IIT Roorkee campus, marking a significant milestone in achieving a self-sustaining, zero-emission green campus. This project not only aids India's renewable energy ambitions but also serves as a global benchmark for efficient energy management and grid resilience in smart campuses.

Dr. Narayana Prasad Padhy, Chairperson (I/C), Board of Governors MNIT Jaipur, Director, MNIT Jaipur and Director, IIIT Kota, Chair Professor IIT Palakkad Technology IHub Foundation (IPTIF), is a Professor (HAG) of Electrical Engineering and has served as Dean of Academic Affairs, IIT Roorkee, Founder Head Mehta Family School of Data Science and Artificial Intelligence, Professor In-Charge Training and Placement, NEEPCO and Institute Chair Professor at IIT Roorkee.

Dr. Padhy is a prolific researcher in the area of Power Systems Engineering and Smart Grid, and has authored over 220 papers in international journals, with an overall citation of 9500 plus, h-index of 50, and i-index of 143. He has contributed significantly to the domain by way of securing 13 international/national research grants with a funding of over INR 200 million, successfully supervised23 research scholars and currently supervising 22 research scholars, publishing research articles, writing six books, securing patents, laboratory development and developing online NPTEL courses.

Dr. Padhy has held visiting positions at eminent international universities, and established research engagements and institutional tie-ups with universities from the US, UK, Denmark, Italy and Canada. He has collaborated with top industries and government agencies for research, laboratory development, and as a knowledge partner. He is well recognised in international and national professional bodies being Fellow of Indian National Academy of Engineering, The Institution of Engineering and Technology, UK, Institution of Engineers, India, Institution of Electronics and Telecommunication Engineers, India, and has actively contributed in various capacities towards their growth. He also represents the country at various international research platforms.

Dr. Padhy is a well awarded and honoured scholarly professional at the national and international levels, including Alexander von Humboldt Experienced Research Fellowship, Samanta Chandra Sekhar Award, IEEE Smart City Jury Award, IEEE PES Chapter Outstanding Engineer Award from IEEE Power & Energy Society, USA, and BOYSCAST Research Fellow Award for Young Scientist.

Plenary Lecture - 2

Day - 1 (22.05.2025), 11:45-12:30

Analog Integrated Circuits and Signal Processing: A brief Historical Overview

Prof. Raj Senani
NSUT, Delhi

Starting from the pioneering work of the world renowned Analog Designer Barrie Gilbert, who introduced the concept of Translinear principle and translinear circuits in 1975, a brief overview of some developments in the areas of Analog Integrated Circuits and Analog Signal processing, during the last five decades, would be outlined.

Raj Senani was born in 1950 at Budaun. He obtained B.Sc. in 1966 from Lucknow University, B.Sc. (Engineering) in 1971 from HBTI, Kanpur, M.E. (Electrical Engineering) with Honours in 1974 from MNREC (now MNNIT), Allahabad and Ph.D. in Electrical Engineering in 1988, from University of Allahabad.

Dr. Senani held the positions of Lecturer (1975-1986), Reader (1987-1988) at the EE Department of MNREC, Allahabad. He joined the ECE Department of the Delhi Institute of Technology, Delhi (later renamed as Netaji Subhas Institute of Technology) in 1988 as an Assistant Professor and became a Full Professor in 1990. Since then, he has served as Head of the ECE Department and several others, Dean Research (1993-1996), Dean Academic (1996-1997), Dean Administration (1997-1999), Dean Post Graduate Studies (1997-2001) and Director during 1996-1997 and 2003-2004. He served as Dean, Faculty of Technology, Delhi Universit,y during 2011-2014 and as regular Director of NSIT during 2008-2014.

Professor Senani has authored/co-authored over 190 research papers- all published in refereed international journals, receiving (as per Google Scholar) over 7000 citations with an h-index of 47 and i-10 index of over 140 so far. He has also written 05 book chapters and 05 research monographs during 2013-2017. He is an Editorial reviewer for over 50 international journals. He served as Editor-in-Chief for IETE Journal of Education during 2012-2017 and now since September 2022-onwards. He has been an Associate Editor for Circuits, Systems and Signal Processing (USA) since 2003. In addition, he is on the Editorial Boards of 5 other International Journals. Professor Senani is a Member of Sigma Xi (USA), Senior Member of IEEE (USA), Fellow of IE (India), Life Fellow of IETE (India), an elected (2008) Fellow of the National Academy of Sciences, India (NASI), the recipient of the Second Laureate of the 25th Khwarizmi International Award for 2012. Professor Senani was ranked among the Top 1% Reviewers of the World for Engineering and Interdisciplinary areas, for three consecutive years 2017, 2018 and 2019 by Publon (Web of Science).

In 2020, 2021, 2022, 2023, and 2024-for five consecutive years, he featured in the list of Top 2% Scientists of the World (career-wise) in the areas of Science, Engineering and Technology, as per a study conducted by a team of Stanford University and Elsevier researchers. On 15th September, 2021- the Engineers Day, he was conferred with the `Eminent Engineer of the year 2021 Award` by the Institution of Engineers, India (IEI) with a scroll of honour for significant contribution to the advancement and application of practice of Engineering in India. On September 14, 2024, he was given the `IETE Life-time Achievement Award' for his outstanding research in the area of Circuits and Systems spanning over five decades.Professor Senani is listed in Marquis' Who's Who in the World (USA) and a number of other international biographical directories, since 1990.

Plenary Lecture - 3

Day - 2 (23.05.2025), 10:00-11:00

EV Charging Infrastructure: Design and Challenges

Prof. Shantanu Mishra
IIT Delhi.

EV charging is becoming an important bottleneck for adaptation of EVs into our society. This talk will focus on the basics of EV charging from the prospective of a consumer as well as a designer. Fundamental building blocks of the charger and their implementation will be discussed. Various challenges of putting numerous EV chargers in our distribution grid will be outlined. It will also cover the existing policies for EV charger installation.

Santanu K. Mishra (S'00-M'04-SM'12) received a B.Tech. degree in Electrical Engineering from the College of Engineering and Technology, Bhubaneswar, India, in 1998, an M.Tech. degree in Energy Systems Engineering from Indian Institute of Technology, Chennai, India, in 2000, and the Ph.D. degree from the Department of Electrical and Computer Engineering, University of Florida, Gainesville, FL, USA, in 2006.

He worked as a Staff Application Engineer with the International Rectifier Corporation in Rhode Island, USA, from 2004 to 2008. He was a professor of Electrical Engineering at the Indian Institute of Technology, Kanpur, India, From 2008 till April 2023. Currently, he is a Professor at the Centre for Automotive Research and Tribology, IIT Delhi. His research interests include EV Charging Infrastructure design, power converter design, implementation, control, and their applications in rural scenario. He serves as an associate editor of IEEE Transactions on Industry Applications and IEEE Transactions on Power Electronics. He is a fellow of IET, INAE, and IETE.

Plenary Lecture - 4

Day - 2 (23.05.2025), 14:30-15:15

Linkages of National and International Networking of Quality Infrastructure for Nation Building

Dr. Sanjay Yadav
Former Chief Scientist and Prof. Physical Sciences, AcSIR,
CSIR-National Physical Laboratory (CSIR-NPL), New Delhi
Vice President, MSI and USI, New Delhi

A robust and integrated National Quality Infrastructure (NQI) is essential for a country's industrial development, trade facilitation, and overall economic growth. In India, the NQI rests on five core pillars: metrology, standardization, accreditation, conformity assessment, and market surveillance. Together, these components help ensure that goods and services meet established quality benchmarks, thereby building trust among consumers and facilitating both domestic and international trade. At the heart of this system is metrology—the science of measurement—which plays a crucial role in enabling precision across scientific, industrial, and regulatory domains. The CSIR-National Physical Laboratory (CSIR-NPL), as India's National Metrology Institute (NMI), is responsible for maintaining national measurement standards and ensuring their traceability to the International System of Units (SI), as mandated under the Legal Metrology Act, 2011. In doing so, it helps align India's measurement system with international practices, supporting sectors that range from health and energy to food safety and environmental monitoring.

This paper highlights the importance of strengthening both national institutions—such as CSIR-NPL, the Legal Metrology Department, the Bureau of Indian Standards (BIS), and the Quality Council of India (QCI)—and their linkages with international organizations like the Bureau International des Poids et Mesures (BIPM), International Organization for Standardization (ISO), International Electrotechnical Commission (IEC), and International Laboratory Accreditation Cooperation (ILAC). These linkages ensure that India's quality infrastructure is not only globally recognized but also capable of contributing to international scientific and trade frameworks. A well-functioning and internationally connected quality infrastructure reduces technical barriers to trade, supports innovation, enhances

industrial productivity, and contributes to public safety and environmental sustainability. Strengthening these national and global networks is therefore not just a technical requirement—it is a strategic imperative for inclusive and resilient nation building.

Dr. Sanjay Yadav, born on 5th August 1962, is an accomplished metrologist with over three decades of pioneering contributions in physico-mechanical metrology. After completing M.Sc. (Electronics) in 1985 and earning a Ph.D. from Meerut University in 1990, He began his career as a Research Scholar at CSIR-NPL and served the Department of Science & Technology, Haryana, as Project Officer before joining CSIR-NPL as Scientist B in 1995, where he rose to Chief Scientist and Head of its largest division. He superannuated in 2022 and post-retirement, he was conferred the Emeritus Scientist (2024) by CSIR.

With over 35 years of experience, Dr. Yadav has significantly contributed to the development of metrological standards in pressure metrology, establishing several primary and secondary pressure standards and participating in international key comparisons. His efforts led to the registration of 17 Calibration and Measurement Capabilities (CMCs) in BIPM's KCDB, bringing India's metrology to global standards. He has represented India in metrological collaborations across France, Japan, Saudi Arabia, Indonesia, Bhutan, and Malaysia.

He has published over 525 publications, including 177 journal papers, 190 conference papers, 30 book chapters, 25 books, and delivered 97 invited talks. He holds 15 patents and copyrights and has guided 12 Ph.D. scholars. He also served as Professor, AcSIR, teaching Advanced Measurement Systems and PMQC.

Dr. Yadav has held multiple leadership roles: Vice President of the Metrology Society of India and Ultrasonic Society of India, Chairman of IQAS and Member Governing Council of FCRI, Editorin- Chief of MAPAN, and Member of several national and international committees (BIS, NCERT, IMEKO, BIPM, APMP, etc.).

He has led and organized over 27 training programs and numerous national/international conferences, including the 33rd APMP GA (2017). Dr. Yadav has raised significant project funding (>Rs. 60 Cr), including national initiatives in pressure metrology, biomedical instrumentation, PPE testing, and quantum standards.

A visionary and team-oriented leader, Dr. Yadav advocates for scientific freedom, international collaboration, and infrastructure modernization. He remains dedicated to advancing India's metrological capabilities and global standing in science and technology. He advocates synergy between senior and young researchers and remains committed to elevating India's metrological landscape to global standards.

He has received many awards/recognitions in India and abroad. Some of them are Col. G.N. Bajpayee Award of Institution of Engineers, India (1989);Commendation Certificates from Haryana Government (1991 & 1992); JICA Fellowship of JAPAN (1998), Commendation Certificates from SASO, Saudi Arabia (2003); nominated as Member of APMP Technical Committee of Mass Related Quantities(TCM), Australia (2013-2019); Nominated as Country Representative in APMP (2019); Members, Standards Advisory Committee, BIS (2019); nominated as member representative in National Advisory Committee of NCERT (2020); Recipient of prestigious APMP Award of Developing Economies 2020, China and Japan; Member of IMEKO Governing Board 2020; Member IMEKO TC6 and TC21, Hungary; Member WGPAV and WGM BIPM 2021, France; nominated as Member Governing Council, FCRI, Kerala; and very recently selected as Emeritus Scientist, CSIR, New Delhi, 2024.

Abstracts

(Paper ID: 3-635)

Paper ID: 3

Title: IMPLEMENTATION OF A SINGLE VDBA-BASED ELECTRONICALLY TUNABLE UNIVERSAL FILTER USING COMMERCIAL ICS

Authors: Tvisha Pushkar*, Rajeshwari Pandey

Email: tvishapushkar01@gmail.com

Abstract: *This study introduces a voltage-mode universal filter design that utilizes a single Voltage Differencing Buffered Amplifier (VDBA) along with two grounded capacitors and one resistor. The design is capable of implementing all five standard second-order filter types—low-pass, high-pass, band-pass, band-reject, and all-pass—without requiring any modifications to the circuit layout. Additionally, the circuit does not depend on component matching. Its functionality has been validated through simulations conducted using PSPICE with a 0.18 µm CMOS technology model.*

Keywords: Voltage Differencing Buffed Amplifier, Voltage-Mode, Active Filter, and Universal Filter.

Paper ID: 10

Title: MODIFIED SELF-TUNED FILTER BASED CONTROL FOR SINGLE -PHASE MICROGRID SYSTEM

Authors: Tripurari Nath Gupta, Pratham Pundir*, Shiv Shambu Choudhary, Abhinay Gautam

Email: ranapratham176@gmail.com

Abstract: *This paper presents a novel control algorithm based on a modified self- tuned filter frequency locked loop (MSTF-FLL) aimed at improving power quality in single-phase microgrid systems interfaced with solar photovoltaic (PV) sources. In single-phase PV systems, maintaining grid stability and ensuring high power quality is a challenge due to fluctuating solar generation and load variations. The proposed STF-FLL algorithm enhances the system's ability to synchronize with the grid voltage and accurately extract fundamental components. By modifying the conventional STF-FLL, the algorithm ensures faster dynamic response and improved harmonic suppression, resulting in better voltage regulation and reduced total harmonic distortion (THD). The algorithm's performance is validated through simulations under various grid conditions, demonstrating its effectiveness in maintaining stable and high-quality power output from the PV system. This work contributes to the advancement of reliable and efficient renewable energy integration into microgrids, addressing key challenges in power quality enhancement*

Keywords: VSC, Solar PV System, GCPV, Power Quality Improvement and MPPT.

Paper ID: 11

Title: ANALYZING THE PERFORMANCE OF CZTGS/CZTS TANDEM SOLAR CELL THEORETICALLY USING SCAPS- 1D

Authors: Jeepa K J*, Rajeesh J, T D Subash, Akshara Babu, Abhijith M, Akhila Raj, Chinchu Anna Varke

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Abstract: *Numerical modelling and simulation of a monolithic CZTGS/CZTS tandem solar cell arrangement were executed in this inquiry confining the SCAPS-1D software. The objective is to strengthen the efficiency of CZTS solar cells through the use of a dual junction CZTGS/CZTS tandem configuration. In order to adjust the copper zinc tin germanium sulfide's (CZTGS) band gap, the upper cell of tandem design contains a harmless element- germanium is used, with the lower cell being made of CZTS. The simulation results provide insights into the optical and electrical characteristics of the tandem structure, aiming to optimize its efficiency and overall performance.*

Keywords: Tandem structure, Fill factor, short circuit current density

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Paper ID: 13

Title: DESIGN OF FIFO SYNCHRONIZER FOR EFFICIENT CLOCK DOMAIN CROSSING USING VERILOG HDL

Authors: Keshav Sharma*, Indr Sharma, Divyansh Kumar Upadhyay and Shailendra Bisariya

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Abstract: *The research paper delves into the increasing complexity of managing clock domain crossings (CDCs) in modern chip designs, particularly in the context of Globally Asynchronous Locally Synchronous (GALS) architectures. It emphasizes the challenges posed by metastability and the necessity for synchronizers to address this issue. With the average number of clock domains in modern devices ranging from 15-20, the paper explores the role of First-in-First-out (FIFO) buffers in mitigating CDC-related integrity issues, data corruption, and incoherence. It provides a detailed analysis of the synchronization challenges in advanced digital systems, shedding light on the critical role of FIFO buffers in managing CDCs and addressing metastability.*

Keywords: Clock Domain Crossings, GALS, Metastability, Synchronizers, Data Corruption, Incoherence

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Paper ID: 15

Title: CURRENT MODE SINGLE CAPACITANCE CONTROLLED OSCILLATOR

Authors: Dharmesh K Srivastava, Manoj T S, Vinay P T, Puneeth S*

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Abstract: *Single-Resistance-Controlled Oscillators (SRCOs) have long been a subject of significant research due to their practical applications in areas like VCO design, FSK generation, instrumentation and measurements and many more. However, the Single-Capacitance-Controlled Oscillator (SCCO) design has not been as widely explored, even though it also offers distinct advantages in certain applications. In this paper, six new configurations of SCCOs all realizable from only a single current feedback operational amplifier (CFOA) have been proposed. The functionality of all the proposed circuits has been verified through simulation results using a commercially available CFOA IC844 and some results are appended.*

Keywords: Oscillators, CFOA, single element-controlled oscillators, voltage-controlled oscillator

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Paper ID: 17

Title: DETECTION AND CLASSIFICATION OF BRAIN TUMORS IN MRI USING CNN AND VGG-16

Authors: Akshay S*, Aruna V S, Gunashree M, and Prakruthi U

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Abstract: *Brain tumors have been the most critical and life-threatening conditions that affect individuals worldwide. Effective treatment of brain tumors depends critically on early diagnosis and identification. We propose a deep learning system for the detection and classification of brain cancers from MRI images that is based on Convolutional Neural Networks (CNNs). Pre-processing techniques like grayscaling and thresholding are used in our methodology to improve the accuracy of feature extraction and categorization. In contrast to conventional techniques, the proposed method makes use of improved color-based characteristics to recognize and distinguish tumor areas in MRI images, giving way for more accurate localization and classification. To assess how well our proposed system works,a comparison of the performance of CNN and VGG–16 has been done , with CNN achieving a higher classification precision above 95% compared to 93% for VGG-16.*

Keywords: Convolutional Neural Networks (CNN), brain tumors, segmentation, Magnetic Resource Imaging (MRI)

Paper ID: 18

Title: PRIORITY-BASED RESOURCE ALLOCATION IN UAV-BASED TELEMEDICINE SYSTEMS FOR REAL-TIME CRITICAL PATIENT CARE

Authors: Kalpana Naidu*, Aditi Thakur, and Vijay Kumar Dharavath

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Abstract: *This paper develops an unmanned aerial vehicles (UAVs)-based telemedicine system with efficient resource allocation, enhancing healthcare accessibility in remote areas. The model maximizes data throughput, minimizes latency, and prioritizes critical patients. Our closedform solution incorporates latency-aware power adjustments and equal bandwidth distribution while satisfying power, bandwidth, data rate constraints and signal-to-noise ratio thresholds. Simulation results validate the model's efficiency in optimizing the utility function while meeting all constraints.*

Keywords: Resource Allocation · UAV-based Telemedicine · CriticalPatients Priority · Real-Time Communication

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Paper ID: 22

Title: FEW-SHOT BIO-MEDICAL IMAGE ANALYSIS USING DIFFUSION MODEL AND MODIFIED EXPLAINABLE AI.

Authors: Premanand Ghadekar, Tejas R. Wasekar*, Shubham R. Tambe, Shruti Sood, Akanksha Wagaskar and Aniket Thenge

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Abstract: *In the area of medical images and specifically pneumonia detection, our work brings to the fore a new kind of machine learning method in healthcare analysis. With the help of diffusion models, image segmentation algorithms are advanced for the isolation of regions of interest which are not chest X-ray focused and yield an interpretability rate, thus allowing healthcare professionals to make diagnoses. This system comes up with an accuracy of 67.15% and has the added benefit of providing a graphical intuitive interpretation that is helpful in the clinical decision-making process while ensuring reliable results. Our approach, focusing on the enhancement of diagnostic productivity, is even for making a scan that is much more responsive to the needs of the patient. A modified Explainable AI approach was utilized to capture a meaningful representation of the X-ray images, capturing essential features, improving critical region visualization. Finally, the time has come when technology is set to change the health care industry through improving the formulation processes of diagnosis.*

Keywords: Arbitrary and consistent diffusion, biomedical imaging, explainable artificial intelligence (XAI), medical image segmentation, model interpretability, few-shot learning and meta-learning, artificial intelligence in healthcare, visual explanations such as Grad-CAM and LIME, disease detection, semi-supervised learning.

Paper ID: 24

Title: AUTOMATED FISH DISEASE DETECTION USING TRANSFER LEARNING FOR SUSTAINABLE AQUACULTURE MANAGEMENT

Authors: Vijayalakshmi M*, Sasithradevi A, Spoorthi J S and Divya Sankari S

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Abstract: *The Indian fish industry plays a crucial role in food security, nutrition, and economic development, significantly contributing to national exports. However, fish diseases pose a major challenge, impacting productivity and profitability. Early and accurate disease detection is essential to ensuring sustainable industry growth. This study presents an automated fish disease detection and classification system using computer vision and transfer learning techniques. The proposed approach classifies fish into seven categories, including healthy specimens and those affected by bacterial infections such as Aeromoniasis, Bacterial Gill Disease, and Bacterial Red Disease, alongside White Tail Syndrome, fungal infections like Saprolegniasis, and parasitic diseases. Multiple pre-trained deep learning models were evaluated, with MobileNet achieving the highest classification accuracy of 97.13%. The proposed solution improves quality control in aquaculture by overcoming the challenges of manual diagnosis and enabling real-time disease monitoring, making it suitable for remote settings.*

Keywords: Transfer Learning, Fish Disease Diagnosis, Convolutional Neural Networks, Aquaculture Health, Export quality assurance

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Paper ID: 25

Title: ENHANCED U-NET ARCHITECTURE WITH SCALE-PYRAMID DYNAMIC ATROUS CONVOLUTION AND ATTENTION MECHANISM FOR PRECISE METAL ARTIFACT SEGMENTATION IN DENTAL CBCT IMAGING

Authors: Pradeep T S* and Arul Linsely J

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Abstract: *This paper introduces an Enhanced U-Net (EU-Net) architecture to address challenges such as irregular shapes, limited training data, and class imbalances in metal segmentation, crucial for reducing artifacts in dental CBCT images. EU-Net integrates a ResNeXt-50 backbone, Triple Attention (TA) modules for multi-scale feature extraction, and Scale-Pyramid Dynamic Atrous Convolution (SDAConv) for adaptive dilation. Monte Carlo simulations generated diverse, realistic datasets for training. EU-Net demonstrated superior segmentation accuracy and computational efficiency compared to existing methods, offering a robust solution for artifact reduction in clinical settings.*

Keywords: Metal Artifact reduction, CBCT, EU-Net, Triple Attention, Scale-Pyramid Dynamic Atrous Convolution.

Paper ID: 30

Title: ULTRA-GAIN SINGLE -INDUCTOR BOOST DC-DC CONVERTER

Authors: Vadthya Jagan*, Kalagotla Chenchireddy, Sapavath Sreenu, G.Sandeep Rao, Shabbier Ahmed Sydu, and Mohammed Areefa Begum

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Abstract: *An ultra-high gain boost converter with ten times output gain at 200-400W loads is suggested in this research. Two high-frequency MOSFET switches are used in the converter. Only one boosting inductor charges and discharges according to MOSFET switch duty cycle in the converter. The converter's input might be lower voltage sources like photovoltaic (PV) panels or batteries that require high voltage gain to operate loads. The solar irradiation amplitude controls the lower voltage PV panel. The PV array's voltage and current must be stabilized since solar irradiation is erratic. Constant voltage (CV) feedback control with voltage reference does this. The converter's output voltage is stabilized by the CV control. MATLAB/Simulink generates graphs and results under varied operating circumstances.*

Keywords: DC-DC converter, Ultra Gain, Boost Factor, Single-inductor.

Paper ID: 31

Title: A SURVEY ON AYURVEDIC PLANT IDENTIFICATION USING IMAGE PROCESSING AND MACHINE LEARNING TECHNIQUES

Authors: Samrudhi Barhate, Aditya Jadhav, Nisha Bhade*, Tanishq Khadkikar and Maya Bembde

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Abstract: *Accurate identification of Ayurvedic plants is essential for preserving traditional medicinal knowledge and promoting safe usage of plant-based remedies. This survey provides an in-depth analysis of machine learning and image processing approaches applied to automated identification of Ayurvedic plants. It examines techniques such as feature extraction, convolutional neural networks (CNNs), and support vector machines (SVMs), emphasizing improvements in accuracy and efficiency.The survey also examines challenges, including limited datasets and variations in plant appearance, and discusses potential future directions such as cross-regional datasets and real-time identification tools for applications. This work provides valuable insights for researchers in plant taxonomy, machine learning, and Ayurvedic medicine.*

Keywords: Ayurvedic plant identification, machine learning, deep learning, image processing, plant recognition, real-time identification, medicinal plants, taxonomy.

Paper ID: 32

Title: FUZZY BASED POWER DECOUPLER METHOD FOR REGULATING ELECTRIC SPRINGS TO IMPROVE POWER FACTOR AND STABILITY

Authors: Vadthya Jagan*, Kalagotla Chenchireddy, Waseem Sultan, Kodumur Meesala Ravi Eswar, Kodumur Meesala Ravi Eswar, Shabbier Ahmed Sydu and Kottha Aruna

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Abstract: *Electric springs (ES) offer a novel, cost-effective, and highly configurable solution to the power quality problems created by the unpredictability of wind and photovoltaic (PV) generation. In order to enhance the present power control for ES-2's dynamic responses, this project proposes a new fuzzy logic control-based power decoupler that decouples the active and reactive powers. A mathematical model of ES-2 in the dq axis synchronous rotating reference frame was established by incorporating the fictitious-axis emulator (FAE) and second-order generalized integrator phase locked loop (SOGI-PLL) into the control algorithm. Later on, the control system is set up with three closed loops: the active power loop, the current loop, and the ES voltage loop. For precise control of the ES-2 output voltage, one of the three loops the ES voltage loop uses a damped proportional resonance (PR) controller. This fuzzy control technique will enhance power-factor and voltage stability. Fuzzification is a control strategy that helps stabilize the system's voltage and power factor at loads. Lastly, simulations confirm that the suggested decoupled power control is effective.*

Keywords: Electric springs, Fuzzy logic, PV, PLL, Decoupler, Reliability, Grid.

Paper ID: 37

Title: CRAYFISH OPTIMIZATION BASED TUNING OF FRACTIONAL PI CONTROLLER FOR BLOOD PRESSURE REGULATION

Authors: Aryan Kumar, Akanksha Dwivedi* and Ahmad Ali

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Abstract: *This paper introduces a fractional-order PI controller for the regulation of mean arterial blood pressure, a critical concern for post operative patients who frequently experience abnormal blood pressure. The controller's parameters are obtained by minimizing integral time absolute error (ITAE) using the crayfish optimization algorithm. System's closed-loop performance is evaluated, demonstrating improvements in undershoot, settling time, and ITAE. Simulation results validate the effectiveness of the proposed tuning method.*

Keywords: Mean arterial blood pressure, Fractional order PI, Sodium nitroprusside, Crayfish optimization, Sensitivity.

Paper ID: 40

Title: CONVOLUTIONAL NEURAL NETWORKS AND MEL-FREQUENCY CEPSTRAL COEFFICIENTS BASED IMAGE AND AUDIO PROCESSING FOR PEST DETECTION IN PRECISION AGRICULTURE

Authors: Adwitiya Mukhopadhyay*, Ruchitha S, Chinmayee S, Bhavana M, Akshay S, Narendran S M, and Soumik Das

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Abstract: *Agriculture, the bedrock of economic development, plays a pivotal role in global stability, particularly in developing nations. While modern agricultural practices have enhanced yield and productivity, pest control remains a significant challenge. To address this crucial issue, a novel Machine Learning (ML)-based pest detection model is proposed. The proposed system leverages the combined power of image and audio processing to achieve highly accurate and timely pest identification. Convolutional Neural Networks (CNNs) extract intricate features from visual data, enabling precise pest recognition within images. Mel-Frequency Cepstral Coefficients (MFCCs) also analyze audio recordings, capturing characteristic pest sounds for enhanced detection capabilities. This synergistic approach significantly improves the system's robustness and versatility. This advancement particularly impacts developing countries, where robust pest control strategies are crucial for ensuring food security and economic stability.*

Keywords: Agriculture; CNN; MFCC; Pest Detection

Paper ID: 50

Title: STRENGTHENING HIGH-PERFORMANCE COMPUTING SYSTEMS WITH ADVANCED AI-DRIVEN CYBERSECURITY SOLUTIONS FOR ENHANCED PROTECTION

Authors: Iqra Rizvi*, Shreya Pandey, Ashish Tiwari and Anil Kumar

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Abstract: *This research explores the dynamic landscape of High-Performance Computing (HPC), examining its current applications and future potential while analyzing its intersections with emerging technologies such as Artificial Intelligence (AI), Cloud Computing, Cybersecurity, and the Internet of Things (IoT). The study highlights the benefits of cloud-based HPC, including scalability, adaptability to varying workloads, and cost-efficiency, while also addressing challenges like data security and network latency in cloud environments. It delves into HPC's critical role in training complex AI models, conducting cybersecurity simulations, and enhancing threat analysis. Additionally, the paper explores how HPC facilitates real-time data processing and decision-making for massive datasets generated by IoT devices. Looking toward the future, the study discusses key trends shaping HPC's evolution, including its increasing integration with AI, the democratization of access through cloud computing, advancements in hardware architectures, and the advent of exascale computing. While recognizing challenges such as heat generation and data security in exascale systems, the research underscores HPC's transformative potential in driving scientific breakthroughs and technological innovation.*

Keywords: Cloud Computing, Models, Cloud Security, Service management, AI in Cloud, IoT in Cloud.

Paper ID: 51

Title: DENSENET-POWERED TEXTLINE SEGMENTATION IN DEGRADED TAMIL MANUSCRIPTS FOR DIGITAL PRESERVATION

Authors: P.Prakash*, Kasthuri P, Swetha M, Dainty Christiana J C R and Sakthi M

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Abstract: *In Tamil Nadu, ancient manuscripts are an important part of the region's culture and history. However, over time, many of these manuscripts have become degraded, making it difficult to read and preserve them. This issue, along with the lack of efforts to digitize these texts, puts them at risk of being lost forever. It is important to protect and digitally document these manuscripts so that future generations can learn and appreciate their value. In this work, we attempted on detecting text lines in Tamil manuscripts by using advanced pre-processing and segmentation techniques. Our approach focuses on improving the clarity of the text and separating it into clear, distinct lines. We used a large dataset of Tamil manuscripts, including texts like Agathiyar Vaithiyam, Ilakka-na Vilakkam, Kambaramaayanam, Kantha Puraanam, and many others, which contain repeated characters in different styles. To handle the challenges of style variations and noise, we trained a DenseNet model on this dataset. The results show that the model achieved a validation accuracy of 89.02%, This model supports cultural preservation and can be employed for future developments, such as character recognition and translating ancient Tamil texts into modern formats, ensuring that all valuable scripts are accessible to more people.*

Keywords: Text detection algorithm, DenseNet preprocessing techniques, segmentation algorithms, ancient script digitization.

Paper ID: 52

Title: HARNESSING EYEBROW FEATURES TO DETERMINE STRESS: AN INTEGRATED ML-DL FRAMEWORK

Authors: Akshay S*, Archana C, Anu Rose, Kavya Bijith P, Adwitiya Mukhopadhyay, and Manoj Kumar

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Abstract: *Stress detection is essential for proactive mental health management. Proactive mental health management cannot function well without some means of stress detection, yet traditional ways more often than not yield results that are either inaccurate or are not real-time type. This study proposes a layered ML and DL framework based on features around the eyebrows as important indicators for stress detection. This model was tested on several facial features using the NUIG EyeGaze01 dataset. One of the remarkable points it made sure it highlighted throughout the eyeball was the degree of improvement of the classification of the feature set when eyebrows were included. CNN has an accuracy of 96% when working with features around the eyebrows and 62% when it is not. This reinforces the relevance of eyebrow features in the realm of facial expression analysis and paves the way for more advanced AI to stress detection tools.*

Keywords: Facial Expression · Stress Detection · Deep Learning · Eyebrow features

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Paper ID: 63

Title: FPGA IMPLEMENTATION FIR FILTER USING SELF TUNABLE APPROXIMATE COMPUTING

Authors: Harini Shriram. S*, Krishna Meera. V, Chandralekha. R, V.Srirenga Nachiyar, Bavithramsri S K

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Abstract: *These are automatic management systems that use mechanically dynamic parameters or mechanically figure out their best configuration in the VLSI style to adapt to random dynamic settings. The study aims to support the design of an approximate self-adaptive FIR filter, enabling an automobile to self-tune during operation. There are two completely different management systems that are suggested. One uses internal light-weight checkers (LWCs), and the other uses an everyday heartbeat (HB) to reset the approximation circuits on a regular basis. When these checkers see a change in the job, they reset the pruning of the approximations. Removing circuit blocks and the wires that connect them may be a technique known as probabilistic pruning, which trades computing accuracy for overhead-free power, space, and delay savings. The application's error tolerance determines how many prunings are necessary. The decision to prune a node is based on two factors: the activity or toggle count and the node's relevance, which may be a structural characteristic. The nodes with the lowest significance- activity product (SAP) across time are the first to undergo pruning. Pruning nodes primarily aims to lower the mistake rate.*

Keywords: Finite impulse response(FIR), Regular Heartbeat (HB) and lightweight checkers (LWCs),Gate level Pruning, Probabilistic pruning.

Paper ID: 75

Title: A HYBRID MACHINE LEARNING APPROACH FOR FAULT DETECTION AND PERFORMANCE PREDICTION IN PV SYSTEMS

Authors: Nikhil Kushwaha, Vinod Kumar Yadav and Radheshyam Saha

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Abstract: *This research presents a hybrid machine learning approach for fault detection and performance prediction in photovoltaic (PV) systems. The proposed methodology effectively diagnoses faults and predicts power output under various shading conditions, demonstrating its versatility and robustness. The training process achieved stable convergence, evidenced by a consistently decreasing training loss, which underscores the optimization of model parameters. The model's performance characterized by low error percentages across diverse environmental scenarios, indicating its reliability. Furthermore, high accuracy rates in fault detection across multiple fault types enhance the operational efficiency of PV systems. The close alignment between actual and predicted power outputs further validates the effectiveness of the hybrid machine learning (ML) model, highlighting its potential for practical applications in real-world PV installations. Overall, the findings underscore the promise of integrating advanced machine learning techniques in improving the reliability and efficiency of photovoltaic energy systems.*

Keywords: Hybrid Machine Learning, Fault Detection, Performance Prediction, Shading Conditions.

Paper ID: 78

Title: STABILITY REGION BASED GRAPHICAL METHODS FOR CONTROLLER DESIGN: AN OUTLINE SURVEY

Authors: Md Imran Kalim*, Akanksha Dwivedi, and Ahmad Ali

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Abstract: *Proportional-Integral-Derivative (PID) controllers are indispensable in industrial control systems, yet their tuning often relies on empirical methods that lack precision and pose safety risks, particularly for unstable or oscillatory processes. This manuscript addresses these limitations by presenting a comprehensive review of methods to compute the complete set of stabilizing PID controller gains which consequently enables for the selection of optimal parameters to obtain the desired performance. Various techniques such as the generalized Hermite-Biehler theorem, Nyquist stability criterion, and D-decomposition are reviewed for their effectiveness in determining stabilizing regions. Beyond stability, the study emphasizes the integration of performance and robustness specifications, including gain and phase margins, settling time, and damping ratio. Advanced approaches such as convex optimization and centroid-based tuning are explored for deriving specific controller parameters. The applicability of these methods is demonstrated across diverse domains such as motor control, power systems and process industries. This work provides a thorough synthesis of stability region-based PID controller design which offers both theoretical insights and practical tools for achieving robust and high-performance control systems.*

Keywords: PID controller, Performance portraits, All stability region.

Paper ID: 79

Title: ADAPTIVE DIFFERENTIAL EVOLUTION STRATEGY TO IMPROVE AGC PERFORMANCE OF HVDC INTEGRATED POWER SYSTEM

Authors: Udaya Kumar Moyyi, Ch. Venkateswara Rao, and Tulasichandra Sekhar Gorripotu*

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Abstract: *This has been one of the very core mechanisms, even so critical, in power systems toward maintenance and balancing of electricity generation to consume without disturbing the system frequency. The advancement of optimization techniques has enabled the enhancement of AGC performance by minimizing frequency deviations and reducing generation mismatches toward efficiency in operation. The present work aims at presenting the applications of adaptive differential evolution (ADE) technique that include HVDC link for AGC systems. In this article, a methodology has been followed by describing how system is working with and without HVDC link in AGC deregulated frameworks. The performance is evaluated with regards to key metrics such as the frequency regulation improvement, and tie-line power. The implementation analysis in the support of figures and tables shows the advantage of HVDC link inclusion with fuzzy PIDF controller to possess superior performance in minimizing frequency deviations and, consequently, improving the system stability. The study gives valuable insights into optimization techniques and their potential to advance the operations of AGC and improve power system reliability.*

Paper ID: 81

Title: CANCER CELL DETECTION USING TERAHERTZ METAMATERIAL ABSORBER BASED BIOSENSOR

Authors: Nilesh Kumar Shrivastava, Mohit Kumar Da, Prabir Ghosh, Abhik Gorai, Pravin Kumar Samanta*, and Wriddhi Bhowmik

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Abstract: *A multiband terahertz metamaterial absorber (TMA) has been designed to operate within the 1 THz to 6 THz range, serving as a biosensor for early cancer detection. The proposed TMA consists of three semicircular rings with varying radii, where the outermost ring features multiple trapezoidal-shaped microstrip patches. While multiple absorption spectra have been observed, two prominent resonance peaks achieve an absorptivity of 99%, while the remaining peaks exhibit absorptivity close to 80%. For cancer cell detection, the resonance peaks at 2.216 THz and 5.665 THz, both with 99% absorptivity, are utilized. The biosensor exhibits sensitivities of 0.296 and 0.592 THz/RIU as the surrounding material's refractive index change incrementally from 1.3 to 1.5 in intervals of 0.2. This range is particularly relevant for biomedical samples, including normal and cancerous breast and skin cells. Additionally, an in-depth investigation has been conducted to differentiate between normal and cancerous breast and skin cells based on their exact refractive indices. Notable changes in the absorption spectrum have been observed, even with slight variations in the refractive index of cancerous cells compared to normal cells, highlighting the sensor's effectiveness in early cancer detection.*

Keywords: Absorption spectrum, early cancer detection, sensitivity, breast cells, skin cells.

Paper ID: 82

Title: PERFORMANCE EVALUATION OF A GRID-CONNECTED OFFSHORE WIND FARM USING LOAD FLOW, SHORT-CIRCUIT AND TRANSIENT STABILITY ANALYSIS

Authors: A.Rathinavel and R. Ramya*

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Abstract: *The First Offshore Wind Project of India (FOWPI), a key initiative under the EU-funded Clean Energy Cooperation with India (CECI) program, aims to enhance India's capacity for low-carbon energy while improving energy efficiency. It strengthens energy security and fosters strategic cooperation between the EU and India, contributing to climate change mitigation. FOWPI is set to be India's first 200 MW offshore wind farm, located 25 km off the coast of Jafarabad, Gujarat. This study analyzes the performance of a grid-connected 200 MW offshore wind power plant using ETAP, focusing on load flow, short-circuit, and transient stability analyses. Load flow analysis ensures optimal power distribution, voltage regulation, and reactive power compensation. Short-circuit analysis evaluates fault currents, protection coordination, and the impact of various fault types. Transient stability analysis assesses the system's ability to recover from disturbances, examining voltage and frequency stability as well as Fault Ride-Through (FRT) capability. The findings contribute to improving the reliability, protection, and stability of offshore wind integration. Future research can explore the integration of Battery Energy Storage Systems (BESS) to mitigate renewable energy intermittency and enhance grid stability.*

Keywords: Load flow, Short circuit, Transient Analysis

Paper ID: 83

Title: ADAPTIVE PROTECTION SCHEME FOR HYBRID MICROGRID: CASE STUDY ON FAULT ANALYSIS AND RELAY COORDINATION IN IEEE 9-BUS SYSTEM

Authors: Mohamed Nawfal, Gowtham Prakash, Raaz Haneef, Ramya R*

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Abstract: *This research focuses on creating a protection scheme for a hybrid microgrid that combines renewable energy sources, such as wind, solar with a conventional generator. The IEEE 9-bus system is simulated and analysed using ETAP software, this provided a means of foundation for studying load flow analysis, short circuits study, and relay coordination under various fault scenarios. The fluctuating nature of renewable energy sources adds challenges to maintaining stable and reliable system performance. To tackle these complexities, the study employs directional overcurrent relays (DOCRs) along with IEEE Very Inverse curves to optimize fault detection and isolation. The proposed approach incorporates both primary and backup protection to ensure selective fault clearing while preserving the system's integrity. Simulation results highlight the scheme's ability to effectively isolate faults at key locations, including transformers, generators, and renewable energy source. Additionally, relay coordination reduces unnecessary trips, shortens fault clearance times, and boosts overall system reliability, even under changing grid conditions. This work underscores the importance of customized protection strategies in hybrid microgrid, especially as distributed generation continues to expand.*

Keywords: Hybrid Microgrid, Relay Coordination, IEEE 9-bus System, ETAP Simulation, Fault Analysis, Overcurrent Relay, Renewable Energy Integration.

Paper ID: 84

Title: EV ENERGY MANAGEMENT: FORECASTING ENERGY CONSUMPTION AND RANGE USING MACHINE LEARNING

Authors: Mondru Chiranjeevi*, T. Venkateswara Rao

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Abstract: *Predicting energy consumption and range in electric vehicles (EVs) is essential for improving performance, planning infrastructure, and reducing range anxiety. This study utilizes various machine learning methods, such as Multiple Linear Regression (MLR), Decision Trees (DT), Random Forests (RF), and Neural Networks (NN), to forecast energy consumption using diverse datasets that include telematics, charging station usage, and environmental factors. The models are trained and validated through cross-validation, with their performance evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2) metrics. The results show that Neural Networks deliver the best performance, achieving an R^2 value of 0.98, an MAE of 12%, and an RMSE of 14%. Random Forests also perform well, with an R^2 of 0.95. Analysis of energy consumption under various driving conditions reveals that speed and environmental factors significantly influence energy usage, with highway driving consuming the most energy. This research highlights the potential of machine learning to improve EV energy management, optimize charging strategies, and support sustainable transportation solutions.*

Keywords: Electric Vehicles, Energy Consumption, Machine Learning, Prediction Model, Range Forecasting, and Charging Behavior.

Paper ID: 85

Title: HIKING OPTIMIZATION ALGORITHM BASED OPTIMAL TUNING OF PI CONTROLLER FOR MECHANICAL VENTILATORS

Authors: Pritam Raj*, Sumit Ranjan Kumar and Ahmad Ali

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Abstract: *Mechanical ventilators (MVs) require precise pressure con trol to safely support patients with respiratory failure. Traditional control methods often fail to balance responsiveness, stability, or robustness, which can cause lung injury. This paper proposes a novel tuning strategy for PI controller to enhance pressure regulation in MVs. In the first step, all-stabilizing region is identified by integrating Routh-Hurwitz stability criteria with constraints on maximum sensitivity to ensure guaranteed gain and phase margins. Centroid of the all-stabilizing region is chosen as the baseline controller. In the second step, the Hiking Optimization Algo rithm (HOA) is used to fine tune the controller parameters to minimize settling time, overshoot, and tracking error. Simulations demonstrate that the HOA-tuned controller outperformed the benchmark IMC-PI controller with improved performance metrics.*

Keywords: Mechanical Ventilators, PI Controller, All-stabilizing region, Hiking Optimization Algorithm (HOA), Pressure Control Ventilation (PCV)

Paper ID: 89

Title: TECHNO-ECONOMIC ANALYSIS OF RENEWABLE SYSTEM INTEGRATED WITH AC GENERATOR IN REMOTE LOCATION

Authors: S.Vinoth John Prakash, Vempalle Rafi, R.Durga Devi, R.A. Priya , K. Barathi and S. Amosedinakaran

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Abstract: *Green energy is developing recent years to overcome the effect of environmental pollution. The main purpose of this work is to conduct a techno-economic analysis for hybrid system in car Island. The suitable hybrid system consists of solar/AC generator (ACG), solar/biomass generator (BG) and solar/ACG/BG are designed at low cost and less emission. The main problem is supplying electrical power to remote location where there is no extension of transmission line. The proposed solution is to install hybrid standalone renewable system for power generation with low cost. To analyze the system, residential AC and DC load demand is considered. The average solar radiation is considered from NASA data. Mathematical modeling for various sources such as solar, battery, generator and converter have been done. The result of optimization process determines that the net present cost is low in solar/ACG/BG configuration. The novelty of the research work is that the analysis is performed in particular remote area with real time data of renewable sources and optimization results. The new thing proposed in this research is that the optimization is performed with power generation using renewable including biomass plant in remote location to reduce emission with low cost. The benefits over existing strategy is cost of energy is low as 0.159 $/kWh.*

Keywords: Solar, Biomass, AC generator.

Paper ID: 95

Title: VOLTAGE STABILITY ASSESSMENT IN DISTRIBUTED GENERATION PHOTOVOLTAIC SYSTEM USING STABILITY INDEX COMPUTATION

Authors: Jarabala Ranga, Senthil Kumar M, Sundar R, Sabarish P, Hubert Tony Raj L, Saravanan S, Rajakumar P*

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Abstract: *Distributed generation (DG) is increasingly adopted in rural places since power is generated at or near the load site. Also, DG does not require transmission/distribution lines for power transfer. In the current scenario, all the power utilities, including DG, operate in maximum loading conditions to meet the power demand considering the economic operation. This causes a voltage deviation to fall below a secure limit and induces a voltage instability problem. Consequently, it leads to voltage collapse or blackout in the power networks. Hence, voltage stability assessment has become an integral part of the electrical power systems to initiate necessary countermeasures. In this study, the voltage stability (VS) of a DG photovoltaic (PV) system is assessed through stability index (SI) evaluation. The voltage stability is measured for different loading conditions. The proposed VS investigation is carried out via simulation and experimental setup on a 5 kWp PV system.*

Keywords: Distributed Generation, Photovoltaic System, Voltage Stability, Voltage Collapse, Stability.

Paper ID: 101

Title: RELIABILITY ASSESSMENT OF A GRID-CONNECTED SOLAR PHOTOVOLTAIC SYSTEM CONSIDERING A PARALLEL BACKUP SUBSYSTEM

Authors: Sudhanshu*, Sachin Kumar, and Sachin Negi

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Abstract: *This paper delves into the factors that influence the reliability of solar photovoltaic systems and overall power generation. One critical element discussed is the significance of real-time monitoring, which allows immediate detection of issues, enabling prompt interventions to maintain system efficiency and uptime. The study specifically focuses on the physical configuration and arrangement of a grid-connected solar PV system with its six subsystems: the PV module, the battery storage system, the inverter, the converter, the controller, and the protection mechanisms. These subsystems work in tandem to ensure the smooth operation of the entire system. The paper also looks at three different failure cases. Case 1 (maximum failure rate), Case 2 (minimum failure rate), and Case 3 (use of a parallel backup subsystem) with various failure rates: maximum and lowest. It provides a complete analysis of how each case affects system reliability and power generation. A novel aspect of the study is considering a parallel protection backup system with a new failure rate. The study shows that over time the solar photovoltaic system becomes more reliable, especially when failures of subassemblies are considered and fixed. The reliability results are improved when compared to the existing literature.*

Keywords: Battery, Converter, Grid protection, Inverter, PV module, Reliability, Solar PV system.

Paper ID: 103

Title: ENHANCED DYNAMIC PERFORMANCE OF A SOLAR-POWERED SRM DRIVE THROUGH ADVANCED CONTROL STRATEGIES

Authors: P. R. Jeevarathinam*, K. Vinoth

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Abstract: *This research introduces a combined system designed to improve EV performance and efficiency. The system connects a solar panel setup to a special Integrated SEPIC-Luo DC-DC converter to maximize power from the solar panels, using a smart Artificial Neural Network (ANN) controller optimized by the African Buffalo Algorithm (ABA) to constantly pull the maximum power, even with changing sunlight. Extra Energy from Two-way Converter is stored using Supercapacitor to power the EV. SRM motors whose speed is precisely controlled by a Proportional-Integral (PI) controller with a Hysteresis controller. Surplus energy can also be sent back to the grid via a single-phase VSI. The integrated design aims for optimal use of solar power, better EV efficiency, and support for sustainable energy and grid stability, achieving a tracking efficiency of 98.85% for the ABA-ANN controller and a converter efficiency of 90% in MATLAB simulations.*

Keywords: Key components of the proposed system are an integrated SEPIC-Luo converter for optimized power conversion, an ABA-optimized ANN controller for maximum power point tracking, a three-phase SRMN for efficient propulsion, and a hysteresis-based PI controller for precise speed regulation.

Paper ID: 105

Title: A LIGHT-WEIGHT CNN-TRANSFORMER BASED FRAMEWORK FOR DYNAMIC SIGN LANGUAGE RECOGNITION

Authors: Ajay Kumar Naik G, Imon Kumar Biswas, Ch. V. Rama Rao,

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Abstract: *Deaf individuals primarily use sign language to communicate, but interaction with hearing individuals can be challenging due to translation barriers. This research focuses on Sign Language Recognition (SLR) using a deep learning model that targets dynamic signs, which are commonly used in everyday communication. The model combines a compact Convolutional Neural Network (CNN) with a Multi-Headed Attention (MHA) mechanism for effective feature extraction from video frames. It has a size of 2.14 MB and contains only 177356 trainable parameters, making it resource-efficient. Here, American Sign Language (ASL) and Indian Sign Language (ISL) datasets are used for Continuous Sign Language Translation and Recognition (CSLTR), achieving 99.89% accuracy for word-level data and 95% for sentence-level data. The model's performance is competitive with existing SLR approaches.*

Keywords: Convolutional Neural Network (CNN), Transformer, Sign Language Recognition, Spatial Features, and Temporal Features.

Paper ID: 108

Title: MODELING AND OPTIMIZATION OF EMERGING HYBRID POWER SYSTEMS

Authors: Sravanthi Pagidipala and Sandeep Vuddanti*

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Abstract: *Increased power demand, concerns over air pollution, and global warming have pushed power systems to move from conventional power generation to renewable power generation. This paper presents the modeling and optimization of emerging hybrid power systems (EHPSs), including renewable energy resources (RERs) – wind and solar PV, battery energy storage systems (BESSs), electric vehicles (EVs), electric vehicle charging stations (EVCSs), and demand response (DR). The transition to EVs marks a pivotal shift in the automotive and energy sectors, driven by the urgent need for sustainable transportation and renewable energy (RE) solutions. Energy storage systems (ESSs) and spinning reserves can solve several operational problems in EHPSs to mitigate the intermittent nature of RERs and EVs while providing various other benefits, including the provision of ancillary services, security, and reliability improvement etc. This paper also presents various optimization techniques and uncertainty-handling approaches for solving the issues and challenges associated with EHPSs.*

Keywords: Hybrid power system, Energy management, Energy storage, Electric vehicles, Renewable energy, Uncertainty.

Paper ID: 113

Title: EFFECTIVE HYBRID DEEP LEARNING MODEL FOR WIND POWER PREDICTION

Authors: Bikash Chandra Saha* and Avinash Kumar

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Abstract: *Non-conventional energy sources, particularly wind power, play an important role in generating sustainable energy. The annual growth rate of wind farms is directly proportional to the number of nations making the switch from conventional to renewable energy. Wind turbines are usually recognized as one of the most efficient ways of producing renewable energy. To address the unpredictability of energy systems, wind power prediction is essential. The intricate design and challenging operating conditions of wind power plants further complicate the task of balancing energy supply and demand. This research aims to create an effective Deep Learning (DL) model for forecasting wind power generation. The study uses hybrid DL models such as Convolutional Neural Network - Long Short-Term Memory (CNN-LSTM) and Autoencoder - Long Short-Term Memory (AE-LSTM). The study makes use of Supervisory Control and Data Acquisition (SCADA) data from Turkish wind turbines, which have undergone the appropriate pre-processing. The processed data is sent to both hybrid models and a standard Long Short-Term Memory (LSTM) model for validation. The evaluation uses error measurements, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Error (MAE). Computational time is also considered. The simulation results show that the proposed AE-LSTM model lowers wind power forecasting errors more effectively than the CNN-LSTM and LSTM models. The AE-LSTM model achieved an MSE of 0.0532, RMSE of 0.2306, MAPE of 13.56%, and MAE of 0.145, demonstrating its effectiveness in improving the accuracy of wind power forecasts.*

Keywords: Wind Power, Missing Value, Time Forecast, Deep Learning, Long Short-Term Memory, Error Metrics.

Paper ID: 116

Title: POWER QUALITY AND STABILITY ANALYSIS OF A RENEWABLE MICRO GRID USING SMALL-SIGNAL STABILITY ANALYSIS WITH PQ BASED ARTIFICIAL NEURAL NETWORK CONTROL SCHEME

Authors: Prakash Chandra Patra*, Pratap Kumar Panigrahi, Anshuman Satpathy, Niranjan Nayak

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Abstract: *Micro grids (MGs), comprising diverse micro sources like solar PV, wind generating system, Battery energy storage system power electronics, and constant power loads (CPLs), faces stability challenges. Stability analysis is vital during both design and operation. This paper explores a modelling method for CPLs in AC systems. A small-signal state-space model (SSM) of a micro grid is developed, including sub-models for the power controller and network/loads. The model is simulated to investigate PID, artificial neural network (ANN) and P-Q ANN controllers. The study examines the impact of factors such as P-Q gains, CPLs, and line impedance on micro grid stability. A model of an islanded micro grid in MATLAB/SIMULINK is developed and simulated for different time spans. Different case studies like power quality issues, constant power load and irradiance variations are investigated. Finally the Eigen value analysis is done in this work.*

Keywords: Power quality issues, Micro Grid (MGs), Renewable energy sources (RES), PID, Artificial neural network (ANN) and P-Q ANN controllers.

Paper ID: 117

Title: FIREBOT: AI-POWERED AUTONOMOUS FIRE-FIGHTING ROBOT WITH REAL-TIME DETECTION AND GSM ALERTS

Authors: Kanimozhi S, Harshad S*, Vishranth M S K and Nitin Pranav T S

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Abstract: *Fire-Bot is a machine-learning-based autonomous firefighting robot aimed at real-time fire detection, extinguishing, and emergency alerting. It features infrared flame sensors, smoke sensors, and machine-learning-based image recognition for precise fire detection. With a servo-driven water pump, it extinguishes fires effectively, and a GSM-based module provides instant emergency alerts. The mobility of Fire-Bot is controlled through DC motors and an ultrasonic sensor-based collision avoidance system for self-navigation. One of its notable features is the Reusable Solar-Battery System, which supports prolonged operation without recharging from outside. Experimental tests indicate 95.4% accuracy in detection, fire extinguishing in 9.9 seconds, and emergency notifications within 2.5 seconds. The use of AI, sensor-based authentication, and renewable energy increases efficiency, reliability, and lifespan, and FireBot is a scalable solution for industrial, commercial, and residential fire protection. Future developments involve IoT integration, thermal imaging, and deep-learning-based fire severity estimation for enhanced adaptability and performance.*

Keywords: AI based fire detection, real-time firefighting, GSM emergency alerts, robotic firefighting systems, intelligent fire prevention, sensor-based fire detection, IoT- enabled fire safety, AI driven emergency response, smart firefighting technology, renew- able energy powered robots

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Paper ID: 118

Title: A MACHINE LEARNING FRAMEWORK FOR MODE IDENTIFICATION IN POWER SYSTEMS WITH CORRUPTED PMU DATA

Authors: Mani Shankar Kumar* and Shekha Rai

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Abstract: *This paper presents a robust GCN-TCN-based method for estimating poorly damped modes in wide-area smart grid monitoring, natively addressing issues related to degraded Phasor Measurement Unit (PMU) data. This method is specifically designed to mitigate the impact of missing data and outliers, which may arise from causes such as communication failure, PMU or Phasor Data Concentrator (PDC) failures, and cyber-attack that impact mode estimation. To address this using the Graph Convolutional Network (GCN) algorithm is used for outlier detection and removal, while Temporal Convolutional Networks (TCN) are employed for imputing missing data. Lastly, Total-Least Square Estimation of Signal Parameters via Rotational Invariance Technique (TLS-ESPRIT) is used for mode parameters estimation. This method ensures accurate mode estimation, thereby enhancing system stability and reliability. This approach guarantees precise using two-area data, real-time Western Electricity Coordinating Council (WECC) measurements, and simulated IEEE 39-bus system data within a Real-Time Digital Simulator (RTDS).*

Keywords: PMU, Bad Data, TLS-ESPRIT, GCN-TCN, Modes Estimation

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Paper ID: 119

Title: DETECTION OF 9 VARIETIES OF SKIN CANCER USING A DEEP LEARNING ENSEMBLE

Authors: Shipra, Sweta Kumari*, Shiv Kumar Choubey, Gauri Sinha

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Abstract: *Skin cancer is a serious and often life-threatening ailment that impacts numerous lives each year. The visual similarity between various melanocytic lesions, such as melanoma and dermatofibroma, complicates the discovery and diagnosis process. Given the varying degrees of severity, early detection is crucial to prevent the progression of cancer, which can be life-threatening. This highlights the need for robust classification systems to accurately differentiate between types of skin cancer. The research proposes an automated classification algorithm designed to boost the speed, performance, and timeliness of dermal cancer diagnosis. This approach utilizes a pre-trained neural network, initially developed for a different task, to classify skin cancer images into nine distinct categories. The method employs BigTransfer (BiT), a ResNet-based transfer learning technique, to achieve high classification accuracy. The complete cancer detection system is implemented using Power Apps, ensuring a seamless endwise solution. The behavior of the proposed method is benchmarked against current techniques, demonstrating superior classification rates and improved diagnostic capabilities.*

Keywords: Biomedical imaging, darmatology, deep learning, Machine learning

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Paper ID: 123

Title: COMPARATIVE STUDY OF ISLANDING DETECTION FOR A SINGLE-PHASE PHOTOVOLTAIC SYSTEM USING MACHINE LEARNING

Authors: J. Deo, S. S. Panigrahi, B. Panda, S. Behera*

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Abstract: *The rapid growth of photovoltaic (PV) systems necessitates effective islanding detection to ensure grid stability and safety. Traditional methods have limitations in speed and accuracy, but machine learning (ML) offers a promising solution to handle data samples from signals. In this work, an ML-based islanding detection system for a single-phase PV (SPPV) system is developed. ML models like Decision Tree (DT), Support Vector Machine (SVM), and fine K- Nearest Neighbour (KNN) are utilized here. The ML performance is improved by thoroughly observing the effects of hyperparameters and setting the best of it and presented. The three ML models are compared in performance based on accuracy and training time. Datasets of seven important electrical parameters are generated from the simulation of the model in MATLAB/Simulink and the ML models are trained. The fine KNN excels in detection accuracy when contrasted with recent work.*

Keywords: Islanding Detection, Machine Learning, fine K- Nearest Neighbour

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Paper ID: 127

Title: FPGA BASED VENDING MACHINE WITH EXCESS AMOUNT RETURN FEATURE

Authors: Aditya Mishra, Sweta Kumari*, Shipra and Shiv Kumar Choubey

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Abstract: *The vending machine is an automated device that dispenses a variety of products to customers when cash or a credit card is placed. The products range over a variety of products like food, beverages, newspapers, tickets, and more. In contrast to traditional purchase methods, vending machines are more accessible these days. The goal of this paper is to create a vending machine that can dispense different products at varying costs with the added functionality of "return change" when a coin with a higher denomination is provided/inserted and "return money" when a request is cancelled. The design of this vending machines uses the finite state machine (FSM) methodology and is accomplished through using behavioral modelling to create the Verilog code for the FSM- based vending machine.*

Keywords: Vending Machine, FSM, Verilog, FPGA, Mealy Machine

Paper ID: 128

Title: OPTIMIZATION OF ROBOTIC GRIPPER CONFIGURATIONS USING KINEMATIC MODELLING AND GENETIC ALGORITHMS FOR ENHANCED STABILITY AND PERFORMANCE

Authors: Rajalakshmi Murugesan*, Santha Kumar Mohan and Karthik Chandran

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Abstract: *The design and optimization of robotic grippers are vital to enhancing the precision and functionality of robotic systems in industrial automation and rehabilitation. This paper develops a methodology for optimizing the gripper performance using genetic algorithms (GA), forward kinematics, and force-torque analysis. The mechanical representation is accurate and well-represented through a 3D model via STL files. Forward kinematics, derived from Denavit-Hartenberg (DH) parameters, calculates the fingertip position for different joint configurations. The GA seeks to minimize operational torque and thus optimize the gripper's configuration for stability and performance. A fitness function that measures configurations regarding their fingertip positions and applied forces is presented through a force-torque schematic of the acting forces and torques. It has been found that there is a significant improvement in the joint angles, fingertip positions, and torque values, which bring down energy consumption and improve stability. A convergence plot shows how effectively the GA works, and comparative analyses reveal where performance gains will be. This study provides valuable information on the robotic gripper design and its application in industrial and healthcare robotics.*

Keywords: Robotic gripper, genetic algorithm, forward kinematics, force-torque analysis, optimization, energy efficiency

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Paper ID: 137

Title: SINGLE VCII BASED BIQUADRATIC FILTER CONFIGURATION

Authors: D R Bhaskar, Kirti Kaharwar*, Ram Bhagat and Pragati Kumar

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Abstract: *Active filters are often employed in various areas of signal processing, control systems, instrumentation, and communication to eliminate the distortion in a signal. This paper presents a generalized structure of a second-generation voltage conveyor (VCII) based multifunction filter configuration employing one VCII+ and five admittances. The proposed configuration can realize, low pass filter (LPF), a high pass filter (HPF), and a band pass filter (BPF) from the same circuit configuration by choosing the appropriate combination of the different admittances. Simulation results using PSPICE employing 180nm TSMC CMOS technology characteristics are used to validate the application of the suggested filter functions. The sensitivity, Monte Carlo, and temperature analyses are performed to test the usefulness of the proposed filter structure. The obtained simulation results coincide with the theoretical predictions.*

Keywords: Analog Filters, Analog Signal Processing, Active Filters, VCII

Paper ID: 140

Title: MULTI-INPUT SINGLE OUTPUT ELECTRONICALLY TUNABLE UNIVERSAL FILTER USING TWO COMMERCIALLY AVAILABLE ICS LT1228

Authors: Vivek Kumar, Ram Bhagat and D. R. Bhaskar

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Abstract: *In this paper, a voltage-mode (VM) multi input single output (MISO) biquad filter circuit is proposed using two commercial LT1228 ICs, as the active element, one resistor and two capacitors. The presented circuit provides five voltage mode filter responses such as high pass (HP), low pass (LP), band pass (BP), band stop (BS) and all pass (AP). All filter responses provide orthogonal electronic tuning of the natural frequency and bandwidth. The functionality of new proposed circuit is tested through PSpice simulation tool using macro-model of LT1228.*

Keywords: voltage mode, MISO, LT1228, universal filter.

Paper ID: 141

Title: VOLTAGE MODE THIRD-ORDER QUADRATURE SINUSOIDAL OSCILLATORS USING UVC

Authors: Tvisha Pushkar, and Rajeshwari Pandey

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Abstract: *This paper introduces a novel third-order quadrature sinusoidal oscillator (TOQSO) utilizing two Universal Voltage Conveyors (UVCs), three capacitors, and three resistors to achieve high precision and low distortion sinusoidal wave generation. The proposed TOQSO design provides two output voltages in perfect quadrature (90° phase difference), making it highly suitable for applications in communication systems, instrumentation, and signal processing. A key advantage of this design is the independent control of the Condition of Oscillation (CO) and Frequency of Oscillation (FO), where CO is adjustable via resistor R_1, and FO is controlled independently using resistor R_2. This flexibility ensures enhanced tuning capability without affecting other circuit parameters. The circuit exhibits low passive sensitivity, contributing to greater stability and improved performance. Extensive simulations were performed using SPICE with TSMC 0.18µm process parameters to validate the proposed design, confirming its theoretical feasibility and practical efficiency. The simulation results demonstrate stable oscillations, low harmonic distortion, and reliable frequency response, proving the circuit's effectiveness for advanced analog signal processing applications.*

Keywords: Universal Voltage Conveyor, Voltage-Mode, TOQSO, and Analog Circuit Design.

Paper ID: 144

Title: A CFOA BASED INVERSE BAND PASS FILTER AND ITS APPLICATION AS PID CONTROLLER

Authors: Kaushal Kumar*, Piyush Jaiswal, Rishabh Jha, Ram Bhagat and Manoj Kumar

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Abstract: *This study describes the design and analysis of a CFOA-based inverse band-pass filter (IBPF) and its implementation as a Proportional-Integral-Derivative (PID) controller in a closed-loop system. In addition, the plant is equipped with a CFOA-based low-pass filter (LPF), illustrating the circuits' usefulness in control applications. Both filters make use of the AD844 IC, which has the distinct properties of Current Feedback Operational Amplifiers (CFOAs), such as broad bandwidth, high slew rate, and gain-bandwidth independence. The IBPF has inherent proportional, integral, and derivative actions, allowing it to function as an analog PID controller. Theoretical research and simulations confirm the system's ability to produce precise control, increased stability, and lower steady-state error. This study highlights the adaptability of CFOA-based circuits in high-speed, high precision control systems and signal processing.*

Keywords: CFOA, Inverse Band-Pass Filter, Low-Pass Filter, PID Controller, AD844 IC, Control Systems.

Paper ID: 145

Title: SMART INTEGRATION OF WIND AND BESS ENABLED SOLAR POWER TO GRID

Authors: Sonu Kumar**,** Ruchi Agarwal, Sanjay Agrawal**,** Lokesh Kumar Yadav**,** Nikhil Chaudhary, Praveen Kumar, Shivani*, Pankaj Pratap

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Abstract: *The integration of renewable energy resources (RER) of hybrid solar and wind power into the existing grids has increased significantly in the last few decades. However, the grid integration have many challenges in operation and control that affect the reliability and stability of the grids. Wind and solar power are available in huge amounts on earth. This paper presents the hybrid operation of solar power and wind power in reliable and stable mode with Battery Energy Storage System (BESS). The proposed multi-level inverter worked with superior control. For its stability in connection to the grid, the system first analyzes the mathematical models of wind energy production, solar energy production and Battery Energy Storage System. Then build a complementary power generation system and design a management strategy. Finally, the simulation is verified by MATLAB software. The simulation results show that the proposed control strategy improves the stability in grid operation and the ability to deliver maximum power to the network. During uncertainties of wind and solar generation, the system is losses are reduced which reducing the voltage deviation in the grid.*

Keywords: Renewable Energy Resources. BESS, Solar Power, Wind Power.

Paper ID: 147

Title: VDCC-BASED VOLTAGE MODE UNIVERSAL FILTER

Authors: Ayush Bhardwaj*, Ram Bhagat and Anup Mandpura

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Abstract: *This paper presents a voltage mode (VM), triple input and single output (TISO) biquad filter using single voltage differencing current conveyor (VDCC) and three passive components. The proposed configuration is capable of realizing second-order high pass filter (HPF), low pass filter (LPF), band pass filter (BPF), band stop filter (BSF) and all pass filter (APF). The proposed biquad configuration has tunability of center frequency (ω), pole quality factor (Q). The proposed filters are implemented in PSPICE and the filter responses are simulated. The simulation results are then verified with theoretical analysis.*

Keywords: Biquad, VM, VDCC, Filter, Analog signal processing, TISO, Single input double output**.**

Paper ID: 149

Title: RELIABILITY ASSESSMENT OF RADIAL DISTRIBUTION NETWORK WITH PROBABILISTIC FAILURE OF PROTECTION DEVICES

Authors: Manish Kumar Madhav*, K.B. Yadav and Rajesh Kumar Yadav

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Abstract: *The reliability of the radial distribution network (RDN) is crucial for ensuring a regular and uninterrupted power supply to consumers. In radial distribution networks, the performance of protective devices significantly influences system reliability, with failure probabilities directly affecting interruption frequency and duration. This study investigates the impact of protection device failures on the reliability of radial distribution networks. A heuristic approach using particle swarm optimization (PSO) is applied to optimize the placement of protection devices, aiming to enhance the distribution network reliability. By strategically minimizing interruption frequency and duration, the proposed method improves load point adequacy. The proposed method is tested on a modified IEEE-34 bus system, considering 100% and 80% reliable switches. Reliability indices go down about 5% due to protective device failure. The analysis highlights the effectiveness of PSO in optimizing reliability while considering failure probabilities of protection devices. The findings provide valuable insights for designing resilient distribution systems, demonstrating the practical applicability of PSO in modern power distribution networks.*

Keywords: Distribution Network, Failure Probability, Reliability Indices, Protection Devices, Reliability.

Paper ID: 150

Title: BIDIRECTIONAL INDIAN SIGN LANGUAGE TRANSLATOR

Authors: R Pavan Kumar, Kurakula Sai Sree Akshaya*, Vipin Kamble and Ankit Bhurane

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Abstract: *Communication for individuals with hearing and speech impairments remains a challenge due to the lack of efficient translation systems. This paper presents a Bidirectional Indian Sign Language Translator, enabling seamless communication between sign language users and non-signers. The system supports real-time conversion of text or speech (input in English or Hindi) into sign language, generating animated sign output using SiGML files. The text-to-sign translation process involves language translation, stopword removal, grammatical reordering, and stemming, using NLP and pyreparsing. Additionally, the system con- verts sign language into text from uploaded videos or real-time webcam input using a Transformer-based deep learning model and mediapipe. A web application integrates both functionalities, ensuring accessibility and ease of use. This research aims to enhance communication efficiency, fostering a more inclusive and accessible interaction for individuals with hearing impairments.*

Keywords: Bidirectional Indian Sign Language Translator, SiGML, NLP, Pyreparsing, Transformer, Mediapipe.

Paper ID: 154

Title: AUTOMATIC DETECTION OF DIABETIC FOOT ULCERS: ENHANCING EARLY DIAGNOSIS USING DEEP LEARNING

Authors: Nandana Suresh*, Nirupama R Pillai, Sana Mathew, Vinaya Prasad P, Akshara Babu, and Justy Jameson

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Abstract: *Diabetic foot ulcers (DFUs) are a serious complication of diabetes that, if not detected early, can lead to infections, tissue damage, and even amputations. Timely diagnosis is crucial to preventing severe outcomes, yet access to specialized care remains a challenge, particularly in under-resourced areas. This presents an AI-powered deep learning model designed to identify DFUs from foot images using convolutional neural networks (CNNs). The system accurately distinguishes between healthy and ulcerated skin, enabling early detection and intervention. The model outperformed all others with an accuracy of 89.62% which means that the model generalized well on most of the unseen data. With a simple and accessible interface, both patients and healthcare professionals can upload images for real-time analysis, making diagnosis more efficient. This feature ensures that even individuals without medical expertise can access early diagnostic support, making it particularly beneficial in areas where specialized care is unavailable. By enhancing accuracy and accessibility, this technology reduces healthcare costs while improving diabetes management. It empowers individuals to take proactive steps in their care and supports healthcare providers in making informed decisions. Ultimately, this AI-driven solution aims to improve patient outcomes and overall quality of life.*

Keywords: Diabetic foot ulcers (DFUs), Convolutional Neural Networks (CNNs), early diagnosis, AI-driven solution

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Paper ID: 165

Title: DEVELOPMENT OF SOLAR ENERGY CONVERSION SYSTEM FOR LINEAR LOAD APPLICATION

Authors: Sreedevi Kunumalla*, Durgam Rajababu, A V V Sudhakar, B Sathyavani,

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Abstract: *Connecting a load abruptly to a Solar Energy Conversion System (SECS) can affect both the voltage and frequency. These impacts can be reduced by employing a suitable controller and PWM technique in the SECS. The system incorporates a Load Current Observer (LCO) to monitor transient behavior of currents, a Unified Space Vector Pulse Width Modulator (USVPWM) serves to control inverter output voltage and frequency while an Adaptive Voltage Controller (AVC) minimizes Total Harmonic Distortion (THD). This study simulates the SECS using MATLAB/Simulink, integrating LCO, AVC, and the USVPWM technique, and examines the resulting data.*

Keywords: Adaptive Voltage Controller, DC-DC Converter, Load Current Observer, Total Harmonic Distortion, Transient time, Unified Space Vector Pulse Width Modulation technique

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Paper ID: 166

Title: HAND GESTURE -CONTROLLED ROBOTIC SYSTEM FOR CLEANING APPLICATION: A HYBRID APPROACH USING GREY WOLF OPTIMIZER AND EFFICIENTNET

Authors: Roshan Chitranshi*, Swaroop Mallick, Mohammad Shabeeh, Mohd. Shariq Mahboob, Saurabh Pandey and Shweta Singh

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Abstract: *The study is a development of a new technical approach where a hand gesture-based robot is designed to be utilized in cleaning tasks. The model is hybrid in nature because it uses a Grey Wolf type Optimizer (GWO) and EfficientNet in a combined way. The main goal of the model developed here is to improve the accuracy and efficacy of hand gesture recognition and robot control by the means of parameter optimization. The GWO is used to tune the hyperparameters of the gesture recognition model, thus the outcome is the system's performance in the real-time hand gesture detection is improved. EfficientNet, one of the best in its class of deep learning models, is used for feature extraction and it is a thin yet highly accurate way for understanding complex hand gestures. The biggest achievement of the model is in the combination of these two methods in order to create a strong, efficient, and responsive cleaning robot which on the other hand can interpret a wide range of hand gestures, thus enabling user control to be intuitive. The suggested system is a great step forward in human-robot interaction, specifically in the household or industrial cleaning environments where the robotic automation offers a practical, as well as easy-to-use, solution.*

Keywords: Hand gesture Recognitions, robotic model, cleaning mechanism, Grey Wolf type Optimizer, EfficientNet, smart learning, human-robotic interaction, gesture controller, relevant feature extraction, realistic-time detections.

Paper ID: 168

Title: SYNERGISTIC STABILIZATION OF N-DIMENSIONAL FRACTIONAL ORDER CHAOTIC SYSTEMS VIA FIXED POWER REACHING PRINCIPLE BASED FRACTIONAL ORDER SLIDING MODE CONTROL

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Abstract: *This paper introduces a new kind of fractional order control methodology for the chaos suppression of nth order perturbed fractional order version of chaotic systems with fixed time power reaching principle. The proposed control approach devises a fractional order version of fast terminal sliding mode strategy to accelerate the chaos suppression of* n^{th} *dimensional fractional order version of chaos. In addition, a fixed time power reaching principle is being utilized for the convergence of the proposed sliding surface to origin, independent of its initial conditions. By suitable tuning control gains, this introduced strategy facilitates strong robustness against time varying nonlinear uncertainties, reduced chattering effects, reduced control effort, fixed time convergence of sliding variables independent of initial conditions and better time domain tracking capabilities. Moreover, the experiment* of validating *the proposed technique has been conducted whose simulated results have further been clarified with respect to the conventional fractional order version of sliding mode technique.*

Keywords: Fractional order version of control, Fixed time control, Chaos, Sliding Mode Control.

Paper ID: 186

Title: MODELLING OF PMSM FOR PERFORMANCE EVALUATIONS USING ANSYS MAXWELL RMXPRT

Authors: D D Padma Priya, S. Amosedinakaran*, P.Babu Rao, B.Vinod, P.Marishkumar and S.Vinoth John Prakash

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Abstract: *Permanent Magnet Synchronous Machines (PMSMs) require precise modeling and simulation in order to be optimized, perform fault analysis, and assess performance. A thorough method for modeling a PMSM with ANSYS Maxwell RMxprt, a potent tool for electromagnetic design and analysis, is presented in this work. By including important design characteristics such rotor shape, magnet positioning, stator winding arrangement, and material qualities, the proposed study aims to provide a thorough model of a PMSM. The modeling procedure is done in RMxprt, which makes it easier to quickly prototype and verify the design before doing finite element analysis (FEA) in Maxwell 2D/3D. The PMSM's performance features, such as torque-speed curves, cogging torque, and electromagnetic losses, are assessed and confirmed using theoretical computations. Additionally, the effect of design variants on machine performance is examined, including winding configurations, pole-slot combinations, and magnet thickness. In order to improve the PMSM's overall efficiency and refine its architecture, the outcomes of the RMxprt model are crucial. Engineers may use the results to optimize PMSM performance for a range of industrial applications, such as robots, electric cars, and renewable energy systems.*

Keywords: PMSM, Performance, FEA, EV, RMxprt.

Paper ID: 195

Title: AUTONOMOUS MARINE OIL COLLECTOR: A ROBOTIC SOLUTION FOR EFFICIENT OIL SPILL CLEANUP

Authors: Shreenivas Margamwar*, Paramjeet Rathod, Aditya Dapke, Pradyumn Bhanegaonkar, Aarti Salunke, Prajakt Narwade

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Abstract: *Oil spills pose a significant threat to marine ecosystems and cause severe environmental and economic damage. Conventional oil spill mitigation techniques, such as oil booms, sorbents, and in-situ burning, have limitations in efficiency, cost, and environmental impact. This research presents an autonomous robotic oil skimmer designed to collect oil from water surfaces using a hydrophobic conveyor belt. The system integrates realtime monitoring via ESP32 Wi-Fi modules, ultrasonic sensors for oil level detection, and a motor-driven navigation system. Initial testing demonstrated a 96% oil recovery rate. The proposed system offers an efficient, sustainable and scalable solution to mitigate marine oil pollution.*

Keywords: Oil Spill Cleanup, Autonomous Oil Collector, Robotic System, Hydrophobic Conveyor Belt, Real-time Monitoring, ESP32, Environmental Pollution, Marine Pollution Control, Sustainable Technology

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Paper ID: 196

Title: STUDY OF RELIABILITY OF POWER GRID INTEGRATED WITH DISTRIBUTED RENEWABLE ENERGY SOURCES: A REVIEW

Authors: Sneha K. Awaze* and Govind Rai Goyal

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Abstract: *Need for carbon emissions reduction and climate change has made the transition to renewable energy sources (RES), including wind and solar, essential for developing sustainable, low-carbon energy systems. Yet, these phenomena introduce substantial challenges in integrating RES to existing power systems since their output is not stable unlike that of conventional fossil fuel-based generation. This variability makes balancing supply and demand difficult, with the potential for grid instability and power interruptions. Since BESS and smart grid solutions are crucial technologies to eliminate these issues. BESS charge excess energy during periods of high renewable generation and discharge energy during times of peak demand to help smooth fluctuations. Smart grid technologies encompass real-time monitoring, communication networks, and adaptive control algorithms to optimize renewable energy utilization and enhance system reliability. These technologies also facilitate dynamic load balancing, improve resilience and offer flexibility to adapt to renew- able energy variability. Moreover, Distributed Generation (DG) systems including but not limited to small scale solar and wind installations also support energy efficiency and grid stability. But to avoid imbalances, their integration needs sophisticated reliability metrics and control methods. With a focus on forecasting, integration, and grid aspects, this study seeks to devise methodologies and techniques leading to bolstered reliable and resilient renewable-embedded power systems, facilitating the transmission towards a sustainable energy future.*

Keywords: Renewable energy sources, Renewable-integrated power systems, Reliability

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Paper ID: 198

Title: APPLICATION OF HUNTER PREY OPTIMIZER ALGORITHM FOR MULTI-OBJECTIVE DG ALLOCATION

Authors: Vijayakumar G, Nandakumar E, Kamatchi Kannan V, Sankara Subramanian A T, Ram Kumar C, Rajakumar P* and Balasubramaniam P M

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Abstract: *This paper investigates the efficacy of the hunter prey optimization (HPO) algorithm for optimizing multiple distributed generation (DG) units in a radial power distribution system (RDPS). Multiple DGs are optimized to minimize total real power losses (TRPL), improve bus voltages (BV), and enhance the voltage stability of RDPS. The effectiveness of the HPO algorithm is tested for three units of type I DG placement and allocation in the IEEE 69-bus benchmark power network. Type I DG is modelled as a constant real power source. The multiple DG positions and sizes are optimized considering relevant equality and inequality constraints. Finally, HPO optimized DG allocation simulation results related to the test results of different algorithms available in the literature. The comparison reveals more favourable results for the HPO algorithm than other algorithms.*

Keywords: Distributed Generation, Radial Power Distribution System, Real Power Loss, Bus Voltage, Stability, Optimization

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Paper ID: 201

Title: DATA DRIVEN CONTROL OF NONLINEAR SYSTEM WITH NOISY MEASUREMENTS

Authors: Aastha Yadav*, Dipayan Guha and Richa Negi

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Abstract: *In recent years, the usability of data in the realization of network-based control systems has significantly increased. In this work, an LQR-based state-feedback control strategy for a discrete-time nonlinear time-invariant system having noisy measurements has been designed to deal with exogenous perturbations. Optimal solutions for the state-feedback controller have been computed, solving data-dependent linear matrix inequalities. The employment of the data-driven control strategy explicitly avoids the identification of plant dynamics for realizing the control law. Extensive stability assessment using the Lyapunov approach is illustrated to ensure controllers' effectiveness. Additionally, the robustness of the closed-loop framework to noisy-corrupted measurements has been demonstrated to affirm the applicability of the proposed approach for stabilizing unstable equilibria in nonlinear systems.*

Keywords: Data Driven Control; Robustness; Noise; Optimality; Stability.

Paper ID: 202

Title: DETERMINATION OF PROXIMITY TO VOLTAGE INSTABILITY UNDER SEVERE LINE OUTAGE USING CUSTOMIZED ARTIFICIAL NEURAL NETWORK

Authors: S.D. Naik*

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Abstract: *The voltage instability predominantly occurs in systems with heavy loads under large scale disturbances. In the deregulated environment and in open access operation, the transmission lines operate under stressful conditions. There are instances of instability of voltage occurring in some countries recently. This poses a challenge before the authorities to keep system voltages under control. The loss of voltage stability can result in regional load loss or an unprecedented opening of HV transmission lines and transformers by their switchgear systems leading to widespread outage and collapse of voltage in the system. The function of operation control center or the Energy Management System (EMS) of the power company is that of system security assessment process and calculation of limits of system operation before and after the contingency. A customized ANN is designed for voltage stability study. For a given power system, under most severe line outage, the Maximum loadability and bus voltages for any loading condition can be found out without performing the time-consuming load flow repeatedly. The results are verified for IEEE 6 Bus system.*

Keywords: Voltage stability; Loadability; Active power; Reactive power; Customized ANN

Paper ID: 203

Title: COMPENSATING V & F DEVIATIONS THROUGH TRANSIENT DROOP CHARACTERISTICS ENABLED MICRO SOURCE INVERTER

Authors: Sameer Bhambri*, Vivek Shrivastava and Manoj Kumawat

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Abstract: *This paper proposes a strategy which compensates deviations in the output voltage and frequency caused by an inverter output impedance, droop coefficient (m, n and Ke) terms in the robust droop control. This is achieved through adopting the transient droop characteristics in an already developed robust droop control for single phase inverter. A parallel communicationless system of two micro source inverters is considered which is equipped with the proposed control strategy to get proportionate power sharing , nominal output voltage and nominal system frequency at the steady state. The effectiveness of the proposed strategy is evaluated and compared, with and without incorporating the transient droop characteristics through simulation study in the MATLAB/SIMULINK platform. In the next step, the proposed structure is modified to improve the dynamics of various responses contributing novelty in the proposed work.*

Keywords: Compensation, Controller, Droop, Error, Power sharing.

Paper ID: 207

Title: SCALABLE HARDWARE OPTIMIZATION AND PERFORMANCE COMPARISON OF AES ENCRYPTION VARIANTS ACROSS DIFFERENT TECHNOLOGY NODES

Authors: Shreenivas G. Margamwar*, Chaitanya N. Kadadas, Shubhashri G. Joshi, Shashank Kumar Singh, Yogesh Sejwal, and Jayaraj U. Kidav

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Abstract: *Ensuring the secure transmission and storage of digital information is essential for the effective operation of modern computing systems. The Advanced Encryption Standard (AES) is widely adopted due to its robust security, ease of implementation, and compatibility with both hardware and software platforms. Among its variants, AES-256 offers the highest level of security, albeit with increased computational and resource overhead. This paper presents a comparative analysis of AES-128, AES-192, and AES-256 hardware implementations across 180nm, 90nm, and 45nm CMOS technology nodes using Cadence Genus and Innovus tools. Key performance metrics such as area utilization, power consumption, and timing are evaluated. To improve efficiency, optimization techniques including pipelining and loop unrolling are applied. A novel "security-per-watt" metric is introduced to assess the trade-off between energy efficiency and cryptographic strength. Additionally, the post-quantum resilience of AES and implications for IoT and embedded applications are discussed. This comprehensive study provides a practical foundation for developing secure, high-performance, and energy-efficient cryptographic solutions suitable for deployment in resource-constrained environments.*

Keywords: AES-256 , AES-192 , AES-128, CMOS Technology Scaling, Power Efficiency, Hardware Optimization, Security-per-Watt , PostQuantum Cryptography

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Paper ID: 210

Title: SMART SPEED CONTROL OF SINGLE PHASE INDUCTION MOTOR VIA WI-FI ENABLED ESP8266

Authors: Chithras T, Vijayakumar G, Kesavan D, Sakthidasan A, Jeba Rexciya M, Rajakumar P and Pranav Kumar Rowlo*

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Abstract: *Induction motors (IM), particularly single-phase types, are almost exclusively employed for household applications. Speed, direction, and temperature control in single-phase induction motors have become essential for different operations. In this work, a wireless real-time speed controlling system is implemented for a single-phase induction motor. The control system includes a temperature, voltage, and current monitoring system. Temperature sensors measure the temperature of the winding and bearing. Voltage sensors are used to measure the induction motor's voltage, while current sensors are used to measure the current level. The speed of a single-phase induction motor is detected with the help of an infrared (IR) sensor. All of this sensor data is received by the ESP8266 microcontroller, which then processes it and communicates data for necessary action if required. Furthermore, wireless speed control is achieved through the Blynk mobile application via the cloud platform. Wireless control is achieved with the aid of a smart Wi-Fi-enabled ESP8266 microcontroller. All the parameters, such as speed, temperature, voltage, and current, will be available in the Blynk mobile application for real-time monitoring.*

Keywords: Blynk Application, Communication, Induction Motor, Real-time Monitoring, Speed Control, Wireless Control.

Paper ID: 228

Title: COMPARISON AND ANALYSIS BETWEEN LINE START PERMANENT MAGNET SYNCHRONOUS MOTOR AND AXIAL FLUX PERMANENT MAGNET SYNCHRONOUS MOTOR AS A DRIVE SYSTEM IN ELECTRIC VEHICLE

Authors: Payel Bhattacharya*, Arindam Kumar Sil, Mousumi Jana Bala

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Abstract: *In Electric vehicles new driving systems should be developed in response to concerns about environmental pollution and the depletion of fossil-fuel resources caused by combustion engines. Therefore, one promising technology for guaranteeing a green future is the electric automobile. One of the key components of this electric vehicles (EVs) is an electrical motor. The efficiency and stability of EV performance can be enhanced with the right motor design and selection. As electric motor drive machinery continues to advance, electric vehicles (EVs) have been revived in recent decades. This paper provides a comparison analysis of two kinds of electrical motor drive technologies used in EV application. The comparison between the performances of Axial Flux Permanent Magnet Synchronous Motor (AFPMSM) and Line Start Per-manent Magnet Synchronous Motor (LSPMSM) of same power rating with two dif-ferent Pole configuration have been drawn here. All the four motor configurations in this study are designed to perform as a possible EV application. The studied motors have predicted under steady-state condition. These motors have been compared based on their torque, efficiency and losses.*

Keywords: Axial Flux Permanent Magnet Synchronous Motor (AFPMSM), Line Start Permanent Magnet Synchronous Motor(LSPMSM), Electric vehicle (EV), Rated Torque, Efficiency

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Paper ID: 234

Title: AUTISM SPECTRUM DISORDER IDENTIFICATION FROM FACE IMAGES USING DENSENET 121

Authors: K. Anuradha* and M. Sucharitha

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Abstract: *Autism Spectrum Disorder (ASD) is a common neurodevelopmental disease affecting children, with an incidence rate of 0.62% worldwide. ASD impacts brain development, resulting in difficulties in behavior, social interactions, and communication. Children with ASD who experience morphological differences exhibit distinct characteristics compared to Typically Developing (TD) children. Early detection of ASD is essential to ensure that caregivers can provide appropriate support and intervention. Existing ASD classification models face limitations such as small and imbalanced datasets, high computational costs, and lack of interpretability, making real-world applications challenging. In this study, we explore using Deep Learning(DL) models, which include Convolutional Neural Networks (CNNs), ResNet50, MobileNetV2, InceptionV3, VGG19, and DenseNet121, for autism classification. The model is trained on the publicly available Autism Image Data dataset from Kaggle, which includes 3,040 images—2,940 designated for training and 100 for testing. By leveraging transfer learning with the DenseNet121 architecture, Our model attains an accuracy of 84% on the given dataset, highlighting its potential for effective ASD classification.*

Keywords: ASD, facial images, CNN, transfer Learning, DensNet121

Paper ID: 237

Title: DESIGN AND IMPLEMENTATION OF FRACTIONAL ORDER PID CONTROLLER FOR CONTROLLING REGULATION IN BLOOD PRESSURE

Authors: Abhishek Jain*, Mohammad Atif Siddiqui and Md Nishat Anwar

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Abstract: *The article presents the designing of Fractional Order Proportional Integral Derivative (FROPID) controller, to regulate the Blood Pressure (BP) and reject the load disturbances. The parameters of the FROPID are optimized by minimizing the objective function for Integral Time Absolute Error (ITAE), Integral Absolute Error (IAE) and Integral Square Error (ISE) using Grasshopper Optimization Algorithm (GOA). Simulations are performed for the proposed system and the results are compared with PI and 2 DOF H_2 optimal PI controllers that are tuned by using Dragonfly Algorithm (DA) and a FROPID controller tuned by using Genetic Algorithm (GA). The proposed Fractional-Order Robust PID (FROPID) controller exhibits superior performance in terms of the IAE, achieving a faster rise time and a shorter settling time compared to conventional controllers.*

Keywords: Blood Pressure, FROPID controller, Grasshopper Optimization Algorithm.

Paper ID: 240

Title: ADVANCING SEPIC CONVERTER DESIGN FOR OPTIMAL PERFORMANCE IN POWER ELECTRONICs

Authors: Rahul Kumar, Rohan Kumar Gupta, Suneel, Pritam Kumar*, Chinmay Bera, Rashmi Kumari, Amitesh Kumar

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Abstract: *In the contemporary landscape of portable electronics, the demand for efficient and compact power converters has never been greater. These devices must adeptly manage a wide range of input and output voltages to ensure optimal performance and longevity. Traditional power converters, particularly buck-boost configurations, often fall short in achieving high efficiency across diverse operational ranges. They are plagued by issues such as inconsistent input current, inverted output voltage, and increased voltage stress, which compromise their reliability and applicability in varied scenarios. To address these challenges, the Single-Ended Primary Inductor Converter (SEPIC) emerges as a versatile and efficient solution. Capable of both stepping up and stepping down voltage levels, the SEPIC converter boasts a straightforward control mechanism that ensures low-noise operation, making it an ideal choice for a wide array of applications. This paper presents a comprehensive methodology for designing the passive components of a SEPIC converter, ensuring optimal performance across a broad spectrum of input and output voltages. By leveraging the SEPIC converter's unique capabilities and adhering to the design principles detailed herein, engineers can develop power conversion solutions that meet the stringent efficiency, size, and reliability requirements of today's portable electronic devices.*

Keyword: Capacitor, Duty Ratio, Inductor, Ripple, Single-Ended Primary Inductor Converter (SEPIC).

Paper ID: 253

Title: A MACHINE LEARNING APPROACH FOR HELMET DETECTION IMPLEMENTATION AND BIKE SPEED CONTROL SYSTEM

Authors: P. Lakshmi Prasanna, P. Kamalesh Karthik*, P. Surya, V. Keerthi and K. Vijaya Vardhan

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Abstract: *Motorbikes are more common and affordable than other types of vehicles, many individuals in our country use them for a variety of reasons. In addition to taking up less parking space, they offer greater mobility and flexibility in both urban and rural settings. Riding a bike is both fun and dangerous. Despite being cautioned by law, many drivers choose not to wear helmets. Mortality rates have been gradually rising in certain developing nations. To protect the public and identify drivers who are not wearing helmets, helmet detection technology is required. We will limit the speed by determining whether the biker is wearing a helmet to adhere to regulations and ensure the safety of the rider. We employ a Machine Learning (ML) algorithm to limit the bike's maximum speed to meet these standards; if not, it will permit the minimal or necessary speed in an emergency, depending on the situation or the riders' needs.*

Keywords: MCU, ML algorithm, DAC, DC motor

Paper ID: 262

Title: HYBRID FUZZY-ANN CONTROL STRATEGY FOR POWER QUALITY IMPROVEMENT USING DSTATCOM

Authors: Vadlamudi Bindu, Vennela Voodi, Manaswi Chintapenta*, Nakka Srinivas, Mohammad Riyaz

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Abstract: *Growing electrical demand, along with economic and social development as well as improved technology, raises the need for a stable and reliable power supply of high quality. However, the integration of renewable energy into distribution systems increases the difficulty associated with maintaining quality power due to its intermittent character. This project focuses on improving power quality at the distribution level by integrating advanced control strategies using fuzzy logic and ANNs with a DSTATCOM. The fuzzy controller adaptively regulates the output of the DSTATCOM using threshold-based logic, dynamically adjusting parameters based on system conditions to compensate for power disturbances. In the meantime, the ANN determined dynamic weights for the load's active and reactive power components. Thereby, it provides accurate system state representation that fine-tunes fuzzy controller outputs in terms of the optimum compensation associated with specific disturbances. Simulations in the MATLAB environment proved that the suggested hybrid control strategy is effective enough to help in reducing voltage sags and swells hence allowing reliable power quality and uninterrupted operation of sensitive loads.*

Keywords: dstatcom, fuzzy controller, Ann control strategy, distribution generation system.

Paper ID: 269

Title: FOPI CONTROLLER DESIGN FOR BLOOD PRESSURE REGULATION USING HIPPOPOTAMUS OPTIMIZATION ALGORITHM

Authors: Gaurav Kumar Gupta*, Sumit Ranjan Kumar, and Ahmad Ali

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Abstract: *Blood pressure readings indicate the force that heart exerts on arterial walls while pumping blood. Maintaining mean arterial blood pressure during surgeries is crucial for controlling excessive bleeding. Vasodilating drugs like sodium nitroprusside are used to lower blood pressure, but the required dosage and effects vary significantly depending on the condition of the patient. This necessitates continuous monitoring and precise drug delivery. To achieve this, a fractional-order proportional-integral controller is tuned by first obtaining the stable region in the plane of controller parameters which satisfy the specified constraint on the maximum sensitivity. Then, the hippopotamus optimization algorithm is used to select the suboptimal parameters. Constraint on the maximum sensitivity ensures adequate phase margin (ϕm) and gain margin (Gm) so that the closed-loop system is stable. MATLAB simulations demonstrate superior performance of the proposed controller compared to existing control strategies.*

Keywords: Mean Artial Blood Pressure, Hippopotamus Algorithm, Fractional-Order Proportional Integral, H∞ criteria, Performance Index

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Paper ID: 280

Title: A DESIGN OF HIGHLY EFFICIENT PV BASED WIRELESS CHARGER FOR EV AT DIFFERENT ATMOSPHERIC CONDITIONS

Authors: Preety Kumari*, Ravikant Kumar, U K Sinha

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Abstract: *Transportation that is electrified will contribute to lower greenhouse gas emissions and higher petroleum prices. Wireless electric vehicle charging systems (WEVCS) are one potential non-plug alternative method for charging EVs. Here, MATLAB/Simulink software has been used to design and simulate a 5kw charging station. Modern electric cars (EVs) are replacing conventional internal combustion (IC) vehicles in an effort to lower greenhouse gas (GHG) emissions. Our proposed architecture incorporates an optimized coil design, advanced power electronics with silicon carbide switching devices, and an adaptive frequency control scheme to maintain high efficiency across variable coupling conditions. Laboratory and field tests demonstrate system efficiency exceeding 92% at the nominal 6.6 kW power level, with sustained performance across the full operating range of 4.9 to 8.09 kW. Wireless charging solutions for electrical vehicles (EVs) are superior to cable-based systems because of their robustness, safety, and ease of use. Photovoltaic array and WPT system can be combined to provide a renewable energy source for electric vehicle charging. First, a photovoltaic source model is examined, then, a DC/DC converter with MPPT control scheme has been used to guarantee best PV generator power. Wireless EV charging system's simulation findings are shown in this article.*

Keywords: PV Array, MPPT, DC-DC Boost Converter, Inverter, Transmitter coil, Receiver coil.

Paper ID: 282

Title: CLASSIFICATION OF PROSTATE CANCER WITH DEEP LEARNING ENSEMBLE MODEL

Authors: Samana Jafri* and Gajanan Birajdar

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Abstract: *Prostate cancer is perhaps one of the most widespread cancers affecting men; therefore early and correct diagnosis of this disease is of paramount importance for its successful treatment. Deep learning methods have been applied to handle the identification process as well as the classification system which relies on PI-RADS scores for prostate cancer. The implementation of such methods contains specific limitations. As a result, this paper employs deep learning with the idea of ensemble learning, which is built on three well-known CNN architectures: Resnet50, MobilenetV2, and VGG16. Three separate pre-trained models of ImageNet are used. These models were trained and validated on the publicly available Fast MRI Prostate dataset which includes MRI images annotated with PI-RADS scoring from 1 to 5. The collected dataset was preprocessed intensively, where all incoming images were resized and flattened to prepare it for model training. PI-RADS provided the network with a dense output layer that classified MRI images into five risk categories. Performance results from the proposed model exceed the individual outcomes achieved by separate Deep Learning architectures alongside recent study approaches in this field. The proposed system reached the highest test accuracy rating at 95.03%.*

Keywords: Deep learning, T2W MRI images, prostate cancer diagnosis and ensemble learning.

Paper ID: 283

Title: REAL TIME IMPLEMENTATION OF DEEP LEARNING FRAMEWORK FOR MEDICAL DATA ANALYSIS

Authors: Pradeep Enukonda* and Latha P

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Abstract: *Brain tumor detection using deep learning has significantly improved diagnostic accuracy in medical imaging. This project implements a real-time CNN-based brain tumor detection system on the PYNQ-Z2 FPGA board to enhance speed, efficiency, and power consumption. The trained TensorFlow/Keras model is first converted into an HLS-friendly format using hls4ml to enable FPGA-based execution. The model is synthesized into hardware using Vivado HLS, and the resulting bitstream is deployed on the PYNQ-Z2 board. Using Direct Memory Access (DMA), MRI images are transferred to the FPGA for inference, significantly reducing latency compared to CPU-based processing. The optimized FPGA implementation achieves real-time performance, making it suitable for clinical applications. The system efficiently classifies MRI images into four categories: Glioma, Meningioma, No Tumor, and Pituitary Tumor, enabling fast and accurate tumor detection for medical diagnostics.*

Keywords: Brain Tumor Detection, CNN, Deep Learning, FPGA, PYNQ-Z2, hls4ml, Vivado HLS, Medical Imaging, Real-Time Inference, MRI Classification, Bitstream, Hardware Acceleration.

Paper ID: 288

Title: FEATURE EXTRACTION OF ECG SIGNALS BASED ON POLYPHASE FILTER WITH DISTRIBUTED ARITHMETIC

Authors: Dhakshina Murthy M, Sharon Sthuthi E, Santhosh Kumar S and Sudhanya P*

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Abstract: *This paper presents the feature extraction of ECG signals using an optimized signal filtering technique integrating polyphase structure with Distributed Arithmetic (DA) for Finite Impulse Response (FIR) filters. The proposed methodology enhances real-time ECG signal processing by reducing computational complexity, power consumption, and hardware resource utilization. ECG data from the MIT-BIH Arrhythmia Database is extracted and preprocessed. The polyphase structure decomposes the FIR filter into parallel subfilters, minimizing latency, while DA eliminates the need for multipliers by utilizing precomputed lookup tables (LUTs), significantly improving efficiency. The filtered ECG signals undergo feature extraction to identify QRS complexes. The results demonstrate improved accuracy and resource optimization, highlighting the potential of this approach for advanced cardiac monitoring systems.*

Keywords: Field Programmable Gate Array (FPGA), Electrocardiogram (ECG), Distributed Algorithm (DA).

Paper ID: 289

Title: A NOVEL DEEP LEARNING FRAMEWORK FOR AUTOMATED SICKLE CELL DISEASE DETECTION

Authors: V.Ceronmani Sharmila, Utkarsh Agrawal, Rohan Rajeev Kumar, Abinash Kodamsingh*

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Abstract: *This study introduces a novel deep learning framework for automated detection of Sickle Cell Disease (SCD) by analyzing erythrocyte morphology in blood smear images. The proposed system integrates robust data preparation, advanced segmentation techniques, and two deep learning models: a Customized Deep Convolutional Neural Network (CDCNN) and a fine-tuned VGG16. Initially, the framework processes raw images by removing extraneous space and isolating individual cells, ensuring high-quality input for classification. The CNN models are trained to differentiate between normal, sickle, and abnormal cell shapes using optimized hyperparameters and adaptive learning algorithms. Experimental results indicate that the CDCNN outperforms VGG16, achieving superior accuracy, precision, recall, and F1-score. The approach minimizes manual intervention, enhances diagnostic reliability, and demonstrates potential for scalable medical diagnostics. Overall, the system provides an efficient, automated solution for early detection and analysis of SCD, offering a promising tool for improving patient outcomes in clinical settings and very high scalability.*

Keywords: Sickle Cell Disease, CNN, Erythrocytes, and Deep Learning.

Paper ID: 290

Title: UNDERWATER ACOUSTIC SOURCE LOCALIZATION USING TIME-DIFFERENCE OF ARRIVAL (TDOA) WITH KALMAN FILTER

Authors: Ajay Chikkam* and Fayaz Ahamed Shaik and Karthik Abburi and ChandraKanth Avula and Vijaya Sankar A and Surya Narayana Gunnam

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Abstract: *Accurately locating underwater acoustic sources is essential for many important fields, including marine research, environmental monitoring, and national defense. One of the most widely used techniques for this task is called Time Difference of Arrival (TDOA). This method determines how long it takes for sound to travel to different hydrophones, helping to pinpoint the source's location. However, the underwater environment can be tricky. Factors like background noise, signal distortion, and inaccuracies in sensor readings can lead to significant errors in these estimates. To tackle these challenges, the project focuses on improving TDOA-based localization by integrating a Kalman filter. This powerful mathematical tool helps refine predictions by reducing errors and managing uncertainties. By applying the Kalman filter, noise can be better handled and sensor drift accounted for, resulting in more accurate and dependable localization. Promising results from simulations demonstrate that the Kalman filter performs significantly better than traditional TDOA methods in terms of accuracy. This work highlights the potential of combining TDOA with Kalman filtering for real-time underwater tracking, even in challenging and noisy conditions. This advancement could enhance the understanding and monitoring of underwater environments.*

Keywords: Acoustic Source Localization, Time Difference of Arrival, Kalman Filter.

Paper ID: 293

Title: PSO BASED GENE EXPRESSION NETWORK MODELLING AND PID CONTROL OF SARS-COV-2 BIOPATHWAY FOR ENHANCING DRUG TARGET IDENTIFICATION

Authors: Nivendu Nandy* and Anirban Dey and Kaushik Das Sharma

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Abstract: *The COVID-19 outbreak highlighted the urgent need for effective treatments and accurate gene network models to track disease progression and gene behavior. This research introduces a particle swarm optimization (PSO)-driven parameter estimation method for nonlinear ordinary differential equations (ODEs), facilitating alignment with time series gene expression micro-array datasets related to SARS-CoV-2 infection on human lung tissue. Furthermore, a Proportional-Integral-Derivative (PID) control scheme is implemented on a highly sensitive regulatory node to elucidate the dynamic responsiveness of gene networks. PSO's global search capability enables effective system identification by minimizing the prediction error while addressing the high dimensionality and non-linearity of gene expression data. The PID controller effectively stabilizes the gene expression levels toward desired reference states, aiming to overcome the viral infection and provide insights into the internal gene network response to visualize drug target potential. PSO-based parameter estimation is validated by minimizing state errors and ensuring convergence with experimental data, while PID control is evaluated by tuning parameters to regulate expression, enhancing stability, and optimizing time-domain response. This scheme offers deeper insights into drug target identification by revealing critical gene expression dynamics, enabling more precise therapeutic targeting.*

Keywords: Gene Expression Modelling, Particle Swarm Optimization, SARS-CoV-2.

Paper ID: 302

Title: SPEED CONTROL OF SOLAR POWERED BLDC MOTOR USING PI CONTROLLED BOOST CONVERTER

Authors: Rajani Kandipati*, RamaKoteswaraRao Alla, Lakshmi Charishma Dronadula, Lakshmi Niharika Kothuri, Govardhan Komari and Chandu Chaithanya Kanaparthi

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Abstract: *This work presents a high-efficiency speed control technique for a solar-powered Brushless DC (BLDC) motor using a boost converter to enhance system performance. A photovoltaic (PV) array supplies power, with a Maximum Power Point Tracking (MPPT) technique ensuring optimal energy extraction under varying solar irradiance. A modified Perturb and Observe (P&O) algorithm addresses traditional P&O limitations, minimizing oscillations and improving dynamic response. The boost converter regulates the PV voltage to meet the BLDC motor's requirements, ensuring stable operation. Proportional Integral (PI) controller controls speed by adjusting the boost converter's duty cycle, precisely regulating voltage and current. The system is simulated in MATLAB/Simulink to evaluate performance under different conditions. Results demonstrate that combining the boost converter with modified P&O-based MPPT enhances power extraction, while PI-based speed control ensures smooth motor operation. This approach significantly improves efficiency, making it ideal for renewable energy-based motor applications.*

Keywords: Solar-powered BLDC motor, Boost converter, MPPT, Modified Perturb and Observe.

Paper ID: 306

Title: VILLA: VISION-LANGUAGE LAYOUT ANALYZER FOR FLOOR PLAN ANALYSIS

Authors: Sabari Nathan, Sasithradevi A*, and P Prakash

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Abstract: *Floor planning plays a crucial role in architectural design, impacting functionality, space optimization, and human-centric usability. Traditional approaches for analyzing floor plans rely on rule-based systems, CAD software, or manual inspection, which can be time-consuming and error-prone. In this work, we propose ViLLA (Vision-Language Large Architecture), a hybrid AI-driven framework that integrates a Large Language Model (LLM) with a Vision Transformer (ViT) for automatic floor plan analysis. The model leverages ViT for image-based feature extraction and an LLM, fine-tuned via LoRA, to generate structured textual insights, including room classifications, layout summaries, and functional zoning. The LLM embedding module bridges visual encodings with semantic reasoning, enabling robust spatial understanding. Our approach is evaluated on diverse floor plans, demonstrating its ability to accurately extract key architectural attributes, assess design coherence, and generate high-level summaries. By automating floor plan analysis, ViLLA enhances efficiency in architectural design review, space planning, and real estate assessment.*

Keywords: Floor Plan Analysis, Large Language Models, Vision Transformers, Architectural AI, Generative AI, Spatial Understanding

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Paper ID: 307

Title: DOUBLY EVEN ORDER MAGIC SQUARE (DEOMS) BASED RECONFIGURATION SCHEME FOR THE MITIGATION OF POWER LOSS IN SOLAR PHOTOVOLTAIC SYSTEMS

Authors: Karthick B*, M. Senthil Raja, Hariharan Narayanaswamy and M. Sudhagar

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Abstract: *Partial shading significantly reduces photovoltaic (PV) system performance, lowering energy output and causing hotspots that may damage modules. As renewable energy demand grows, optimizing solar efficiency under inconsistent shading is vital. This study proposes a novel method using the Doubly Even Order Magic Square (DEOMS) to rearrange PV arrays and mitigate shading effects without altering electrical connections. MATLAB simulations on an (8×8) PV array under various shading patterns compare DEOMS to traditional layouts like Total-Cross-Tied (TCT) and Chaos Baker Map (CBM). Results show DEOMS achieves better Global Maximum Power Point (GMPP) tracking, reduces power loss to 500 W, increases the fill factor to 42.75%, reaches a 97% execution ratio, and produces 18,590 W at GMPP—demonstrating its effectiveness in enhancing PV performance under shading.*

Keywords: PV Array Reconfiguration, Total-cross-Tied, Double Even Order Magic Square.

Paper ID: 314

Title: FREQUENCY ANALYSIS OF HIGHLY PENETRATED RENEWABLE ISLANDED MICROGRID USING CASCADED CONTROLLER

Authors: Bisaya Bhoi*, Rajaram Behera and Sagarika Rout

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Abstract: *Modern energy system is shifting towards renewable energy sources (RES) for clean energy, sustainability and energy sovereignty, which distributes native economy and lessens transmission and distribution costs. However, it introduces instability within islanded microgrid when integrated, due to its intermittent nature as it is extensively dependent on weather conditions. It drives the risk of disruptions and induces inclination to perturbations. This critical failure is enough to trigger frequency drastically drop causing system collapse. In islanded microgrid due to low inertia and limited generation reserves make the system even worse. Therefore, load frequency control (LFC) is presented here in this hybrid system to resolve the issues for optimized power utilization and to suppress the system from instability. Load frequency control tackles these issues by managing generation-consumption divergence. In this survey, cascaded proportional integral with fractional order PID (PP-FOPID) based ant lion optimization (ALO) algorithm is proposed with least Integral squared error (ISE) ensuring robust frequency regulation.*

Keywords: sustainability, Energy storage capability, Efficient power management

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Paper ID: 322

Title: OPTIMAL ALLOCATION OF ELECTRIC VEHICLE CHARGING STATION AND DISTRIBUTED GENERATION: AN APPLICATION OF HUNTER PREY OPTIMIZER ALGORITHM

Authors: Sivakumar J, Ravi V, Mohanapriyaa N, Silambarasan V, Balasubramaniam P M, Sundar R, and Rajakumar P*

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Abstract: *Most of the existing transportation systems around the globe are fossil fuel-based, and hence they release enormous amounts of greenhouse gases into the environment. The pollution increases the surrounding temperature and causes global warming. The situation demands an immediate alternative transportation system to minimize the pollution. Electric vehicles (EVs) are a recent invention in a transportation system for controlling environmental pollution. The battery is the integral part of the EV system, and it requires regular charging. Charging stations need to be adequately placed in the power grid system for longer transportation vehicles. However, inappropriate placement of EV charging stations (EVCS) in the power grid will increase the power losses (PL) and offer an economic burden to utilities. In the present work, distribution network power losses are minimized by optimizing the placement of EVCS and distributed generation (DG). EVCS placement and DG allocation are optimized by applying a novel hunter-prey optimizer (HPO) algorithm. The effectiveness of the proposed HPO technique is realized on the 33-bus IEEE radial distribution system (RDS).*

Keywords: Charging Station, Electric Vehicle, Distributed Generation, Optimization, Power Losses.

Paper ID: 324

Title: POWER FACTOR CORRECTION OF BIDIRECTIONAL WIRELESS POWER TRANSFER FOR EV APPLICATION

Author: Roshan saju and Narayanamoorthi Rajamanickam*

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Abstract: *The rapid adoption of EVs creates an imperative need for efficient and flexible charging systems that can move beyond simple conductive methods. This project deals with designing and simulating a bidirectional system and efficient energy transfer from the grid to the EV. The suggested bi-directional capability allows the possibility to charge the battery and is applicable for V2G operations to support good promotion toward renewable energy sources and grid stability. The power transfer is of high efficiency by using a resonant inductive coupling. The designs are properly developed to minimize energy losses while maintaining safety. Optimal management of the power flow direction under load variation conditions was exhibited by the control strategies in the simulation. The simulation results indicate that bidirectional WPT is feasible as a viable sustainable charging solution for EVs and supports the development of a more resilient energy ecosystem..*

Keywords: wireless power transfer system, EV charger, High-frequency inverter, Renewable energy source

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Paper ID: 329

Title: A COMPARATIVE ANALYSIS OF SMC, STSMC WITH NN FOR GRID STABILITY

Author: T. Vyshnavi* and Divya R

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Abstract: *This work suggests an advanced control scheme for solar power generation based on a Super-Twisting Sliding Mode Controller (STSMC) to achieve improved power stability and eliminate chattering. Though Sliding Mode Control (SMC) is robust, it is plagued by high frequency oscillations. STSMC eliminates this drawback with a smoother response while ensuring robustness. Furthermore, a Neural Network (NN)- based current controller is implemented in MATLAB/Simulink to adapt to system uncertainties and enhance dynamic performance. Simulation results indicate that STSMC-NN demonstrates better power control, minimized leakage currents and Total Harmonic Distortion (THD), improving power quality and grid stability in solar energy systems.*

Keywords: Solar Power, MPPT, SMC, STSMC, Power Smoothing.

Paper ID: 331

Title: PERFORMANCE ANALYSIS OF AUTONOMOUS EMERGENCY BRAKING SYSTEM USING CAN

Authors: Jayasudha*, Lekshmi R Chandran, and Akash K

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Abstract: *The main causes of road accidents are human error, delayed brakes, and inattention, which presents a severe risk. For this reason, collision avoidance systems were created along with collision prediction. One of the main features of a collision avoidance system is the autonomous emergency braking (AEB) system. The system automatically provides drivers with the necessary braking during emergencies and warns drivers to anticipate a collision. This report tests the real-time behavior of an AEB system integrated with Controller Area Network (CAN). This study has been done using MATLAB simulation according to the Euro NCAP test process to look for the effect of communication delays on the system's performance. This study presents the effectiveness and reliability of CAN protocols in several scenarios with an AEB system to enhance responsiveness and security. The majority of previous work did not illustrate a real-world situation. The suggested model incorporates a communication protocol and shows a real-world scenario.*

Keywords: AEB, CAN, Euro NCAP test protocols.

Paper ID: 339

Title: DESIGN OF GAIN SCHEDULED REINFORCEMENT LEARNING ALGORITHM BASED CONTROLLER FOR CONICAL TANK

Authors: N Santhosh, Sathish Kumar S and S Kangalakshmi*

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Abstract: *This paper proposes a novel gain scheduled Reinforcement learning algorithm based controller for non-linear process like conical tank. The conical tank is modeled using the first principles. The total height of the tank is broken down into multiple subranges and for each of the subranges, an agent is trained. Agent training yielded policy is backed by deep neural networks, which are effective for complex and dynamic environments. The Deep Deterministic Policy Gradient (DDPG) based agents, an off-policy based actor-critic methodology of reinforcement learning is superior for continuous action spaces. The gain scheduler selects the agent based on its set point during the simulated model deployment phase. The efficacy of the proposed controller is compared with PI controller and their metrics are calculated.*

Keywords: Gain Scheduling, Reinforcement Learning, Conical Tank, DDPG, Non-linear process

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Paper ID: 347

Title: IMPLICATION OF CNN MODEL IN IOT BASED HETEROGENEOUS DISTRIBUTED SYSTEM

Authors: Vivek Veeraiah, Ritu Dahiya, Gitanjali Shrivastava*, C.Naga Ganesh, Ankur Gupta and Jay Kumar Pandey

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Abstract: *The growing complexity and quantity of data produced by IoT devices create major problems for accuracy, scalability, and real-time decision-making in distributed systems. In many settings, conventional approaches often lack consistent and scalable performance. The present study improves prediction accuracy and system efficiency by solving this issue using CNN methods paired with IoT-based heterogeneous distributed systems. Using CNNs trained on historical data, the suggested approach carries out important activities including predictive maintenance, virus identification, garbage sorting, and fraud detection. A layered middleware architecture is shown to isolate procedural calls and allow smooth operation across various hardware and software components. MATLAB was used to run experimental simulations comparing filtered and unfiltered CNN methods. While the filtered CNN model performed better with an accuracy of 91.85%, the unfiltered CNN model had an overall accuracy of 86.45%. These findings show how well the suggested method allows real-time reactions in various IoT settings and improves prediction accuracy. The findings of the research indicate that combining CNN with IoT systems backed by middleware will greatly enhance scalability, interoperability, and smart automation across many application areas.*

Keywords: CNN model, IoT, Heterogeneous distributed system.

Paper ID: 350

Title: AN ACCURATE FUEL LEVEL MEASUREMENT SYSTEM UNDER SLOSHING CONDITIONS USING DATA FUSION APPROACH

Authors: Prakash Ranjan* and Sathiya S

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Abstract: *The accuracy and reliability of fuel level sensing in automotive industries is mainly dependent on level sensors where the traditional sensors such as capacitive sensors and differential pressure transmitters work well at static conditions. But, during running state of the vehicle, the acceleration induced sloshing makes the level measurement by conventional sensors inaccurate. This work aims to develop a reliable and accurate fluid level measurement system under acceleration induced sloshing conditions. The system utilises 3 ultrasonic sensors to measure the liquid level of the tank from 3 pre-determined positions. The known accelerations are induced in the tank to create slosh conditions, which is measured using accelerometers. The signals of fuel level with induced sloshes are acquired from 3 different ultrasonic sensors, and fed to Kalman filter to reduce the environmental noise. The filtered data from 3 sensors are fused to estimate the actual fuel level of the tank. The results show that the proposed system can efficiently filter the noises and estimate the fuel level with better accuracy under sloshing conditions.*

Keywords: Fuel-Level · Kalman filter · Slosh · Ultrasonic sensor.

Paper ID: 351

Title: DESIGN OF PI CONTROLLER FOR ARTIFICIALLY VENTILATED HUMAN RESPIRATORY SYSTEM USING CRAYFISH OPTIMIZATION

Authors: Gaurav Singh, Akanksha Dwivedi* and Ahmad Ali

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Abstract: *An artificially ventilated human respiratory system is designed to assist patients experiencing respiratory difficulties by ensuring stable oxygen levels. This is achieved through precise regulation of airway pressure using a medical ventilator. In pressure-controlled ventilation, airway pressure is managed via a motor-driven piston mechanism, which adjusts airflow into the lungs. Various patient-specific factors, including physical condition, oxygen requirements, and age influence the optimal tuning of the controller parameters in a respiratory ventilator system. Given these complexities, computer-assisted algorithms have emerged as a preferred solution for artificial ventilation systems, offering enhanced performance, improved efficiency, and a more intuitive control mechanism. In this paper, a PI controller is designed to optimize ventilator performance. Controller parameters are fine-tuned by minimizing the integral time absolute error using the crayfish optimization algorithm.*

Keywords: Artificial ventilation system, Crayfish optimization, Pressure control ventilator, PI controller.

Paper ID: 352

Title: PERFORMANCE ANALYSIS OF GG-NMOS FOR ESD APPLICATIONS

Authors: Prabhat Kumar*, Dheeraj Kumar Sinha, and Sanjay Kumar

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Abstract: *This paper investigates the physical details of n-FET-based electrostatic discharge (ESD) protection devices, such as Gate-Grounded NMOS (GG-NMOS), under ultra-high current pulsing or ESD conditions to comprehend their performance and its current-voltage (I-V) characteristics. The two-dimensional Technology Computer Aided Design (TCAD) device simulation results, which offer a predictive analytical model of snapback I-V characteristics, have made it possible to clearly understand the charge movement in device structure regions and their mutual electrostatic coupling under such circumstances. Additionally, in order to define terminal features, the process of parasitic bipolar activity in the GG-NMOS structure has been explored. A systematic parametric analysis is used to analyse the impact of Snapback devices and its I-V characteristics for ESD design windows. Additionally, 180nm GG-NMOS transient device simulations are carried out at various Drain Contact to Gate Spacing (DCGS) ranges 0nm to 20nm with a predefined drain current pulse of Human body model pulses (1A/ns).*

Keywords: Electrostatic discharge (ESD), ESD protection, GG-NMOS, Integrated Circuits, Snapback Devices.

Paper ID: 353

Title: AN AUTOMATIC AIR QUALITY MONITORING AND PURIFYING SYSTEM

Authors: Kodakkal Amritha*, Bodige Meghana, Putta Charishma, Lande Niharika and NerellaChandana

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Abstract: *The paper is based on the project 'An Automatic Air Quality Monitoring and Purifying System' which mainly deals with monitoring and reporting some of the significant environmental parameters like temperature, humidity, heat index and gas concentrations. The network of sensors is utilized for the measurement of pollutants such as CO_2 and volatile organic compounds (VOCs). The main goal of the work is to design a reliable sensor network that can acquire accurate data efficiently from diverse environments. The system uses Thing-Speak for visualization and analysis. This system is integrated with air purifiers, set automatically to activate when the gas levels exceed 0.5V(greater than 100ppm). The hardware results and the real-time graphs obtained from Thing-speak are displayed in the paper. Thus, the project promotes the awareness of air quality in the surroundings and gives an option of creating healthier living spaces by improving the air quality through automated monitoring and purification.*

Keywords: IoT, Air Quality, Air Purifiers, Automated monitoring.

Paper ID: 370

Title: PERFORMANCE COMPARISION OF FIR FILTER USING DIFFERENT ADDERS IN 90nm CADENCE CMOS TECHNOLOGY

Authors: Divya Beebireddy, Srihithapatel Mamidi*, Karthikeya Irukula, Lahari Karimikonda

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Abstract: *FIR filters are commonly employed for various wireless frequency responses, typically constructed using compressors, external structures, and a sequence of delays to generate their outputs. These filters are essential for implementing digital frequency responses, utilizing multipliers, adders, and delays in their design. Adders are fundamental components in FIR filter implementations, with two main types: serial adders perform bit-by-bit addition, prioritizing cost-effectiveness if speed is not critical, while parallel adders execute additions simultaneously across bits, offering faster performance. Finite Impulse Response (FIR) filters are widely employed in digital signal processing applications due to their inherent stability and linear phase characteristics. However, their implementation efficiency significantly depends on the choice of arithmetic units, particularly adders, which directly influence speed, area, and power consumption. This project presents a performance comparison of FIR filters implemented using three different types of adders: Ripple Carry Adder (RCA), Carry Select Adder (CSLA), and Carry Skip Adder (CSKA), synthesized and simulated using 90nm Cadence CMOS technology. The primary objective is to analyse and evaluate the trade-offs in terms of delay, area, and power consumption for each adder configuration within the FIR filter architecture. Simulation results demonstrate that while RCA offers a minimal area footprint, it suffers from higher delay compared to CSLA and CSKA. CSLA exhibits superior speed performance at the cost of increased area, whereas CSKA provides a balanced trade-off between delay and area. This comparative analysis aids in selecting the most suitable adder type for FIR filter implementations in low-power and high-speed digital signal processing applications.*

Keywords: FIR filter, carry select adder, Ripple carry adder,carry save adder, carry skip adder, Vedic multiplier.

Paper ID: 378

Title: NON-INVASIVE EEG-DRIVEN AUTONOMOUS WHEELCHAIR

Authors: R. Rajalakshmi, K. Kishor Kumar*, S. Prajesh, S. Bhoomika, and C.Vikaash

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Abstract: *This paper proposes a affordable Brain-Computer Interface (BCI), non-invasive autonomous wheelchair system that is specifically designed for people who possess severe motor disabilities. Its underlying principle lies in the usage of signals alongside fault-free eye blinks, and facial movements for navigation. Everything is done on the basis of integrating an EXG Pill sensor and using a Raspberry Pi 5 for signal processing along with an ESP32 microcontroller for motor control. Remarkable changes can be seen from a low cost, adaptability, and safety aspects. The tests of the prototype showed a high level of precision during the driving and a quick reaction time to the input that suggests that it may be employed as a tool to convey mobility and independence to the users with disabilities.*

Keywords: Brain-Computer Interface (BCI), Electroencephalogram (EEG), Autonomous Wheelchair, Raspberry Pi 5, Edge Computing

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Paper ID: 386

Title: VOLTAGE MODE ACTIVE FILTERS USING CURRENT FEEDBACK OPERATIONAL AMPLIFIERS

Authors: Arunav Dhaka*, Aaryaveer Bhardwaj, Aditya Jain, Ram Bhagat

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Abstract: *This paper introduces an innovative approach to Voltage Mode active filter design utilizing Current Feedback Operational Amplifiers (CFOAs), with a focus on the AD844 integrated circuit. Our research presents a flexible configuration employing two CFOAs and two/three capacitors and two/three resistors to realize various filters types, including low-pass, high-pass, and band-pass filters. To validate our approach, we present a comprehensive analysis including theoretical calculations, PSPICE simulations, and real-world experiments using AD844 ICs. The results demonstrate a strong correlation between simulated, theoretical and measured responses, underscoring practical viability of design.*

Keywords: Second Order Active Filters, Current Feedback Operational Amplifier, AD844, High Pass Filters, Low Pass Filters, Band Pass Filters.

Paper ID: 388

Title: SMART COUPLER ALIGNMENT SYSTEM USING SWERVE DRIVE FOR WIRELESS DRONE CHARGING STATION

Authors: Boopathi C S, Dominic Savio A, Aadithya Vardan, Harshwardhan Singh, Mo-hammad Adnan, Venkatesan Ramakrishnan*, Balaji C, Narayanamoorthi R

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Abstract: *As the demand for drone technology continues to grow, the need for efficient and reliable charging solutions has become increasingly crucial. Em-powering of autonomous operation, reduced maintenance, and elimination of wear-prone connector wireless charging is the preferred charging method. How-ever, the misalignment between transmitter and receiver causes reduction in power transmission efficiency. In a circular coupler, the flux density peaks at the center; even minor misalignments can cause a reduction in mutual flux, thereby decreasing power transmission. This article proposed a smart alignment system based on swerve drive and sensing coil, which ensures the exact alignment of transmitter and receiver coupler. It is also essential to prioritize charging the most drained drone's battery. An additional system is employed with an algorithm that gives the highest priority to the battery with the maximum depletion. The wireless charging system is validated with MATLAB simulation results obtained with 3.7 kW charging station which achieves 94.6% efficiency while charging.*

Keywords: Wireless power transfer, Drone charging, Swerve drive, Wireless charging, Charging station.

Paper ID: 393

Title: A NOVEL METHOD FOR USER AUTHENTICATION USING EEG SIGNALS FOR ENHANCED SECURITY

Authors: Harini Manivelan*, Ishwarya A, S. Moorthi

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Abstract: *EEG signals are generated by brain activity, a unique biometric authentication which is difficult to spoof. User Authentication with EEG signals has a safe and hygiene approach to security, substituting the traditional physical contact methods like fingerprints. Unlike fingerprints, EEG signals can be captured without physical contact. The EEG system developed in this work establishes authentication of a user to unlock electronic device by reading the unique brain wave pattern. Here the analogy for locking and unlocking an electronic device is shown through relay-LED setup configuration. The system is also verified in real-time by extracting the brain waves pattern from the subject and these signals are given as input to Raspberry Pi which drives the external circuit of relay-LED configuration to show the result. This research paves the way for enhanced security and user authentication using EEG signals implemented on Raspberry Pi.*

Keywords: EEG, Raspberry Pi, user authentication, security, relay

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Paper ID: 395

Title: FPGA IMPLEMENTATION OF 32-BIT ARITHMETIC LOGIC UNIT USING REVERSIBLE LOGIC GATES

Authors: Kommalapati Rajesh*, Uppala Srinivasa Rao, Surendrakumar Painam, P. Janardhan Saikumar, T. Venkata Krishnamoorthy, Suneel Miriyala

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Abstract: *An essential part of the Central Processing Unit (CPU), the Arithmetic Logic Unit (ALU) oversees carrying out arithmetic and logical operations. Irreversible logic, the foundation of traditional ALU architectures, greatly increases power consumption, a key issue in contemporary computing. With its uses in cutting-edge domains like quantum computing, optical computing, and nanotechnology, reversible logic presents a viable route toward processing that uses less energy. To overcome the power consumption issues with traditional ALUs, this work investigates the design of a 32-bit ALU using Reversible Logic Gates (RLGs). The main objective is to develop a low-power ALU that maximizes a few important parameters, including computational performance, power dissipation, circuit complexity, and the production of garbage outputs, which are extra outputs that are included in some reversible logic designs. Reversible logic's potential in real-world applications depends on minimizing these factors. The intention of the suggested 32-bit ALU design is to advance the field of developing digital systems that use less energy. The design will be simulated and synthesized using Xilinx Vivado software and implemented on the Nexys 4 DDR platform, which utilizes the Artix FPGA. This implementation will allow for practical evaluation of the proposed ALU's performance and its capability for low-power operation.*

Keywords: Arithmetic Logic Unit, Reversible Logic Gates, Low Power Consumption, Computational Performance, Digital Systems, Energy Efficiency.

Paper ID: 397

Title: A RENEWABLE ENERGY INTEGRATED IOT-BASED BATTERY MANAGEMENT SYSTEM FOR ELECTRIC MOBILITY

Authors: P.Sivaprakasam, K.Suresh, Balaji C, Monica P, Shrisha kabdwal, Dominic Savio A, Venugopal R*

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Abstract: *Electric vehicle (EV) battery performance is a key focus in modern transportation systems, and the integration of Internet of Things (IoT) technologies offers new opportunities for real-time monitoring and management. One of the persistent challenges in battery systems is the gradual loss of energy capacity, which adversely affects efficiency and operational lifespan. This study introduces an IoT-enabled battery monitoring solution built around the ESP32 microcontroller, complemented by an LCD interface and voltage sensing capabilities. The system is capable of tracking essential battery met-rics in real time, facilitating early detection of performance degradation. Such monitoring allows for prompt notifications and supports proactive maintenance strategies. Given that battery safety concerns continue to limit broader EV adoption and range expansion, this approach aims to enhance both operational reliability and user confidence. Experimental analysis con-firms that the proposed system successfully identifies early signs of battery deterioration, making it a viable tool for improving EV safety and dependability.*

Keywords: Electric vehicle, IoT, battery monitoring system, International Telecommunications Union (ITU).

Paper ID: 410

Title: ARTIFICIAL INTELLIGENCE FOR TRANSFORMER FAULT DETECTION: A COMPREHENSIVE REVIEW OF DISSOLVED GAS ANALYSIS TECHNIQUES

Authors: Swati R. Ghutke* and Nitin K. Dhote

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Abstract: *DGA is an acronym for Dissolved Gas Analysis that can very effectively be used as diagnostic technique for fault detection in transformers where gases are measured and interpreted as to be dissolved in insulation oil. Some traditional DGA methods such as Duval Triangle, IEC Ratios and Key Gas Method are available, but they mostly fall short of flexibility and precision in fault diagnosis under complex scenarios. A whole new breed of effective artificial intelligence data-driven solutions has been– and are continuously being– realized to allow efficient fault classifications with very high accuracy and real-time fault detection abilities, including predictive maintenance functionality. In this study, investigations on the application of artificial intelligence approaches, such as Machine Learning (ML), Deep Learning (DL), and hybrid AI frameworks for transformer fault diagnosis using DGA data, will be presented. The study compares such methods with those widely accepted and used against traditional methods in terms of performance. There are thus inherent threats, such as data dependency, room for model interpretation, and real-time emergence. Some results seen indicate that neural networks and support vector machines are neural networks trained by which fault detection can achieve at least 20% performance improvement over any typical methods. The study further concentrated on how the possible amalgamation of AI with the Internet of Things (IoT) may allow a relatively quick transformation into real-time health monitoring of transformers. Future direction would be improving generalization to AI model improvement, its transparency, and intelligent grid systems designed for fault-tolerant transformer networks.*

Keywords: Artificial Intelligence, Deep Learning, Dissolved Gas Analysis, Fault Detection, Machine Learning, Transformer Monitoring.

Paper ID: 420

Title: LOAD FREQUENCY CONTROL OF A SINGLE AREA DISTRIBUTED GENERATION SYSTEM CONSIDERING CS OPTIMIZED CASCADE CONTROLLER

Authors: Ainala Prem Kumar, Kotani Anusha, Ranbir Sah, K N T Mohan Reddy, Nekanti Rajesh, Adireddy Ramesh, and Naladi Ram Babu*

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Abstract: *This article presents a thorough examination of load frequency control (LFC) in a single area power system. The study utilizes a crow search (CS) optimized tilt integral fractional order PID (TI-FOPID) cascade controller. CS technique is employed to optimize TI-FOPID controller and exhibits its superiority over TID and FOPID controllers. Studies suggest that TI-FOPID controller provides better dynamics in LFC system with non-linearities such as GRC and GDB. Further, investigations reveal that with distributed generation integration, system has shown significant improvement in terms of peak over shoot and undershoot. Furthermore, sensitivity analysis with inertia variations reveal that proposed controller is robust.*

Keywords: Crow search, TI-FOPID controller, non-linearities, distributed generation and sensitivity analysis.

Paper ID: 426

Title: DESIGN OF MULTI-STAGE FRACTIONAL-ORDER CONTROLLER FOR LOAD FREQUENCY CONTROL OF ISLANDED MULTI-MICROGRID SYSTEM

Authors: Manish Kumar*, Deep Shekhar Acharya, and Pankaj Mishra

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Abstract: *The implementation of renewable energy sources (RES) within microgrids enhances sustainability but introduces challenges in frequency stability, especially in islanded multi-microgrid (MMG) systems. This research proposes a novel multi-level fractional-order proportional-derivative (FOPD(1+PI)) regulator for load frequency stabilization (LFC), optimized using the Gray Wolf Optimizer (GWO). A performance comparison with traditional PID and fractional-order PID (FOPID) controllers, optimally adjusted using Swarm Intelligence Optimization (PSO) and Evolutionary Algorithm (GA), demonstrates that the proposed controller achieves superior dynamic performance. Results show that the FOPD(1+PI) controller reduces settling time by 12% and 20% compared to FOPID and IOPID controllers, respectively, while also minimizing frequency deviations. These findings highlight the effectiveness of advanced fractional order control methods for improving the stability and reliability of MMGs under fluctuating load conditions.*

Keywords: multi-microgrid, load frequency control, optimization algorithm, FOPD(1+PI) controller

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Paper ID: 455

Title: HAZY IMAGE QUALITY ENHANCEMENT USING DEEP NEURAL NETWORKS: A COMPARISON BETWEEN TRADITIONAL METHODS AND DEEP LEARNING APPROACHES

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Abstract: *This study explores the enhancement of hazy image quality by comparing traditional and deep learning-based dehazing techniques. Three conventional methods (DCP, CAP, CLAHE) and three deep learning models (ResNet50, ResNet101, VGG16) were evaluated under varying haze conditions. Performance was assessed using PSNR, SSIM, MSE, Blur metrics, model size, and processing time. The ResNet50 model attained the highest PSNR 14.21 dB and SSIM 0.78, while the VGG16 model attained a balance between image quality and efficiency. Among the traditional models, the DCP model obtained the highest PSNR at 12.36 dB, while the CLAHE model obtained the highest SSIM 0.74 and lowest mean square error (MSE) 0.023. Traditional models remarkably surpassed deep learning models in the processing time, the CAP process time takes only 0.017 seconds per image, compared to 0.52 seconds for the ResNet101 model. Also, deep learning models offer superior visual restoration, traditional models offer faster results with acceptable quality, making them more suitable for immediate applications. The study focuses on trade-offs to facilitate model selection based on application requirements.*

Keywords: Image enhancement, dehazing, deep learning, ResNet, VGG16, PSNR, SSIM, CLAHE, DCP, CAP.

Paper ID: 472

Title: A COMPARATIVE ANALYSIS ON NANOWIRE FET BASED DEVICES FOR IMPROVED PERFORMANCE.

Authors: Neha Patel*, Ashish Raman, Sarbdeep Singh

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Abstract: *As semiconductor devices keep on with shrink in size, conventional MOSFETs face more challenges due to short-channel effects (SCEs), which decay their performance. Gate All-Around (GAA) nanowire FETs have come out as a hopeful alternative to FinFETs, offering superior electrostatic control, improved subthreshold characteristics, and reduced leakage currents. This study explores several nanowire FET architectures with major improvements such as dual material gate engineering, gate-stack and doping-less approaches that is charge plasma. The aim of promotion of nanowire FETs for future ultra-scaled applications is offered through a review of findings from recent simulation and experimental experiments.*

Keywords: Cylindrical Charge plasma based nanowire FET, Charge plasma FET with gate staking (HfO2, TiO2) and double material gate.

Paper ID: 473

Title: THE INTERACTION OF CHARGED PARTICLES NEAR THE CATHODE SURFACE AND ITS EFFECT ON THE VOLTAGE DROP AND IONIZATION ENERGY IN THE PLASMA ARC

Authors: Hasan Abedulhadi Kadhim* and Rafid Abbas Ali

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Abstract: *Theoretical analyses and simulation results are shown to investigate the effect of mercury-thallium (Hg-Th) mixture on the arc plasma. This is achieved by doping mercury (Hg) with different concentrations (0.05, 0.1, 0.5) mol of thallium (Tl). The effects of applied voltages at 20 V and 50 V, temperature, and different thallium concentrations (0.05, 0.1, 0.5) mol are investigated. The Near-Cathode Plasma Layer (NCPL) software is used in this study to measure the Schottky correction and ionization potential. The results are significantly affected by the concentration of plasma parameters, especially when doping mercury (Hg) with thallium (Tl) at the highest concentration (0.5 mol), as well as by the effect of the potential difference. It is currently observed that the Schottky correction (ΔA) increases with increasing cathode surface temperature (Tw) from 2000 K to 3500 K, while it decreases linearly from 3500 K to 5000 K for all concentrations, because the electrons lose a large portion of their energy through collisions and ionization, leading to a saturation state. It is also observed that with increasing applied voltage, especially at 50 V, the Schottky correction (ΔA) increases due to the accelerating electric field, which leads to an increase in the energy of the charged particles and, consequently, an increase in collisions between the charged particles.*

Keywords: Arc discharge, Temperature of cathode surface, Schottky correction, Ionization voltage.

Paper ID: 474

Title: THE IMPACT OF PRE-PROCESSING TECHNIQUES ON FACIAL RECOGNITION PERFORMANCE USING CUSTOMIZED CNN ARCHITECTURE

Authors: Ali H. H. Al-Amili*, Fatin E. M. Al-Obaidi and Ali A. D. Al-Zuky

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Abstract: *Facial recognition (FR) as an artificial intelligence technology involves the processing and analysis of facial images via computer algorithms to identify and authenticate individual identities. FR technology suffer from a series of challenges due to differences in lighting, pose, and data quality. This research systematically investigates the impact of pre-processing techniques such as data augmentation (DA), face cropping, and data balancing on the performance of FR recognition systems using a customized convolutional neural network (CNN) model. This research investigates the influence of various preprocessing strategies on recognition accuracy by a customized CNN architecture which was pre-tested for computational efficiency like size and speed utilizing the Labelled Faces in the Wild (LFW) dataset in addition to a locally created dataset. Experimental results show that effective preprocessing significantly improves model accuracy and performance, in which the accuracy reached up to 97.67 for the LFW dataset and 99.33 for the local dataset.*

Keywords: Face Recognition (FR), CNN, Deep Learning, AI.

Paper ID: 475

Title: THE EFFECT OF CONCENTRATION AND VOLTAGE ON THE VOLTAGE DROP IN THE PLASMA LAYERS NEAR THE CATHODE SURFACE IN ARC PLASMA

Authors: Banaz Hasheem Hassan*, Rafid Abbas Ali, Read S. Sabry

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Abstract: *In this research, the effect of temperature, voltage, and concentration changes on barrier potential (∅ b) , degree of ionization (ω) ,and the ratio of ionization length to free path length (α) were studied. The study was conducted at two voltages (U=20 v, U=40 v), and at a temperature (5000-3000) k, where different concentrations of thallium (0.005, 0.05, 0.5) mol, were used and added to pure xenon The results showed that with increasing the concentration of thallium in the mixture, as well as with increasing the temperature of the cathode surface from 3660 to 5000 K, and because the electrons pass easily without the need for an electric field, the degree of ionization increases with increasing the temperature of the cathode surface and with increasing the concentration of thallium. As for increasing the voltage, we did not observe any effect. As for the ratio of ionization length to free path length, it gradually decreases with increasing thallium concentration, and is at its highest values when using pure xenon, due to its higher ability to ionize electrons compared to the mixture. It was also observed that the value of the ratio of ionization length to free path length at 20 v is higher than at 40V.The voltage barrier increases with increasing thallium concentration, and is high at low temperatures, reaching its highest value at 3000 k, due to the effect of elastic and inelastic collisions.*

Keywords: arc discharge, Voltage barrier, Ion energy, electron energy, plasma ionization degree

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Paper ID: 476

Title: MITIGATING HOT SPOT EFFECT ON PV ARRAY USING RECONFIGURATION TECHNIQUE

Authors: Jaya Bharati* and Ruchi Agarwal

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Abstract: *The objective of the paper is to show optimally designed photovoltaic (PV) configuration performs better in contrast to non-optimal PV configuration under partial shade condition (PSC). The performance of PV systems is substantialy affected by partial shading conditions. It results power loss and hot spots issue due to mismatch current within PV arrays. The paper investigates impact of various static PV array configurations in alleviating the effects of PSC on the system performance. The non-optimized configurations i.e. series-parallel (SP), total-cross-tied (TCT), honeycomb (HC) and Sudoku puzzle based optimized configuration are considered to show the analysis. The I-V and P-V characteristic are portrayed under different partial shading conditions. The analysis considers the number of performance indexes such as power output, mismatch loss, fill factor, and efficiency. The comparative analysis in terms of performance index of all the configurations are shown using tabular form so that performance of optimized PV configuration can be highlights.*

Keywords: Partial shading condition; PSC; Solar Photovoltaic (PV), Sudoku puzzle

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Paper ID: 481

Title: A COMPREHENSIVE ANALYSIS OF MALWARE DETECTION ON ANDROID USING MACHINE LEARNING

Authors: M. Ambika, R. Saranya, M. Mahalakshmi, S. P. Santhoshkumar*, D. Vikram and T. Nalini

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Abstract: *Users are more susceptible to malware as the number of mobile devices, particularly Android smartphones, increases. Decision Trees, SVMs, RFs, and KNNs are used in our effort to address this problem and enhance Android security. The goal is a successful Android malware detection and prevention system. We'll test these machine-learning techniques' efficacy against a range of illnesses. The popularity of Android devices has raised security worries, necessitating the development of innovative technology capable of spotting emerging threats. An Android security solution might be ML algorithms that identify patterns and predict future events. In this paper used DT, KNN, RF, and SVM algorithms to create a strong malware detection system. In order to solve security issues and improve Android device security and resilience to new attacks, we rank a number of ML techniques for Android malware detection based on speed, accuracy, and efficiency. Decision trees are transparent, random forests use ensemble learning, support vector machines excel in probabilistic classification, and k-nearest neighbors (KNN) are the best at identifying Android infections based on similarity patterns, according to research. This work culminates in the use of machine learning to strengthen Android security and lays the foundation for mobile device security research.*

Keywords: android security, SVM, K-NN, ML, decision tree, random forest, Malware Detection

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Paper ID: 483

Title: ENHANCING MEDICAL IMAGING THROUGH CNN-BASED SUPER-RESOLUTION

Authors: Seshagiri Rao Sugguna* and Sumesh E.P

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Abstract: *Brain tumour diagnosis and analysis make great use of magnetic resonance imaging (MRI). Standard MRI images, however, frequently have poor spatial resolution, which makes it difficult to identify small anatomical features. Conventional interpolation-based super-resolution (SR) methods, including bicubic interpolation, blurring and artefacts result from their inability to recover high-frequency elements. Image quality has been improved by recent developments in deep learning-based SR techniques such Generative Adversarial Networks (GANs), Super-Resolution Convolutional Neural Networks (SRCNN), and UNet based architectures; nonetheless, fine anatomical features and noise suppression remain difficult. This work presents Hybrid Net, a new CNN-based super-resolution model including Dense Attention Blocks (DAbs) for improved feature extraction. HybridNet uses both channel-wise and spatial attention processes, unlike standard SR techniques, therefore boosting MRI reconstruction accuracy and lowering of artefacts. Mean Squared Error (MSE) loss with Adam optimiser is used to train the model applying brain tumour MRI data. Peak Signal-to– Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Cosine Similarity all help to assess performance. High-quality MRI reconstructions with enhanced structural preservation are produced by Hybrid Net, which shows better performance than SR-CNN, SRGAN, and UNet-based super-resolution approaches, experimental data show. The suggested method lowers noise, so sharpens tumour boundaries, and improves image quality, so enabling more accurate medical diagnosis. Comparative analysis verifies that attention-based deep learning models greatly enhance super-resolution performance, so supporting their possibilities to advance clinical application and medical imaging.*

Keywords: MR images super-resolution, super-resolution, Convolutional Neural Networks (CNN).

Paper ID: 484

Title: AUDIO AMPLIFIER WITH ACTIVE NOISE CANCELLATION IN RLS ALGORITHM ON FPGA

Authors: Manish A. Bhavsar*, Vinod D. Ghuge, Baliram N. Kendre, Ravi Ranjan Kumar, Chaitanya Kadadas, Jayaraj U Kidav

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Abstract: *This project presents the design and implementation of an audio amplifier with Active Noise Cancellation (ANC) using the Recursive Least Squares (RLS) algorithm on the Arty A7- 100T FPGA. The system captures environmental noise via a microphone and combines it with a primary audio signal from an AUX input. By processing these inputs through the RLS algorithm, an anti-noise signal is generated and output through a speaker, effectively reducing unwanted noise in real-time.*

The RLS algorithm is implemented in Verilog to leverage the FPGA's high-speed parallel processing capabilities for low- latency performance. I2C communication enables integration with external peripherals like digital microphones or audio codecs. Simulations were conducted using MATLAB Simulink, and the system achieved a 14.6 dB improvement in Signal- to-Noise Ratio (SNR) with a power consumption of 1.597 W. These results highlight its potential for low-power, real-time audio applications such as hearing aids, smart earbuds, and communication systems.

Keywords: Active Noise Cancellation, FPGA, Recursive Least Squares, Audio Signal Processing, I2C Communication, Verilog, Vivado, Matlab, Software/hardware implementation

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Paper ID: 486

Title: OPTIMIZED SOLAR ENERGY UTILIZATION FOR EV CHARGING: ANN-BASED MPPT WITH LEVENBERG-MARQUARDT ALGORITHM

Authors: Sivakumar J*, Ramesh S, Kannan K.R, Nandakumar E, Ananthi S and Dineshraj V

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Abstract: *The availability of energy is essential for the economic advancement of all growing nations. The increasing severity of environmental concerns and the rise in oil costs have led to a significant increase in the development and deployment of Electric Vehicles (EVs) and Hybrid Electric Vehicles (HEVs) in recent years. As electric vehicles gain prominence, developments in their supporting technologies, such as charging stations, have become more crucial. This article presents an Artificial Neural Network (ANN) based Maximum Power Point Tracking (MPPT) system for optimizing solar energy use in electric vehicle charging stations. The Levenberg - Marquardt method is used to train the neural network, which controls the duty cycle of the DC-DC converter for optimal power extraction. The charging station incorporates solar photovoltaic (PV) technology with an energy storage system to improve efficiency. The efficacy of the suggested approach is assessed with MATLAB/Simulink and compared with the Bayesian Regularization methodology. The findings underscore the enhanced precision of the ANN based MPPT method, demonstrating its efficacy in optimizing energy consumption for electric vehicle charging stations*

Keywords: Electric Vehicles, Artificial Neural Network Maximum Power Point Tracking, Solar Photovoltaic System, Energy Optimization.

Paper ID: 487

Title: LOAD FREQUENCY CONTROL IN FOUR-AREA INTERCONNECTED POWER SYSTEM USING FUZZY PI CONTROLLER WITH PENETRATION OF RENEWABLE ENERGIES

Authors: Praveen Kumar Mishra*, Ashwani Kumar and Ambarisha Kumar Mishra

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Abstract: *In order to ensure system stability, frequency regulation, and power conversion among interconnected areas, load frequency control, or LFC, is a cru-cial component of contemporary interconnected power systems. Because there are several control areas with different generation and load characteristics in a four-area linked power system, LFC plays an even more intricate and important function. The basic ideas, control schemes, difficulties, and developments in the area of Load frequency controller for a four-area linked power system are exam-ined in this study. The study also explores the design considerations, simulation findings, and mathematical modelling that support the successful use of LFC mechanisms.*

Keywords: Load Frequency Control (LFC), Four-area Interconnected Power System, fuzzy PI Controller, Renewable Energy intrusion, Frequency Stability, Power System Dynamics, Fuzzy Logic Control (FLC), Renewable Integration, System Response, Frequency Deviation, Control Effort, Robustness and Relia-bility.

Paper ID: 494

Title: MINIMIZING DELAY, POWER AND PDP IN TERNARY LOGIC CIRCUITS: A GNRFET-RRAM APPROACH

Authors: Shaik Ruksana*, Syed Munawwar, Shaik Javid Basha, M. V. Subramanyam, Y. Mallikarjuna Rao, N. AshokKumar

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Abstract: *This research presents a comparative analysis of ternary logic circuits built using either Graphene Nanoribbon Field-Effect Transistors (GNRFETs) or Carbon Nanotube Field-Effect Transistors (CNTFETs) both integrated with Resistive Random-Access Memory (RRAM). The study focuses on a ternary inverter and a comparator revealing significant performance advantages for the GNRFET-RRAM implementation. Specifically, the GNRFET-RRAM ternary comparator demonstrates improvements of up to 31.73% in delay, 51.00% in power consumption, and 66.37% in power-delay product (PDP) when compared to its CNTFET-RRAM equivalent. These substantial gains highlight the potential of GNRFET-RRAM technology for creating energy-efficient and high-performance ternary computing architectures.*

Keywords: GNRFET, RRAM, STI, Adder, HSPICE and VLSI.

Paper ID: 497

Title: EXPLORING THE MODULATION METHODS FOR UNLEASHING ASYMMETRIC MULTILEVEL INVERTER POTENTIAL FOR RENEWABLE ENERGY SYSTEMS

Authors: R.Seyezhai*, M.Sudhakaran

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Abstract: *This paper presents a comprehensive overview of state-of-the-art modulation techniques for Asymmetric Multilevel Inverters (AMTLI). The discussion encompasses sinusoidal Pulse Width Modulation (PWM) techniques, including phase shift PWM, level shift PWM, carrier overlapping PWM, and switching frequency optimal PWM. The primary emphasis of this research revolves around enhancing the performance of the AMTLI system, with a key objective of minimizing harmonic content in the output voltage. This is achieved through the integration of a three-phase AMTLI coupled with advanced PWM techniques. Simulation studies are conducted to assess various modulation methods, comparing their respective performance parameters. The most suitable method is then selected for implementation in the AMTLI, and the obtained results are subsequently verified for accuracy and efficacy.*

Keywords: AMTLI, Modulation strategies, SPWM, Carrier overlapping, switching frequency optimal PWM

Paper ID: 503

Title: FAULT LOCATION SCHEME FOR AC MICROGRID BASED ON MAGNITUDE OF TRANSIENT VOLTAGE

Authors: Mahesh Ahuja* and Ebha koley

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Abstract: *An intelligent fault location technique for AC microgrids is presented in this research. The technique locates faults by measuring the maximum sensor-detected magnitude of the transient voltage signal caused by the fault. The simulations demonstrate a linear link between the distance between the fault location and the maximum magnitude of the transient voltage signal observed by the sensor. Based on this relationship, sensor position data, and microgrid topology, an algorithm is developed to identify defects in AC microgrids. Simulations have been used to validate the approach, with an average localization error of less than 8 percent.*

Keywords: Fault location pinpoint microgrid, transient voltage signal, LG fault

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Paper ID: 510

Title: LMS ADAPTIVE FILTER BASED THREE-PHASE GRID INTERACTIVE SOLAR PHOTOVOLTAIC SYSTEM

Authors: Dheeraj Kumar*, Sanjeev Singh and Shailendra Kumar

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Abstract: *This paper presents a controlled adaptive filtering technique Least Mean Square (LMS) for three phase two stage grid interactive solar photovoltaic (SPV) systems. Incremental conductance (INC) method is used for maximum power point (MPP) from photovoltaic (PV) array through a boost converter. The gate pulse of voltage source converter (VSC) is generated using LMS control algorithm, with a benefit of reduced complexity and improved dynamic behavior of system. The complete system is modeled and its performance is simulated on MATLAB-SIMULINK software. Obtained results are compared with results of synchronous reference frame theory (SRFT) based controller to demonstrate the superiority of LMS algorithm.*

Keywords*:* LMS, VSC, Grid, Solar PV array and Loads

Paper ID: 513

Title: IMPLICATIONS OF HEIGHT, SPACING, AND SHADING FOR AGRIVOLTAICS SYSTEM

Authors: Shanza Neda Hussain, Carla Hardaker, Saptarshi Das, Aritra Ghosh*

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Abstract: *Agrivoltaics, a symbiotic relationship between solar electricity production and agriculture, presents a promising pathway towards a green economy. This paper examines the critical role of photovoltaic (PV) module height in optimising agrivoltaic systems, focusing on its impact on microclimate, crop yield, and electricity generation. Increased module height enhances surface heat transfer, boosting PV production, while also influencing the photosynthetically active radiation (PAR) reaching crops. Varying heights affect shading patterns, soiling, and ultimately, the suitability for different crops, particularly shade-tolerant varieties, while spacing between the modules is another key element in determining the PAR and the air circulation under PV. Analysing various global installations, this review highlights the trade-offs between height, spacing, and their effects on both agricultural output and energy yields suggesting optimal height and spacing are crucial for maximising the synergistic benefits of agrivoltaics.*

Keywords: Agrivoltaics, Height, Spacing and shading, Shade-loving crops, photosynthetically active radiation.

Paper ID: 515

Title: IMPACT OF STRUCTURAL AND MATERIAL VARIATIONS ON THE PERFORMANCE OF JUNCTIONLESS NANOWIRE FIELD EFFECT TRANSISTOR: A COMPREHENSIVE ANALYSIS

Authors: Dipjyoti Borah*, Ashish Raman, Pawan Kumar Verma and Sarabdeep Singh

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Abstract: *Gate All Around Junctionless Nanowire Field-Effect Transistors (GAA- JLFETs) have gained significant attention as promising candidates for next-generation nanoscale electronic devices due to their simplified manufacturing process and en- hanced electrostatic control. However, mitigating leakage current remains a major chal- lenge, as it directly impacts device performance and power efficiency, among various architectures, Double Gate (DG) and Gate Stack Nanowire Junctionless FETs have demonstrated significant potential for enhancing device performance while addressing short-channel effects (SCEs). This study examines how variations in nanowire diameter (10 nm, 15 nm, and 20 nm) and different spacer materials (SiO_2, HfO_2, and Si_3N_4) in- fluence key performance metrics of Gate-All-Around Junctionless FETs (GAA- JLFETs) with channel length of 10nm. The analysis focuses on critical parametrs, in- cluding subthreshold swing (S/S), drive current (I_{on}), leakage current (Ioff), on/off cur- rent ratio (Ion/Ioff), providing insights into device optimization. This paper aims to help improve the design of transistors that are both energy-efficient and high-performing, making them suitable for future nanoelectronics.*

Keywords: Junctionless, GAANFET, gate stack, dual metal.

Paper ID: 516

Title: HIGH EFFICIENT MULTI-PORT CONVERTER BASED WIRELESS EV CHARGING SYSTEM POWERED BY RENEWABLE ENERGY SOURCES

Authors: Venugopal R*, Balaji C, Harisudha Kuresan, Suba S, Saravana Kannan Thangavelu, Dominic Savio A

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Abstract: *Electric vehicles (EVs) are becoming a crucial component in the transition toward sustainable energy solutions. However, the increasing adop-tion of EVs, coupled with their limited onboard battery capacity, has heightened the demand for widespread charging infrastructure. Fast charging stations, par-ticularly Extreme Fast Charging (EFC), can significantly strain the electrical grid, leading to issues such as peak-hour overload, power fluctuations, and volt-age instability. To address these challenges, integrating renewable energy sources with Hybrid Energy Storage Systems (HESS) has emerged as a viable solution. This paper proposes a high-efficiency multi-port converter-based wire-less EV charging system powered by photovoltaic (PV) and HESS. The multi-port converter facilitates flexible energy management, enabling power transfer between PV, HESS, with the EV charging station. A multiport converter opti-mizes energy flow, ensuring stable operation under varying generation and load conditions. The proposed configuration enhances system efficiency, reduces dependency on the grid, and minimizes overall charging costs. Experimental results validate the effectiveness of the system, demonstrating improved energy utilization, stable power delivery, and reliable wireless charging performance for EVs.*

Keywords: Wireless EV Charging, Multi-Port Converter, Renewable Energy, Hybrid Energy Storage System, Power Management, High Efficiency, Bidirec-tional Converter.

Paper ID: 529

Title: STRUCTURAL ANALYSIS OF GATE-ALL-AROUND NANOWIRE TFET

Authors: Ankit Kumar*, Ashish Raman, Nitesh Kashyap, and Prabhat Singh

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Abstract: *Nowadays, the most emphasized device is the Tunnel Field Effect Transistors (TFET).Main focus is on materials and arrangements that improve the device performance. To enhance TFET properties, the study mainly examines the effects of heterostructure material (SiGe) and use of a Triple Metal gate of different work functions.GaAs/InN/SiGe these are the heterostructure materials and the use of nanowire topolo- gies offer better carrier transport and electrostatic control.By employing three unique metals with varying work functions throughout the channel region the ambipolar conduction is suppressed and the TFET Ion cur- rent is enhanced.[2]By using SiO2 as a spacer which reduces Ioff signifi- cantly.[9]High K dielectric materials, such as HfO2 , improve the device's performance, lower the subthreshold slope,have low leakage current and reduced short channel effects.Because of these developments, TFET is now desirable for low-power and high-performance applications.*

Keywords: Gate-All-Around TFET· Nanowire · TFET.

Paper ID: 530

Title: A HIGH-GAIN INTERLEAVED BOOST CONVERTER FOR FUEL CELL VEHICLES WITH PI-BASED VOLTAGE REGULATION

Authors: Anusha Gaddala, Ram Ragotham Deshmukh, A.V.V Sudhakar

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Abstract: *Electric vehicles and renewable power systems utilize fuel cells as energy source though these systems need efficient power conversion technologies because of their low output voltage characteristics. A closed-loop interleaved boost converter receives detailed attention in this paper for its application to fuel cells because it produces a high and stable output voltage. The feedback system which is controlled by proportional-integral action maintains output voltage stability and provides enhanced transient performance. The designed converter operates within the MATLAB/Simulink software by accepting a 40V fuel cell stack input and producing a stable linear 200V output. Elaborated topological organization promotes higher operational efficiency because it decreases strain on current and suppresses ripple. Parallel simulation tests confirm that the implementation of a PI controller yields better voltage stability than open-loop controls at conventional levels. The research confirms how the converter design meets requirements for automotive fuel cell automobiles and renewable power applications which leads to better high-gain power conversion system development.*

Keywords: Interleaved Boost Converter, Fuel Cell Electric Vehicle (FCEV), current ripple, Proton Exchange Membrane Fuel Cell (PEMFC), High-Gain DC-DC Converter

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Paper ID: 532

Title: HRV-BASED STRESS DETECTION

Authors: Angel Jain, Ifra Ali, Anupam Kumar*, and Ashish Sharma

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Abstract: *Stress significantly impacts mental and physical health, and recent research focuses on using heart rate variability (HRV) metrics for early stress detection. HRV measures the variations in the time intervals between heartbeats and has been identified as an effective indicator of stress, showing an inverse relationship with stress levels. [1] This relationship is attributed to the HRV connection with the autonomic nervous system (ANS), reflecting the balance between its sympathetic branches (stress activating) and parasympathetic (stress reducing). [2] The research emphasizes the potential of HRV as a non-invasive objective biomarker to identify stress patterns, highlighting its importance not only in diagnostics but also in the development of real-time stress monitoring systems.*

Keywords: HRV, supervised machine learning, datasets, Random Forest classifier, regression algorithm.

Paper ID: 538

Title: AN INTELLIGENT GRID MONITORING SYSTEM UTILIZING MACHINE LEARNING FOR HIGH IMPEDANCE FAULT DETECTION

Authors: Lekshmi R Chandran*, Devika Subhagan, Geethika Rajeev, Aparna Raveendran, Ilango K and Manjula G Nair

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Abstract: *High Impedance Faults (HIFs) provide an important threat for power system safety because of their transient character and low fault currents. Conventional protection schemes often fail to detect HIFs, leading to equipment damage, safety hazards, and power outages. The paper proposes an intelligent grid monitoring system that uses machine learning approaches to improve the accuracy and real-time detection of HIFs and understand its characteristics, behavior and impact on the power system. Through the application of decision trees, ensemble methods, and artificial neural networks, our system demonstrates improved accuracy in fault detection, notably with ensemble models achieving up to 92.60% accuracy.*

Keywords: High Impedance Fault, Machine Learning, Fault Detection, HIF Detection.

Paper ID: 539

Title: YOLOV8-BASED ELECTRIC VEHICLE CHARGING PORT DETECTION FOR SMART INFRASTRUCTURE

Authors: V.C. Mahaadevan, R. Narayanamoorthi*, Pridhav Krishna Anoop

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Abstract: *The adoption of electric vehicles is expanding at an increasingly rapid rate. This requires the efficient and precise identification of charging ports in order to implement autonomous charging solutions. This paper introduces a YOLOv8s-powered edge detection model that can identify charging terminals in RGB images. The model was trained on a unified dataset that was processed using a custom pipeline to guarantee improved feature extraction and generalization. The standard object detection metrics were employed to validate the model, and the results are presented below. The model exhibited a robust balance between the precision and recall parameters. The outcomes underscore the efficacy of our methodology in surmounting a variety of challenging situations, such as occlusions, harsh lighting conditions, distorted perspectives etc. This research contributes to the development of autonomous electric vehicle (EV) charging solutions by reducing the necessity for human interaction and facilitating the seamless integration of grids and vehicles.*

Keywords: YOLOv8, Edge detection, EV charging, Machine Learning

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Paper ID: 540

Title: Variable structural control based on ADRC for unstable non-minimum phase processes

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Abstract: The unstable process with non-minimum phase behavior is difficult to control using the conventional active disturbance rejection control (ADRC) technique. In this research, the concept of ADRC strategy is combined with the variable structure control (VSC) to achieve the improved performance and robustness for the above said class of processes. The particle swarm optimization (PSO) is used to tune the controller's settings. The performance of the suggested method is found improved compared to the recently reported approach under both the nominal and perturbed conditions.

Keywords: Variable structure control, Particle swarm optimization, Active disturbance rejection control, Unstable non-minimum phase process

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Paper ID: 542

Title: ANALYSIS OF MUTUAL INDUCTANCE OF CIRCULAR ROGOWSKI COIL USING FULL CURRENT THEORY

Authors: Venkatamaheshbabu Lella* and Kandasamy Chandrasekaran

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Abstract: *The Rogowski coil (RC) is a primitive and affordable device that measure high frequency currents, such as surge currents. It measures partial discharge without the need for complex electronic apparatus. Mutual inductance is one of the parameters that will affect the induced voltage in the RC. So, measurement of mutual inductance is crucial. This paper presents an effective method for measuring mutual inductance. In this study the mutual inductance between the primary conductor and the RC is analyzed using Ampere's law and Full current theory. The primary conductor RC is assumed to straight and placed at the center of RC. The mutual inductance is evaluated for effective inner radius 'a' of RC from the primary conductor to the coil ranging from 0.1m to 0.4m, by keeping the outer radius of the RC constant at 0.5m. The primary current frequency is varied from 1kHz to 1GHz. The results show that the mutual inductance due to Ampere's law 'M_a' and mutual inductance due to Full current theory 'M_f' is same at every inner radius(a) up to 100KHz frequency and mutual inductance 'M_f' is dominating for frequency ranging from 1MHz to 100MHz at every inner radius (a) respectively under this study. A comprehensive evaluation of mutual inductance 'M' is conducted for harmonic currents with frequencies ranging from 1KHz to 1GHz. The error between 'M_a' and 'M_f' is calculated and reported. The results reveal that the %error is zero, regardless of peak current, for frequencies up to 100KHz. However, the % error increases with rising frequencies from 1MHz to 1GHz at every inner radius (a) in case of harmonic currents and lightning channel base currents (CBC) and notices that, the maximum error is 235% on M under study.*

Keywords: Mutual Inductance, Ampere's law, Full current theory, Harmonic current, Rogowski coil, Lightning channel base currents.

Paper ID: 543

Title: IMPROVING PV MODULE EFFICIENCY: AN IOT-DRIVEN DUST CLEANING SYSTEM

Authors: Avnish Jha, Harshit Tiwari* and Saurabh Kumar Rajput

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Abstract: *This study introduces an IoT-based autonomous cleaning system for photovoltaic (PV) modules to alleviate efficiency loss through dust accumulation. The system combines real-time sensing with DHT11, LDR, voltage, current, and dust sensors, and ESP32 microcontroller-based decision-making and cleaning triggering. MQTT-based data transmission facilitates remote monitoring, scheduling cleaning times to minimize maintenance expenses. Experimental findings show that there is a 4.16% loss of efficiency caused by dust accumulation, with an overall power saving of 15.06 W after cleaning. The system proposed here provides better PV performance, real-time monitoring, and low-cost maintenance, which makes the system appropriate for efficient solar energy utilization in harsh environments. The research emphasizes the synergy between renewable energy and IoT for ensuring efficient operation over the long term.*

Keywords: IoT-driven automation, Photovoltaic (PV) module cleaning, Real-time monitoring, ESP32 microcontroller, MQTT communication, Efficiency improvement, Sustainable solar energy

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Paper ID: 545

Title: A NEW SOLAR-POWERED EV CHARGING WITH GRID INTEGRATION AND BIDIRECTIONAL POWER TRANSFER

Authors: Ram Kumar*, Sudhansu Kumar, Gaurav Kumar Jha, Puja Kumari, Mani Kumar, Suddu Kumar, and Mithlesh Kumar

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Abstract: The growing demand for electric vehicle (EV) charging infrastructure has led to integrating renewable energy sources, such as solar photovoltaics (PV) and the conventional AC grid. This paper presents the development of an electric vehicle charging circuit integrating solar power and AC grid power to ensure robust charging capability. The design is solar photovoltaic (PV) energy-centered and employs a perturb and observe Maximum Power Point Tracking (MPPT) algorithm to maximize solar energy harvesting efficiency while allowing switching to grid electricity during low solar generation periods. The harvested power is stored in a 100Ah battery, converted to a 96V DC constant output by a boost converter, and then regulated by a buck converter to meet the charging requirements of the EV battery. The system automatically switches to the grid power during low solar generation periods for seamless charging continuation. The charging scheme also allows bidirectional power flow and thus vehicle-to-grid (V2G) and grid-to-vehicle (G2V) energy transfers to improve efficiency. The system's efficacy has been rigorously simulated and tested using MATLAB/Simulink, confirming its ability to reduce grid dependence and encourage cleaner, more responsible EV charging options.

Keywords: Solar PV Charging, Electric Vehicle (EV) Charging, Maximum Power Point Tracking (MPPT)

Paper ID: 553

Title: CFOA-BASED SECOND ORDER VOLTAGE MODE ACTIVE FILTER CONFIGURATION

Authors: Piyush Kumar*, Parthiv Das,Yashwardhan Singh Rajora, Ram Bhagat and Mini Sreejeth

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Abstract: *This research paper proposes an active filter configuration based on Current Feedback Operational Amplifiers (CFOAs). The circuit realizes second-order low-pass filter (LPF), high-pass filter (HPF), and band-pass filter (BPF) employing two CFOAs, three or four resistors, and with two capacitors that are either grounded or virtually grounded. Theoretical analysis of the circuit has been confirmed by OrCAD PSpice simulations proving its correctness and performance. Furthermore, experimental validation with readily available AD844-type CFOAs demonstrates strong agreement with the theoretical frequency response derived from MATLAB simulations. To evaluate the circuit's practical performance and tolerance to component variations, a comprehensive Monte Carlo analysis was performed using PSpice simulations.*

Keywords: Current Feedback Operational Amplifier, Analog Signal Processing, Analog Filters

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Paper ID: 563

Title: LOCALIZATION OF UNDERWATER MOVING OBJECT USING DISTRIBUTED NEIGHBOURHOOD NETWORK FOR UNDERWATER INTERNET OF NETWORK THINGS

Authors: Tippireddy Srinivasa Reddy*, Prateek, and Rajeev Arya

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Abstract: *The accuracy of the localization process is a critical process of underwater sensor network operation in diverse environments due to the increasing prevalence of coordinated attacks (CoA). In past, research has been conducted on the distribution of underwater sensor networks subjected to CoA. Due to coordinated attacks, it is difficult to maintain reliable communication between anchor nodes, injects false data and impact on localization of moving object. The concept of coordinated attacks is first attempted by formulating a mathematical model that is analyze the effects of anchor nodes on the localization process. To compensate CoA, we propose the Distributed neighborhood localization (DNL) algorithm using the relative triangular neighborhood coordinates. The localization accuracy, transmission range of DNL algorithm is compared with standard CRLB, Gauss Newton, JPDA and SDR. The DNL algorithm improves localization accuracy of moving object by 33.23 % and reduces position error, velocity error by 38.26 %, 35.43 % up to coverage of 400 meters. The implementation of proposed method is utilized in subaquatic internet of sensor networks (SIoSN).*

Keywords: Coordinated attacks, moving object, distributed neighborhood localization, localization error.

Paper ID: 573

Title: DESIGN OF MULTI STAGE FOPD(1+PI) CONTROLLER FOR STABILIZATION OF MAGNETIC LEVITATION PLANT

Authors: Harideo Ram*, Deep Shekhar Acharya, and Shradha Kishore

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Abstract: *This paper demonstrates the design and implementation of a multi-stage Fractional-Order Proportional-Derivative(1+PI),i.e., (FOPD (1+PI)) controller for stabilizing a magnetic levitation plant (MLP). The proposed approach provides a better control over the system dynamics and also improves the accuracy of the closed loop system. The parameters of the proposed controller have been tuned using the Grey Wolf Optimizer (GWO) and the performance of the FOPD(1+PI) have been compared with that of conventional PID, FOPID and PD(1+PI) controllers. It has been found that the proposed control scheme delivers the best performance by eliminating overshoot and reducing the settling time by more than 24%. The accuracy of the closed loop response is also enhanced by reducing the integral square error by at least 16%. The proposed control approach has also been tested for disturbance and noise rejection. It has been observed that the proposed FOPD (1+PI) controller delivers promising performance by significantly rejecting the undesired effects of external influences.*

Keywords: Magnetic levitation, fractional order, multi stage, optimization, FOPD(1+PI)

Paper ID: 575

Title: PERFORMANCE ANALYSIS OF A NOVEL THREE-POINT INTERPOLATED DFT USING CHEBYSHEV POLYNOMIAL APPROXIMATION

Authors: Sweta Prasad* and M. Senthil Kumar

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Abstract: *This paper presents a detailed mathematical model of a novel three-point interpolated Discrete Fourier Transform (TpIpDFT) which is realized with various cosine windows namely Hanning, Blackman and Blackman-Harris. It incorporates a novel correlation of the three greatest magnitudes of the Discrete Fourier transform (DFT) frequency spectrum of the windowed signal to interpolate the magnitude response corresponding to the given input signal. The DFT interpolation function uses Chebyshev polynomial approximation method to estimate the frequency, amplitude and phase angle. It also investigates the performance of TpIpDFT rigorously with different kinds of signals.*

Keywords: Phasor Measurement, frequency estimation, interpolated FT, Window functions, Discrete Fourier Transform.

Paper ID: 576

Title: IMPACT ASSESSMENT ON POWER QUALITY PARAMETERS AT VARIOUS STAGES OF INDUCTION MOTOR DRIVE HARDWARE SETUP

Authors: Paramjeet Singh Jamwa*, Rahul Arora, Dheeraj Kumar and Sanjeev Singh

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Abstract: *For the validation of novel control concepts and power electronics topologies, a laboratory scale hardware setup of Induction Motor Drive (IMD) is created. Such setups usually have a continuously variable auto transformer (CVAT) in power circuit to manually control the input AC Supply. Further, an isolation transformer is used in power circuit to provide electrical isolation between various circuits. In the IMD setup, the use of voltage source converter (VSC) and its operation at high frequency to obtain desired control of parameters at the motor terminals, necessitates use of inductive filters. Therefore, this work presents an impact assessment of inductances used at various stages of a complete drive setup, on the power quality parameters, to figure out the real impact on the drive. A hardware laboratory setup of a 750 W three-phase induction motor (IM) operated by cascaded H-bridge three-level inverter (CHB-3LI) is used and the impact is assessed in the form of total harmonic distortion (THD). The results obtained from the experimental setup are recorded on a power quality analyzer (PQA) and presented to demonstrate a huge impact of inductances on THD in voltage as well as current of the drive.*

Keywords: Diode-Bridge Rectifier, Induction Motor Drive, Power Quality Analyzer, Source Inductance, Total Harmonic Distortion.

Paper ID: 590

Title: MULTIMODAL IRIS FEATURE EXTRACTION AND RECOGNITION USING GABOR FILTERS, WAVELETS AND LOCAL BINARY PATTERNS

Authors: Rahila N A* and Nadera Beevi S

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Abstract: *In order to improve the accuracy of iris biometric authentication, this study introduces a novel multimodal iris feature extraction technique. To provide standardised iris representations, the suggested method uses binary mask segmentation and Daugman's rubber sheet normalization technique. It is followed by a multimodal feature extraction method for iris recognition that combines Gabor filters, Discrete Wavelet Transform (DWT), and Local Binary Patterns (LBP). Gabor filters emphasize minute iris details by using multi-orientation convolution to gather spatial and frequency information. The iris image is broken down into frequency sub-bands (LL, LH, HL, HH) via DWT while maintaining both local and global texture variations. By examining variations in pixel intensity, LBP creates a binary pattern histogram that encodes micro-texture patterns. The system's resilience to changes in light, rotation, and scale is demonstrated through experimental evaluation on the UBIRIS, MMU and IITD iris datasets. Hamming distance metrics are used for identification, and the findings show better performance than single-feature methods and current state of art results. While preserving computing efficiency for real-time deployment scenarios, the modular design makes it easier to adapt to different biometric security applications. The proposed method was evaluated on the UBIRIS, MMU and IITD biometric iris datasets, achieving 100%, 96.51% and 97.48% recognition accuracy, respectively. The execution time for UBIRIS was recorded as 30.3 milliseconds, MMU was recorded as 139.1 milliseconds, while IITD required 709.8 milliseconds, demonstrating high accuracy and computational efficiency in biometric iris recognition.*

Keywords: Iris Recognition, Biometric Authentication, Gabor filters, Discrete Wavelet Transform (DWT), Local Binary Patterns (LBP), Hamming Distance, Feature Extraction, Multimodal Iris Feature Extraction.

Paper ID: 593

Title: TWO PORT OCTAGONAL MIMO ANTENNA FOR KU AND K-BAND APPLICATIONS

Authors: P.V.N. Lakshmi, A. Naveen*, D. Sitaramaiah, B. Srujana, B. Akash

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Abstract: *This paper discusses, A Two Port octagonal slotted MIMO antenna suitable for high-frequency applications, especially Fixed Satellite Service is designed and analysed to increase isolation and overall antenna performance. The suggested design makes use of electromagnetic bandgap (EBG) structure and a novel fractal slot layout. The antenna structure has an optimized defective ground structure (DGS) to enhance impedance matching and reduce mutual coupling and the antenna operates within the frequency range of 17.5 GHz to 19.4GHz. It is constructed on a FR4 substrate with dimensions of 27 × 17 × 1.5 mm3. To assess the antenna's performance, parametric analysis has been done. The results of designed antenna show high diversity gain, a low envelope correlation coefficient (ECC), and an improved isolation, making it utilitarian for Fixed Satellite Service application.*

Keywords: Octagonal MIMO, Isolation, Diversity Gain, EBG, DGS, ECC.

Paper ID: 596

Title: ENHANCED FAULT DETECTION IN TWO-TERMINAL TRANSMISSION LINES USING ITD

Authors: K Harshitha*, S K Nazeer, Kumarraja Andanapalli, V Srinivas and B.K. Chaitanaya

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Abstract: *Transmission lines are responsible for 85-87% of power system defects, making transmission line protection a major subject in power system engineering. The current state of the art in traditional distance relay systems allows for very unambiguous mistake identification and provides excellent discriminating characteristics. However, the relay will fail during this occurrence if there is a critical event on the transmission line, such as a switching event (e.g., a load switching, a capacitor switching, or power fluctuations caused by changes in frequency). Developing a supervisory approach to enhance distance relay characteristics can help prevent this distance relay miss operation. There are new methods for detecting faults and identifying fault sections that are based on Intrinsic Time-scale Decomposition (ITD). Using the instantaneous voltage and current signals measured at each terminal, the actuation signal is a three-phase instantaneous power signal. We next use Intrinsic Time-scale De-composition (ITD) on this signal to get the IMFs, or high-frequency components. Lastly, a threshold value is defined to divide fault occurrences from non-fault events, and an energy index is built from these extracted IMFs. The efficiency of two-terminal 345 kV, 50 Hz transmission circuits is tested under a number of fault situations, each with its own set of parameters that may be varied to increase or decrease the simulation's realism. Evidence from this study demonstrates that the suggested approach maintains sensitivity, speed, and dependability.*

Keywords: Transmission lines, Distance relays, Faults, Switching events, Intrinsic Time Scale Decomposition (ITD).

Paper ID: 598

Title: CAN BUS SYSTEM FOR REAL-TIME AUTOMOTIVE SENSOR MONITORING AND REMOTE WEB VISUALIZATION

Authors: Prachi Salodkar, Sarthak Wankkhede and Sanket Bawistale*

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Abstract: *This paper presents a comprehensive study and implementation of a distributed Controller Area Network (CAN) system for real-time vehicular sensor monitoring and remote visualization. The proposed architecture integrates multiple Arduino-based sensor nodes, each equipped with ultrasonic and pressure sensors, which communicate over a shared CAN Bus to a central master node. The master node aggregates and processes the sensor data before transmitting it via an Ethernet (W5500) module to a web-based interface, enabling remote monitoring with low latency and high reliability. By leveraging the robust error detection and arbitration capabilities of CAN Bus, the system achieves efficient data transmission while significantly reducing wiring complexity. The design includes network bridging to establish a seamless internet connection and support real-time visualization. Experimental evaluations confirm that the system delivers accurate sensor measurements and maintains a rapid update rate, making it suitable for automotive applications. Future work will focus on exploring advanced CAN standards to further enhance data throughput and implement lightweight security measures, including message authentication and encryption, to ensure data integrity and system safety.*

Keywords: IoT, EV, CAN Bus, Communication protocol, Ethernet.

Paper ID: 601

Title: ANALYSIS AND EXPERIMENTAL VALIDATION OF REAL TIME EV LOADING EQUIVALENT TO ELECTRICAL LOADING THROUGH MOTOR GENERATOR SET IN LABORATORY

Authors: Uditveer Singh Rajput*, Sanjeev Singh, Giribabu Dyanamina

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Abstract: *For a realistic performance evaluation of electric vehicles (EVs), the authentication and equivalence of laboratory-based electrical loading and real-world EVs dynamics are most important. While resistive loads are commonly used to test motors EVs motor in the lab through motor-generator set, their correlation with actual EV conditions remains unclear. This study establishes an analogy by analysing the EVs power requirements under varying speeds, weights, and inclinations obtained through performing simulation and comparing them with lab-based resistive load tests. Results show that a 200 kg rider at 30 km/h on a flat road equates to an 800W resistive load, while inclination and added weight significantly increase power demand. However, at 50 km/h, power requirements exceed the lab's resistive load capacity, highlighting its limitations. This analysis provides a critical framework for translating lab experiments into real-world EV performance insights, aiding in efficient motor design and power optimization.*

Keywords: EV load equivalency, mechanical vs. electrical load

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Paper ID: 606

Title: ARTIFICIAL NEURAL NETWORK BASED CHARGE SCHEDULING FOR ELECTRIC VEHICLE IN MICRO-GRID

Authors: Nitish Bhardwaj*, Mayank Singh

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Abstract: *Nowadays, with increase in fuel cost and high carbon emission there is hike in penetration of electromotive vehicles (EVs) in the micro-grid. The unscheduled charging of electric vehicles causes uncontrolled increment in power demand and EV charging cost. This paper presents ANN based optimised charge scheduling for bidirectional power flow between electric vehicle charging stations and grid. This proposed scheme will able to decide the bi-directional mode (G2V or V2G) of charging of EVs on the basis of current state of charge (SoC) and plug-in state during parking time and trip distance. The charging and discharging power of electric vehicles can be estimated using above data and proposed optimization techniques to minimize charging cost for consumers and extra power loading to the grid.*

Keywords: Electric vehicles (EV), State of charge (SoC), Vehicle-to-grid (V2G), Artificial Neural Network (ANN)

Paper ID: 607

Title: EVALUATION OF TEA QUALITY MARKER USING EQUIVALENT CIRCUIT OF VOLTAMMETRIC SYSTEM: A STATISTICAL AND MACHINE LEARNING APPROACH

Authors: Sanjeev Kumar* and Arunangshu Ghosh

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Abstract: *Assessing tea quality is essential for evaluating its market value, flavor profile, and consumer appeal. Traditional methods depend on sensory evaluation and chemical analysis, which are time-consuming and require experienced professionals. This study presents an approach to assessing tea quality markers by utilizing an existing equivalent circuit model of the voltammetric E-Tongue system, integrated with statistical and machine learning techniques. The equivalent electrical circuit parameters such as resistances and capacitances have been extracted for 20 tea samples and correlated with theaflavin (TF) which is an importance biochemical compound (quality marker) found in black tea. The prediction results are obtained with both linear as well as nonlinear regression methods. A comparative analysis of results obtained is also done with that of discrete wavelet transform (DWT) and neighborhood component analysis (NCA). The results show that the proposed method delivers high prediction accuracy, providing a fast and cost-effective alternative to traditional tea evaluation techniques. The concept of considering the equivalent circuit elements as characteristic of the analyze has led to a massive reduction in feature size. This work highlights the potential of integrating the modelling of electrochemical systems with advanced data-driven techniques to establish an efficient and scalable framework for tea quality assessment.*

Keywords: System modelling, E-Tongue, machine learning.

Paper ID: 609

Title: KINEMATIC AND DYNAMIC MODELING OF A 4-DOF LOWER-LIMB EXOSKELETON FOR MOBILITY ASSISTANCE

Authors: Ram Kumar* and D. Ezhilarasi

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Abstract: *The development of lower-limb exoskeletons has gained significant attention due to their potential applications in rehabilitation, assistance, and augmentation of human mobility. This paper presents the kinematic and dynamic modeling of a 4-degree-of-freedom (DoF) lower limb exoskeleton designed to assist human gait. The kinematic analysis uses the Denavit-Hartenberg (D-H) convention to establish the mathematical representation of joint transformations and end-effector positioning. The dynamic model is derived using the Euler-Lagrange formulation, incorporating the effects of human-exoskeleton interaction and ground reaction forces. The study further integrates trajectory planning to ensure smooth and natural motion while considering biomechanical constraints. MATLAB simulation results validate the model by analyzing joint torques, actuator forces, and stability under different gait scenarios. The proposed modeling approach is a foundation for controller design and real-time implementation of lower-limb exoskeletons for mobility assistance and rehabilitation.*

Keywords: Lower-limb exoskeleton, Euler-Lagrange method, Denavit Hartenberg convention, human-exoskeleton interaction, rehabilitation

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Paper ID: 614

Title: AUTOMATIC GENERATION CONTROL IN PV-THERMAL POWER SYSTEM USING PID AND FOPID CONTROLLERS

Authors: Jm Sadique Ahamed, S. Jennathu Beevi*, Mahabuba A

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Abstract: *This paper focuses on designing an optimal controller to regulate frequency in traditional power systems. Proportional Integral Derivative (PID) and Fractional Order Proportional Integral Derivative (FO-PID) controllers are used for frequency control. The study involves a two-area PV-thermal power system. Parameter tuning is achieved through Ant Lion Optimization (ALO) and Moth Flame Optimization (MFO) using the Integral Time Absolute Error (ITAE) as the objective function. The performance of PID and FO-PID controllers is compared, with results showing that the FOPID-ITAE combination delivers better results and improves steady-state performance.*

Keywords: Automatic Generation Control, FO-PID controller, PID controller, optimization, ITAE, ALO algorithm, power system control.

Paper ID: 617

Title: OPTIMAL POWER FLOW USING RIME ALGORITHM

Authors: S ImranKhan *, M. Rama Prasad Reddy

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Abstract: *A major challenge in electrical system optimization is optimal power flow (OPF), which intends to determine efficient operating setups for minimization of objectives while ensuring system limits. This paper proposes a Rime algorithm which is nature influenced formation of rime ice to tackle the OPF challenges for IEEE 30-bus and 57-bus test systems. The Rime technique adopts a layered optimization mechanism that balances exploration and exploitation for more efficient convergence and robustness, much like rime ice gradually develops up under different atmospheric circumstances. To achieve the reduction of fuel expenditure, power losses, and strengthen voltage stability, this technique frequently alters its search process, imitating the variable pattern of the formation of rime ice. In contrast to traditional methods for optimization, simulation results prove that the rime-inspired methodology yields optimal solution, and robustness, emphasizing its ability for use in power system optimization.*

Keywords: Optimal Power Flow, Optimization Techniques, Rime Algorithm.

Paper ID: 618

Title: HIGH-GAIN DC-DC CONVERTER: PERFORMANCE ANALYSIS FOR ELECTRIC VEHICLE CHARGING APPLICATIONS

Authors: R. Vadivelan*, N. Chellammal

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Abstract: *This paper introduces a new topology for a DC-DC converter, exhibiting remarkable performance with its capacity to attain a significant voltage gain. This converter architecture has numerous benefits, including higher efficiency, an extensive output voltage range, and significant voltage gain at a reduced duty cycle – crucial attributes for high-gain DC-DC converters. Traditional boost converters frequently have difficulties in satisfying these continuous needs. This research presents a high-gain DC-DC boost converter to effectively reduce these constraints. The suggested converter has a built-in voltage lift technique that allows for large voltage increases over a wide range of operating conditions. The primary attribute of the converter is its simplicity; it employs merely two power switches regulated by pulse width modulation (PWM), hence enhancing cost-effectiveness and facilitating easier implementation. This capability can be helpful in situations where increasing the voltage is crucial. Moreover, the design is distinguished by its substantial voltage gain with fewer components. The converter is capable of generating a positive output voltage, which is essential for numerous electrical systems. Theoretical results are validated through simulations conducted with Matlab/Simulink software. The converter attained an efficiency of approximately 94.2%, highlighting its viability as a dependable and effective option in contemporary EV charging systems.*

Keywords: DC-DC converter, Enhanced Gain, voltage lift, EV Charging

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Paper ID: 619

Title: ENSEMBLE-BASED ADAPTIVE FILTERING FOR SPEECH ENHANCEMENT IN CAR NOISE CONDITIONS

Authors: Arpit Sharma* and Jeebananda Panda

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Abstract: *In-car noise due to engine vibrations, road friction, and external disturbances considerably impairs the performance of speech communication systems. This paper presents an ensemble-based adaptive filtering system to enhance the quality of speech in such in-car noisy conditions. The suggested method is a combination of the outputs of three adaptive filters-Normalized Least Mean Squares (NLMS), Variable Step-Size LMS (VSLMS) with Mathews' adaptation, and Generalized Maximum Correntropy Criterion (GMCC) by dynamic error-based weighting, in which lower error filters are given greater influence in every iteration. Post-processing methods, such as noise estimate subtraction, pre-emphasis, and de-emphasis, further enhance the output to minimize residual noise. The In-Car noisy speech samples at 5dB & 10dB SNR are sourced from the NOIZEUS speech corpus for performance testing. Objective measures of speech quality, such as PESQ (Perceptual Evaluation of Speech Quality), LLR (Log-Likelihood Ratio), fwsegSNR (frequency-weighted segmental SNR), and COVL (Composite Objective Measure), were employed to measure system performance. The results show that the ensemble-based method robustly outperforms individual filters and static combination methods, providing an effective solution for car noise conditions in real-time speech enhancement.*

Keywords: Adaptive Filter, Ensemble, NOIZEUS, SNR, Pre-Emphasis, De-Emphasis, PESQ, LMS, RLS

Paper ID: 621

Title: ADVANCES IN IMAGE-BASED DISEASE DETECTION: A SURVEY ON DEEP TRANSFER LEARNING APPROACHES

Authors: Faiz e yawer, Souvik Mondal, Sohail Akhtar, and Amrita*

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Abstract: *Plant and crop diseases pose a devastating impact to agricultural productivity, decreasing the quality and quantity of agricultural products and threatening global food security. Due to lack of infrastructure, their identification becomes very challenging. Plant diseases can potentially completely prevent grain harvests in very extreme conditions. Therefore, the automatic detection and diagnosis of plant diseases is widely desired in agriculture. The rapid advancements in Deep Learning (DL) have significantly improved the accuracy and efficiency of image-based disease detection. However, training DL models from scratch requires extensive computational resources and large datasets, which may not always be available, especially for rare and new disease. To address these challenges, Deep Transfer Learning (DTL) has emerged as a powerful approach by leveraging pre-trained Convolutional Neural Networks (CNNs) such as VGGNet, ResNet, InceptionNet, and EfficientNet for efficient disease detection. This paper presents a comprehensive survey on image-based plant disease detection using DTL, analyzing methodologies, datasets, model architectures, and performance metrics. It highlights how pre-trained models fine-tuned using domain-specific data, significantly enhance accuracy while reducing computational costs. Finally, the survey outlines future research directions, emphasizing the need for more generalized models, explainable AI techniques, and datasets for real-time disease detection.*

Keywords: Deep Learning, Convolutional Neural Networks, Deep Transfer Learning, Image Based Disease Detection, Precision Agriculture.

Paper ID: 626

Title: A SURVEY ON SENSORS TECHNOLOGY USED FOR TRAFFIC MONITORING AND SAFETY

Authors: Aaditi Kumari* and Archak Mittal

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Abstract: *Advanced technologies are gaining attention as well as im- portance in modern transportation systems. Intelligent transportation systems(ITS) play a significant role in the advancement of smart cities. The uses of sensors technology are the foundation for smart connec- tivity, enabling the implementation of ITSs. With the increasing com- plexity of road traffic systems and the increasing number of road traf- fic crashes(RTCs), there is a critical need for sophisticated monitoring systems that can handle various traffic scenarios and provide real-time hazard predictions. The integration of innovative approaches such as In- ternet of Things(IoT) with advanced sensors, road infrastructure mon- itoring can be improved[1]. Advance Driver Assistance System (ADAS) and Connected and Automated Vehicles(CAVs) equipped with a variety of sensors to sense their environment. Camera sensors collect informa- tion about the environment, traffic conditions and to detect objects in low visibility conditions. LiDAR measures the distances for both moving and stationary objects. Reliable traffic information is essential for im- plementing intelligent choices to improve traffic conditions on the road. This paper highlights the comparative assessment of different groups of sensors technology and the working principles in order to develop sus- tainable ITS for smart city.*

Keywords: Sensors, Automobiles, Intelligent transportation systems, Traffic monitoring, Road safety.

Paper ID: 634

Title: AN INTRUSION DETECTION SYSTEM USING DEEP LEARNING FOR CYBER THREAT DETECTION IN IOT NETWORKS

Authors: Anantha Venkata Sai Kola*, Lavanya Vemulapalli, Chaitanya Chowdary Ravuri, and Akshara Kanagala

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Abstract: *The rapid expansion of IoT networks requires Intrusion Detection Systems (IDS) to protect critical infrastructure because it faces significant security threats. Traditional IDS struggle against evolving cyber threats due to their reliance on static, rule-based detection. The application of Random Forest and Decision Trees as machine learning-based IDS faces three main challenges with high-dimensional data processing along with accuracy optimization and false alarm reduction. The research introduces an IDS that uses a Fully Connected Neural Network (FCNN) which receives training from the UNSW-NB15 dataset. Our methodology combines feature selection algorithms with hyperparameter adjustment techniques to maximize the classification results. The model provides both binary and multi-attack classification features while enhancing feature extraction during learning, generalization and adaptation beyond traditional machine learning approaches. The FCNN-based IDS exhibits 99.4 percent accuracy for binary classification while reaching 97.6 percent accuracy for multi-attack classification while maintaining a low false positive detection rate. The trained model reaches 99.37 percent binary classification accuracy and 87 percent multi-attack classification accuracy during its evaluation with a new balanced dataset. The research findings show that FCNN successfully boosts intrusion detection capabilities through its scalable and efficient security framework for IoT networks.*

Keywords: IoT network security, Intrusion Detection System (IDS), Fully Connected Neural Network (FCNN), UNSW-NB15 dataset, Binary Classification, Multi-Attack Classification

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Paper ID: 635

Title: IMPROVED BREAST CANCER DETECTION USING OPTIMIZATION-BASED FEATURE SELECTION METHODS

Authors: Pravin Kumar Samanta*, Nirmal Kumar Rout, Ganapati Panda

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Abstract: *Breast cancer remains a significant global health challenge, demanding innovative approaches for early and accurate detection. Machine Learning and deep learning methods are playigan important role for early detection of breast cancer. Feature selection methods are playing important role classifications of positive and negative cases. In this paper three different feature selections methods are implemented wrapping methods, Genetic Algorithm, FOX optimization for classification 6 different machine learning techniques are used for the detection of Breast cancer. Two datasets are used for simulations. Testing and cross validations is also performed for the performance analysis. The performance analysis without feature selection methods is also compared. The results of optimization-based feature selection methods outer-performs. These models can also be implemented for other datasets.*

Keywords: Breast Cancer, Machine Learning, Feature Selection, Meta-heuristic Optimization, Artificial intelligence.

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